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Warm inflation models with bounds inspired by String Theory

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Abstract

Inflation can be the solution to important questions left open by the Standard Big Bang cosmology: how the primordial fluctuations were created, why the Universe is almost perfectly flat and the strong fine-tuning needed on its initial conditions. String Theory is the most serious attempt at a quantum gravity theory, at the moment. The Swampland Program aims to discern, within String Theory, the effective field theories that can be completed (at high energies) in quantum gravity from those that cannot. The conjectures, developed in the Swampland framework, have a direct impact on inflation and cosmology. For example, the need for a flat potential for the inflaton and trans-Planckian inflaton excursions are in tension with Swampland Conjectures and are often difficult to realize also in bottom-up models. So, it is interesting to try to integrate these conditions into inflationary models: in this thesis, it is done for the "minimal warm inflation" one. In particular, we focus on the computation of the maximum number of e -folds allowed by the model. The reason behind this calculation is that we want to see if the cited model would support the idea of a Multiverse that can be realized through the false vacuum decay mechanism. Finally, we performed the computation of a dissipation coefficient (different from the one of the previous model) for a Yukawa interaction between the inflaton and the Standard Model (SM) fermions. This computation is then integrated into an inflation model with a known potential, to see if this, linked to the strong regime condition, would lead to a feasible warm inflation realization.

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Chapter 1

Introduction

The scientific background, on which my thesis is based, is composed of inflation, false vacuum decay, Swampland conjectures and thermal field theory.

Until today the exceptional features of the Universe, namely its almost homogeneity and isotropy, are not well explained by the standard Big Bang theory. For this reason, we dedicate the chapter 2 to review the standard Big Bang cosmology. In that picture, to justify the properties observed in the present Universe we must require a high grade of fine-tuning on the initial conditions concerning non-casually connected spacetime regions. A theory that can avoid this issue is inflation: it consists of a period of exponential expansion in the Early Universe, and it will be the protagonist of chapter 3, which was mainly inspired by [1]. There is a multitude of inflation models, but they can be divided into two macro-categories: standard inflation, in which quantum fluctuations are the mechanism that led to the formation of the cosmological structures observed today, and warm inflation, where the seed of primordial inhomogeneities is due to thermal fluctuations. Until now, the simpler way to implement inflation is by considering a single scalar field minimally coupled to gravity in the slow-roll approximation, which consists of the fact that the potential has not to be too steep allowing the rolling of the inflaton to be slow if compared to the expansion of the Universe.

From the experimental point of view, which will be addressed in section 3.5, we have the Planck 2018 results, which showed measures for cosmic microwave background (CMB), from which we obtained constraints on the power scalar spectrum, the tensor-to-scalar ratio, and the scalar spectral tilt. These data ruled out a lot of the simpler potentials in the standard inflation picture, which are, instead, implementable in the warm inflation models.

Since inflation is a theory that, intrinsically, involves gravitation in its description, it was a natural step to ask if inflation can be seen as an effective field theory, that can be completed in the UV by a quantum gravity theory. And, it is in this framework in which the section 3.6 about String Theory implications on inflation is written. More specifically, that is where the Swampland idea comes in: the most valuable quantum gravity theory is, until today, String Theory, so the aim of the Swampland program is to assume it and try to see if the low energy theories respect the constraints arising from the complete theory. Up to now, a series of conjectures was proposed from examples arising from calculable limits of String Theory [2], giving the first theoretical constraints arising from this theory. These have a direct impact on inflation and cosmology, and for this reason, it is possible to integrate these conditions into interesting inflationary models to see which of these is compatible with String Theory. In this thesis, this procedure will be implemented for the "Minimal warm inflation model" [3], and for this reason, the chapter 4 is

devoted to its study.

In chapter 5, the topic discussed is the theory of false vacuum decay, and it is emphasized how it could be an ingredient for the Multiverse realization. Quantum tunneling between two different vacua is possible in Quantum Field Theory. Since, in String Theory, each vacuum corresponds to different fundamental parameters of the observed four-dimensional Universe, if the process of vacuum decay has taken place several times during inflation, a landscape could have been created with a great number of Universes with different values of the physical constants. In such a way, the fine-tuning needed by our Universe, to obtain the observed values of the Cosmological constant or the Higgs mass, would be explained with a probabilistic justification, which, together with the anthropic principle, could lead to a theoretical, but speculative, solution of the hierarchy problems.

Furthermore, in 5.5, we performed the computation of the number of e -folds necessary to have a sufficient number of bubble nucleation events to support the Multiverse idea.

Finally, the chapter 6 focuses on the computation of a dissipation coefficient to discuss a different model of warm inflation. To do so, we have to introduce the basic concepts of thermal field theory and with its means, we performed the calculation for a Yukawa interaction. Then this result is integrated into a warm inflation model and the section 6.2.3 is dedicated to its study.

Chapter 2

Standard Big Bang cosmology

The Big Bang Theory is a model that succeeds in explaining a wide range of observed phenomena, including the abundance of light elements, the cosmic microwave background (CMB) radiation, and large-scale structure (LSS). The idea is that the Universe expanded from an initial state of high density and temperature, enough to allow for hydrogen nucleosynthetic production, and then cooled down to the present state.

During the early stages of the universe, Big Bang Nucleosynthesis (BBN) produced nuclei other than the Hydrogen-1. These nuclei were produced in the first seconds after the Big Bang when the universe was almost uniform and dominated by radiation. The observed quantities of elements in the Universe are generally consistent with the abundances predicted by BBN¹, which is a compelling argument for the Big Bang theory. Elements that are heavier than lithium are believed to have been produced later in the universe's existence through stellar nucleosynthesis, which occurred during the creation, evolution, and death of stars.

In addition to that, the Big Bang model can explain Hubble's law

$$v = H_0 D \quad (2.1)$$

where v is the recessional velocity of galaxies, D is the proper distance to the object, and H_0 is the Hubble's constant, whose most recent measurement[4] comes from CMB² and amounts to $H_0 = 68.3 \pm 1.5$ km/s/Mpc. The recessional velocity is the rate at which an extragalactic astronomical object recedes from an observer as a result of the expansion of the universe. It can be measured by observing the redshift $z(t)$ which is related to the scale factor $a(t)$ as

$$a(t) \equiv (1 + z(t))^{-1} \quad (2.2)$$

On large scales, our Universe is characterized by an almost perfect homogeneity, i.e. translational invariance, and isotropy, i.e. rotational invariant. If we try to write the most general expression for the metric with these two features, we obtain the so-called Friedmann-Robertson-Walker (FRW) spacetime metric³:

¹To be precise, we are talking about mass abundances of about 75% of hydrogen-1, about 25% helium-4, about 0.01% of deuterium and helium-3, trace amounts of lithium on the order of 10^{-10} , and negligible heavier elements

²For the sake of accuracy, we stress that there is a discrepancy between the value obtained by CMB measurements and one suggested by luminosity arguments (which is higher, roughly 73 km/s/Mpc)

³In this section we use the natural units $\hbar = c = 1$

$$ds^2 = -dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right). \quad (2.3)$$

The convention for the curvature parameter k , given a spacelike hypersurface Σ , is as follows: it is $+1$ for positively curved Σ (that leads to a closed geometry), 0 for flat Σ (flat geometry), and -1 for negatively curved Σ (open geometry). The scale factor $a(t)$ regulates the size of spacelike hypersurfaces Σ at time t .

Initially, the assumptions of statistical isotropy and homogeneity of the Universe allowed us to solve Einstein's equations more easily, but now these features are confirmed by CMB observations. In fact, CMB data show remarkable isotropy, once the dipole component (interpreted as due to the Earth's motion with respect to the CMB frame), and the contribution from the galactic plane are subtracted away. We know that space which is isotropic everywhere automatically implies homogeneity (but the converse is not true) through the "Copernican principle". These features of the Universe allow us to extend the Copernican principle to the so-called "Cosmological principle": there is no privileged observer in the Universe.

Eq. (2.3) can be rewritten, by a coordinate transformation, in the following form:

$$ds^2 = -dt^2 + a^2(t) \left(d\chi^2 + \Phi_k(\chi^2)(d\theta^2 + \sin^2 \theta d\phi^2) \right) \quad (2.4)$$

$$r^2 = \Phi_k(\chi^2) = \begin{cases} \sinh^2 \chi & k = -1 \\ \chi^2 & k = 0 \\ \sin^2 \chi & k = +1 \end{cases} \quad (2.5)$$

We can define the Hubble parameter H as

$$H = \frac{\dot{a}}{a}, \quad (2.6)$$

and the conformal time, which can be interpreted as a clock which slows down with the expansion of the universe, is defined as

$$\tau = \int \frac{dt}{a(t)}. \quad (2.7)$$

The difference of the conformal time computed at two different instants takes the name of comoving particle horizon, so we can write

$$\chi_p(\tau) = \tau - \tau_i = \int_{t_i}^t \frac{dt}{a(t)}. \quad (2.8)$$

The initial time is often taken to be the origin of the universe, $t_i \equiv 0$, defined by the initial singularity, $a(t_i \equiv 0) = 0$. Usually, $t_i = 0$ corresponds to $\tau_i = 0$, but this is not automatic, and in fact depends on the scale factor, and for example it is not the case in inflation.

The cosmological equations of motion are derived from Einstein's equations:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} + \Lambda g_{\mu\nu} \quad (2.9)$$

where the Einstein tensor is defined as $G_{\mu\nu} \equiv \mathcal{R}_{\mu\nu} - \frac{1}{2}\mathcal{R}g_{\mu\nu}$, $\mathcal{R}_{\mu\nu}$ is the Ricci tensor⁴ and the Ricci scalar is $\mathcal{R} = g^{\mu\nu}\mathcal{R}_{\mu\nu}$. Moreover, Λ is the cosmological

⁴We use the comma to denote partial derivatives and semi-colon for total derivatives. The Ricci tensor is defined as $\mathcal{R}_{\mu\nu} = \Gamma_{\mu\nu,\alpha}^\alpha - \Gamma_{\mu\alpha,\nu}^\alpha + \Gamma_{\beta\alpha}^\alpha \Gamma_{\mu\nu}^\beta - \Gamma_{\beta\nu}^\alpha \Gamma_{\mu\alpha}^\beta$ where the Cristoffel symbol is defined as $\Gamma_{\alpha\beta}^\mu \equiv \frac{g^{\mu\nu}}{2}(g_{\alpha\nu,\beta} + g_{\beta\nu,\alpha} - g_{\alpha\beta,\nu})$

constant corresponding to an effective energy-momentum tensor $\Lambda g_{\mu\nu}/8\pi G$ (which we will neglect from now on).

We assume that the matter content of a homogeneous and isotropic Universe can be modeled by a perfect fluid, for which the energy-momentum tensor takes the form⁵:

$$T_{\mu\nu} = -p g_{\mu\nu} + (\rho + p) u_\mu u_\nu \quad (2.10)$$

where $g_{\mu\nu}$ is the space-time metric described by Eq. (2.3), p is the isotropic pressure, ρ is the energy density, and $u = (1, 0, 0, 0)$ is the velocity vector for the isotropic fluid in co-moving coordinates.

From the 00 component of the Einstein equations, (2.9), one can obtain the first Friedmann equation:

$$H^2 = \frac{8\pi G}{3}\rho(a) - \frac{k}{a^2} + \left(\frac{\Lambda}{3}\right) \quad (2.11)$$

If we combine the equation obtained from the trace of Eq. (2.9) with Eq. (2.11) we obtain the second Friedmann equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \left(\frac{\Lambda}{3}\right) \quad (2.12)$$

Finally, the conservation of energy, namely $T_{;\mu}^{\mu\nu} = 0$, leads to the following continuity equation (that can be obtained also combining Eqs. (2.11) and (2.12)).

$$\dot{\rho} = -3H(\rho + p) \quad (2.13)$$

We can express Eq. (2.13) in terms of the state parameter, defined as $w \equiv p/\rho$, as

$$\frac{d \ln \rho}{d \ln a} = -3(1 + w) \quad (2.14)$$

Eq. (2.14) can be integrated to obtain a general formula for the energy density

$$\rho \propto a^{-3(1+w)} \quad (2.15)$$

And, if we combine Eq. (2.15) with the first Friedmann equation (2.11), neglecting the curvature term, we obtain a general expression for the scale factor a

$$a(t) \propto \begin{cases} t^{\frac{2}{3(1+w)}} & w \neq -1 \\ e^{Ht} & w = -1 \end{cases} \quad (2.16)$$

The Universe has lived through different phases, depending on which component was dominating at that moment.

The principal possibilities are (non-relativistic) matter domination (MD), which corresponds to $w = 0$, radiation domination (RD), $w = \frac{1}{3}$, and finally, the cosmological constant case, when $w = -1$. In Table 2.1, we summarize the solutions in the relevant cases

⁵In general, for an imperfect fluid the stress-energy tensor can be written as $T_{\mu\nu} = p g_{\mu\nu} + (\rho + p) u_\mu u_\nu + 2(q_\mu u_\nu + q_\nu u_\mu) + \Sigma_{\mu\nu}$, where q_μ is the energy-flux vector and $\Sigma_{\mu\nu}$ is the symmetric and trace-free anisotropic stress tensor. For the particular case of a perfect fluid, there exists a unique 4-velocity so that $q_\mu = \Sigma_{\mu\nu} = 0$, and the energy-momentum tensor reduces to Eq. (2.10)

	w	$\rho(a)$	$a(t)$	$a(\tau)$	τ_i
MD	0	a^{-4}	$t^{2/3}$	τ^2	0
RD	$\frac{1}{3}$	a^{-3}	$t^{1/2}$	τ	0
Λ	-1	a^0	e^{Ht}	$-\tau^{-1}$	$-\infty$

Table 2.1. FRW solutions for a flat universe dominated by radiation, matter, or a cosmological constant.

The standard framework of Big Bang cosmology is the Λ CDM model, which states that the Universe is composed of three major physical quantities: a cosmological constant, denoted by Λ , associated with Dark Energy, cold dark matter (CDM) and ordinary matter. With this assumption, the mass-energy content of the Universe is divided as 26.8% CDM, 68.2% dark energy, and 5% ordinary matter. Assuming this model, we can rewrite the Friedmann equation (with a cosmological constant), (2.11), in terms of the density parameter as

$$H^2(a) = H_0^2 \left(\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_k a^{-2} + \Omega_\Lambda a^0 \right) \quad (2.17)$$

where we have defined the critical energy density as

$$\rho_{\text{crit}} = \frac{3H^2}{8\pi G} \quad (2.18)$$

and the density parameter for x as

$$\Omega_x = \frac{\rho_x}{\rho_{\text{crit}}} = \frac{8\pi G \rho_x}{3H^2} \quad (2.19)$$

where subscript m stands for all type of matter, r for radiation, k for the curvature and Λ for dark energy

Let's now outline briefly the features of dark matter and dark energy [5].

Cold dark matter

Dark matter constitutes the predominant form of matter in the Universe. It is called dark because it doesn't interact through electromagnetic interaction, so it doesn't absorb/emit radiation, and this fact makes it difficult to detect. Dark matter is massive and accounts for 85%⁶ of the total matter of the Universe. It interacts gravitationally⁷, to not be baryonic and to be collisionless. Cold dark matter refers to the realization in which the particle candidates, that compose it, don't have relativistic speed: this is the simplest way to realize DM because it supports the galaxy/galaxy cluster formation.

Its presence is confirmed by several pieces of observational evidence, we list the most important:

- **Spiral galaxy rotation curves:** if all the matter is luminous, far away from the observable boundaries of the matter distribution, we can approximate the galaxy as a point-like object, and by applying Kepler's Second Law, we expect that the rotation velocities will decrease with distance from the center (as $v \sim \sqrt{R}$), but this is not observed, in fact the velocity remains roughly constant

⁶Since dark energy doesn't contribute to matter, we have $26.8/(26.8+5) \simeq 85\%$.

⁷A second interaction can be considered but it has to be weaker than the SM weak interaction

$v \sim \text{const.}$ A way to solve this discrepancy is by requiring an additional matter contribution different from the standard one, indeed dark matter.

- **Gravitational lensing:** this is one of the consequences of General Relativity, a distribution of matter (such as a galaxy cluster), which is between a light source and an observer can bend light, resulting in a distorted image of the source. By measuring the distorted geometries, one can infer the mass distribution of the gravitational lens, which in this case it is observed to be bigger than the luminous mass of the galaxy cluster alone. Also, in this case, a possible solution is the presence of unseen extra matter, like DM.
- **LSS:** The observed large-scale structures of the Universe cannot be explained only by density perturbations of ordinary matter. In fact, at early times radiation dominated over ordinary matter, not allowing the formation of complex structures. A possible solution is again the presence of DM since it is not affected by radiation, its perturbations could grow first, and the resulting gravitational potential might behave like an attractive potential well for ordinary matter speeding up the structure formation process.

Dark energy

It is an unknown form of energy that affects the Universe on the largest scales, it has the effect of driving the late time accelerating expansion of the Universe. The Dark energy density is very low, roughly 3.7 GeV/m^3 , in contrast to DM or ordinary matter at early times, but it dominates the Universe's mass-energy content at late times because it is constant (it scales as a^0) and uniform across space. The simplest explanation for dark energy is that it is an intrinsic, fundamental energy of space, like the cosmological constant Λ . If we rewrite Eq. (2.12) as a force law we obtain

$$F = -\frac{GM}{R^2} + \frac{\Lambda}{3}R \quad (2.20)$$

which shows that the cosmological constant gives rise to a repulsive force whose value increases with distance: so it can be a solution to the acceleration of the universe. The cosmological constant can be seen as a zero-point vacuum fluctuation, so it respects Lorentz invariance and so can be written as $T_{\mu\nu} = \Lambda g_{\mu\nu}$. In this picture, there is an issue, namely, the vacuum expectation value of the energy-momentum is divergent both for bosonic and fermionic fields, and this gives rise to what is known as the "Cosmological constant problem". If we try to compute Λ , we obtain

$$\frac{\Lambda}{8\pi G} = \langle T_{00} \rangle_{\text{vac}} \propto \int_0^\infty \sqrt{k^2 + m^2} k^2 dk \quad (2.21)$$

Since the integral diverges as k^4 , we get an infinite value for the vacuum energy, if we try to extrapolate the quantum field theory up to infinite energy. More physically, a massive particle or a threshold correction, at the Planck scale, gives a value for $\langle T_{00} \rangle_{\text{vac}}$ that is enormously large with respect to the experimental one: 10^{76} GeV^4 vs 10^{-47} GeV^4 , a difference of 123 orders of magnitude! There is no commonly accepted theoretical solution to this problem, but it is clear that matching the experimental value requires a massive degree of fine-tuning. Furthermore, we point out that the value of the cosmological constant has to be in a very narrow window if we don't want the universe to be a very different place from the one we are used to. To be more quantitative, if $\Lambda/8\pi G < -10^{-43} \text{ GeV}^4$ the universe would recollapse in a

period that is too short for galaxies to form and for life, as we know it, to emerge. While, if $\Lambda/8\pi G > 10^{-43} \text{ GeV}^4$ the universe would have accelerated much before the present epoch, not allowing structure formation and the emergence of life.

Let's now briefly summarise the sources that give indirect evidence of dark energy

- **Type Ia Supernovae:** Type Ia supernovae are widely recognized as good examples of standard candles, that can be used to measure cosmological distances due to their consistent and extreme luminosity. Standard candles are important in astronomy because they allow researchers to determine the intrinsic brightness of objects, which in turn helps to measure the expansion history of the universe. This is achieved by observing the relationship between the distance to an object and its redshift, which indicates how fast it is moving away from us. Hubble's law suggests that this relationship is approximately linear (Eq. (2.1)). While redshift measurements are relatively simple, determining an object's distance is more challenging. However, for standard candles like Type Ia supernovae, the object's distance can be calculated based on its observed brightness. From these distance measurements and their relation to redshift, it was inferred that the Universe has expanded more in the latter half of its life[6]: this acceleration can be imputed to dark energy.
- **CMB:** Based on measurements of cosmic microwave background anisotropies, it appears that the universe is almost flat. In order for this to be true, the mass-energy density of the universe must be equivalent to the critical density ρ_{crit} as defined by Eq. (2.18). Unfortunately, the total amount of matter present in the universe (including both baryons and dark matter) only accounts for approximately 30 % of the critical density as observed through the cosmic microwave background spectrum. This suggests that there must be another type of energy present to make up the remaining 70%, which is believed to be dark energy.
- **LSS:** According to the LSS, only 30% of the critical density is made up of ordinary matter. The WiggleZ survey examined over 200,000 galaxies to establish their redshift. By utilizing the fact that baryon acoustic oscillations (BAO⁸) have created voids of around 150 Mpc in diameter that are surrounded by galaxies, these voids were used as standard rulers to estimate distances to galaxies as far as 2,000 Mpc. This process enabled an accurate calculation of galaxy speeds based on their redshift and distance. The data confirmed cosmic acceleration up to half of the universe's age (7 billion years) and limited its inhomogeneity to 1 part in 10. This showed cosmic acceleration independently of the supernovae argument.

⁸Baryon acoustic oscillations are fluctuations in the density of the visible baryonic matter of the universe, caused by acoustic density waves in the primordial plasma of the early universe

Chapter 3

Inflation

With the term "inflation" we refer to a period of exponentially rapid expansion of spacetime in the early Universe, which could have taken place as early as 10^{-34} s after the Big Bang singularity¹. The inflation hypothesis proposes to explain the cosmic microwave background (CMB) anisotropies and the Universe's large-scale structure (LSS). If we don't want to assume the flatness of the early Universe and its large-scale homogeneity, as was conventionally done in the classical Big Bang theory, inflation could be a solution: in fact, it succeeds in dynamically obtaining the initial conditions necessary for the standard cosmological history (which have to be extremely fine-tuned in the classical approach).

3.1 Problems of standard Big Bang cosmology

Let's introduce the "problems" of the classical Big Bang theory:

The Horizon problem

If we re-express the comoving horizon, Eq. (2.8), as a function of the comoving Hubble radius $(aH)^{-1}$ we obtain that the comoving horizon grows monotonically with time in both RD and MD cases.

$$\tau = \int_0^a \frac{da}{Ha^2} \propto \begin{cases} a^{\frac{1}{2}} & MD \\ a & RD \end{cases} \quad (3.1)$$

This means that comoving scales entering the horizon today have been far outside the horizon at CMB decoupling time, but experimentally from measurements of the CMB we know that the universe was very homogeneous, at the time of last-scattering, on scales encompassing many regions that a priori are causally independent.

Flatness problem

We can rewrite Eq. (2.11) as

$$1 - \Omega_{\text{tot}}(a) = -\frac{k}{(aH)^2} \quad (3.2)$$

¹The exact start time of cosmic inflation is not directly measured, it's a theoretical prediction based on the energy scale of inflation: the higher the energy scale, the shorter the time between the Big Bang and the start of inflation. We refer to [1] for this value.

where we have defined the ratio as

$$\Omega_{\text{tot}}(a) = \frac{\rho(a)}{\rho_{\text{crit}}} \quad (3.3)$$

where the critical energy density ρ_{crit} is defined in Eq. (2.18).

We realize that the quantity $|1 - \Omega_{\text{tot}}(a)|$ diverges with time because, in standard cosmology, the comoving Hubble radius, $(aH)^{-1}$, grows with time. The critical value $\Omega = 1$ is an unstable fixed point. So, to re-obtain the pseudo-flatness observed today ($\Omega_{\text{tot}}(a_0) \approx 1$), without resorting to the inflation hypothesis, we have to fine-tune very carefully the value of $\Omega_{\text{tot}}(a)$ to 1 in the early universe.

The topological defect problem

The hot Big Bang theory predicts that topological defects, such as domain walls, monopoles, cosmic strings, and textures can be created by phase transitions induced by symmetry breaking. Let's briefly discuss the nature of these objects:

- **Domain walls**

They are two-dimensional objects that form, during the phase transition, when discrete symmetry is broken. They characterize a region of space where the potential energy is not minimized, and confine a trapped energy density, which can move but cannot spread out or dissipate. A network of domain walls effectively partitions the Universe into various cells. Furthermore, Domain walls have a repulsive gravitational field.

- **Cosmic strings**

They are 1-dimensional objects that form during an axial/cylindrical² symmetry-breaking phase transition in the early Universe. They are thought to be extremely thin, but at the same time, they have a high density that would make them a possible source of gravitational waves.

- **Monopoles**

These are zero-dimensional (point-like) objects that form when a spherical symmetry is broken. Monopoles are predicted to be supermassive and carry a magnetic charge.

- **Textures**

These form when larger, more complicated symmetry groups are completely broken. Textures are delocalized topological defects that are unstable and therefore, can collapse.

While the presence of monopoles and domain walls would contradict observational evidence about the Universe, cosmic strings and textures can be embedded in some models as a seed of LSS and for CMB anisotropies. These objects would dominate the energy density of the Universe due to their geometry, in fact, in contrast to matter or radiation, they scale as a^{-1} or a^{-2} . Moreover, as pointed out by Kibble[7], if these topological defects could form during cosmological phase transitions they

²it is when the topology of the vacuum manifold associated with the symmetry-breaking is not simply connected

would have formed. So, the issue arises because these types of structures are not observed in today's Universe.

The inflation hypothesis resolves these three problems: the exponential expansion leads to a decrease of the comoving Hubble radius, a fact that is not allowed in conventional cosmology. In fact, we recall that, for a universe dominated by a perfect fluid with equation of state parameter w , the comoving Hubble radius goes as $(aH)^{-1} = H_0^{-1} a^{\frac{1}{2}(1+3w)}$ and, in the standard Big Bang cosmology $w \geq 0$ holds, so the comoving Hubble radius can only increase. If we review Eq. (3.2), we observe that for a decreasing $(aH)^{-1}$, the solution $\Omega(a) = 1$ is now an attractor during inflation (in contrast to the conventional picture, where it was an unstable fixed point), so the Universe tends to flatness.

As we see in Eq. (3.1), a decreasing Hubble radius leads to a decreasing comoving horizon, which implies that large scales entering the present universe were inside the horizon before inflation, so the spatial homogeneity can be explained by causal physics before inflation.

Finally, the non-observation of topological defects can be explained by the fact that the exponential expansion has diluted their presence in the Universe.

Now we want to explore the conditions for the realization of inflation. In the solution of the previous "problems", we have motivated the need for a decreasing comoving Hubble radius, i.e.

$$\frac{d}{dt}(aH)^{-1} < 0 \quad (3.4)$$

From the above condition, we can obtain a second one, namely $\frac{d^2 a}{dt^2} > 0$, simply by taking the time derivative. Finally, we observe that, requiring $\ddot{a} > 0$ in Eq. (2.12), we obtain the relation $p < -\frac{1}{3}\rho$, so a negative pressure is necessary! Such negative pressure can be modeled by the dynamics of a scalar field, which we explore in the next section.

An insightful way to understand the differences between conventional cosmology and inflation is a conformal spacetime diagram: to construct it, we want to express Eq. (2.4) in terms of the conformal time, obtaining

$$ds^2 = a^2(\tau) \left[-d\tau^2 + dx^2 \right] \quad (3.5)$$

We also recall that, in conformal coordinates, null geodesics ($ds^2 = 0$) are at 45° . This is a powerful observation because we can study horizon dynamics in both cases, taking advantage of the fact that light determines the causal structure of spacetime. The following diagram shows the picture of standard Big Bang cosmology, where there is a Big Bang singularity at $\tau_i = 0$, so $a(\tau_i = 0) = 0$.

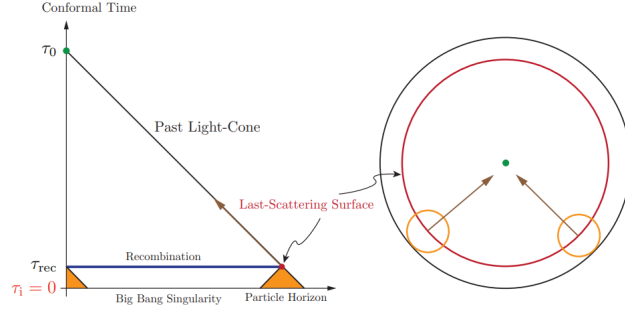


Figure 3.1. Conformal diagram of Big Bang cosmology. Image taken from [2]

We can see that the spacetime points of the last-scattering surface are not in causal contact (their past lightcones do not intersect at the singularity). But this is not a problem anymore in the inflationary picture, since the scale factor during inflation ($H \approx \text{const}$) is $-(H\tau)^{-1}$ and the singularity is now shifted to $\tau_i \rightarrow -\infty$.

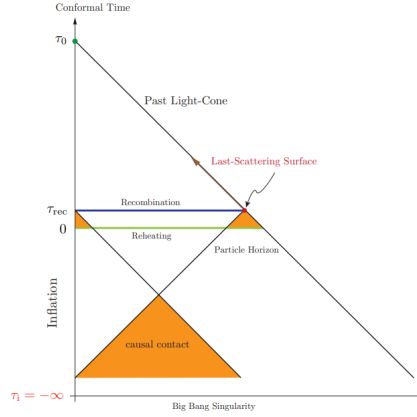


Figure 3.2. Conformal diagram of inflationary cosmology. Image taken from [2]

As we can see the problem of causal connection of the regions at τ_{rec} is brilliantly solved since inflation extends conformal time to negative values.

3.2 Scalar field dynamics

The dynamics of a scalar field minimally coupled to gravity is governed by the following action:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \quad (3.6)$$

where we have defined the Planck mass as $M_{\text{Pl}} = \sqrt{(8\pi G)^{-1}} \sim 2.4 \cdot 10^{18} \text{GeV}$.

The action (3.6) consists of two parts: the gravitational Einstein-Hilbert action, denoted by S_{EH} , and a scalar field action that is composed of a standard kinetic term plus a generic potential V , denoted by S_ϕ . To compute the stress-energy tensor for the scalar field, we use the following equation:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_\phi}{\delta g^{\mu\nu}} = \partial_\mu \phi \partial_\nu \phi - g^{\mu\nu} \left(\frac{1}{2} \partial_\sigma \phi \partial^\sigma \phi + V(\phi) \right) \quad (3.7)$$

From this, we can derive the field equation of motion

$$\frac{\delta S_\phi}{\delta \phi} = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} \partial^\mu \phi) + V_{,\phi} = 0 \quad (3.8)$$

Assuming the FRW metric, Eq. (2.4), and considering the case of a homogeneous field (which only depends on time) the scalar stress-energy tensor $T^{\mu\nu}$ describes a perfect fluid, i.e. Eq. (2.10), with the following expression for the pressure and the energy density:

$$p_\phi = \frac{\dot{\phi}^2}{2} - \frac{1}{6a^2} (\nabla \phi)^2 - V(\phi) \quad (3.9)$$

$$\rho_\phi = \frac{\dot{\phi}^2}{2} + \frac{1}{2a^2} (\nabla \phi)^2 + V(\phi) \quad (3.10)$$

By considering a homogeneous region, we can ignore the gradient terms of Eqs. (3.9) and (3.10). Then we can focus on a scalar field potential that dominates the universe's energy density for some period in the early universe, allowing for a sufficient period of inflation. This usually has to last at least 60 e -folds, which corresponds to an expansion of the Universe by a factor³ of 10^{26} to solve the flatness and the horizon problems.

Substituting Eqs. (3.10) and (3.9) in Eq. (2.13) we obtain

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0 \quad (3.11)$$

where the prime denotes a derivative with respect to ϕ .

Alternatively, it is possible to derive the previous equation from Eq. (3.8).

Now, if we plug Eq. (3.10) into Eq. (2.12) we get:

$$H^2 = \frac{1}{3M_{\text{Pl}}^2} \left(V(\phi) + \frac{1}{2} \dot{\phi}^2 \right) \quad (3.12)$$

The two equations Eqs. (3.11) and (3.12), that we have just derived, determine the dynamics of a homogeneous scalar field in an FRW geometry (with $\Lambda = 0$).

3.3 Slow-roll approximation

In this section, we introduce the concept of slow-roll and the conditions arising from it. We can define the slow-roll parameter ϵ as

$$\epsilon \equiv \frac{\dot{\phi}^2}{2H^2} = -\frac{\dot{H}}{H^2} = -\frac{d(\ln H)}{dN_e} \quad (3.13)$$

where $dN_e = H dt = d(\ln a)$ measures the number of e -folds N_e of inflationary expansion.

³an e -fold is the time necessary for an exponentially growing quantity to increase by a factor of e . The statement holds because $e^{60} \sim 10^{26}$

The slow-roll parameter ϵ can be rewritten as a function of the equation of state parameter w as

$$\epsilon = \frac{3}{2}(w_\phi + 1) \quad (3.14)$$

Accelerated expansion occurs if $\epsilon < 1$, while the de Sitter limit, $p_\phi \rightarrow \rho_\phi$, implies $\epsilon \rightarrow 0$: in this case, the potential dominates over the kinetic energy, namely $\dot{\phi}^2 \ll V(\phi)$.

Accelerated expansion has to last a sufficiently long period, which means $\ddot{\phi}$ also has to be small enough, namely $\ddot{\phi} \ll 3H\dot{\phi}, V_{,\phi}$. We then introduce a second slow-roll parameter:

$$\eta = -\frac{\ddot{\phi}}{H\dot{\phi}} = \epsilon - \frac{1}{2\epsilon} \frac{d\epsilon}{dN} \quad (3.15)$$

The second form of Eq. (3.15) with the condition $\eta < 1$ tells us that the fractional change of ϵ per e -fold is small. Equivalently, we can introduce slow-roll parameters in terms of V and its derivatives:

$$\epsilon_V \equiv \frac{M_{\text{Pl}}^2}{2} \left(\frac{V'}{V} \right)^2 \quad (3.16)$$

$$\eta_V \equiv M_{\text{Pl}}^2 \frac{V''}{V} \quad (3.17)$$

The potential slow-roll parameters are linked to Hubble ones in the following way: $\epsilon_V \approx \epsilon$ and $\eta \approx \eta_V - \epsilon_V$.

In the slow-roll regime $\epsilon_V, \eta_V \ll 1$, so Eq. (3.12) reduces to

$$H^2 \approx \frac{V(\phi)}{3M_{\text{Pl}}^2} \approx \text{const} \quad (3.18)$$

We note that H can be considered roughly constant during inflation due to the almost flatness of the potential.

While for the Eq. (3.11), we have

$$\dot{\phi} = -\frac{V'}{3H} \quad (3.19)$$

So we can point out that the spacetime was approximate de Sitter, namely $a(t) = e^{Ht}$, with de Sitter a cosmological solution to Einstein's field equations of General Relativity. It models the universe as spatially flat, neglecting ordinary matter, so the dynamics of the universe is governed by a positive cosmological constant that sets the expansion rate $H \propto \sqrt{\Lambda}$. Inflation ends when the slow-roll conditions are violated, like $\epsilon(\phi_{\text{end}}) = 1$.

We can also define the number of e -folds as:

$$N_e(\phi) \equiv \ln \left(\frac{a_{\text{end}}}{a} \right) = \int_t^{t_{\text{end}}} H dt = \int_\phi^{\phi_{\text{end}}} \frac{H}{\dot{\phi}} d\phi = \int_{\phi_{\text{end}}}^\phi \frac{1}{M_{\text{Pl}}^2} \frac{V}{V'} d\phi \quad (3.20)$$

The last equality is achieved by manipulating the expression of $N_e(\phi)$ using the background evolution equations, i.e. Eqs. (3.19) and (3.18).

We can determine the field's value when the CMB fluctuations have left the horizon based on experimental measurements. This occurred approximately 40–60 e -folds before the end of inflation, although the exact value depends on factors such as reheating and the universe's post-inflationary thermal history. This information provides us with a mathematical constraint:

$$\int_{\phi_{\text{end}}}^{\phi_{\text{cmb}}} \frac{V}{V'} d\phi \approx 40 - 60 \quad (3.21)$$

3.4 Quantum fluctuations in inflation

This section focuses on quantum fluctuations and how they can lead, in the inflation picture, to the formation of cosmological structures that we observe today.

From observations of the CMB, we know that the early universe was not perfectly homogeneous, it had small inhomogeneities, at the 10^{-5} level, so they can be treated as linear perturbations around a homogenous background. This idea implies that all quantities $X(t, x)$, namely metric $g_{\mu\nu}$ and matter fields like ϕ , ρ , and p , can be written as the sum of a homogeneous background $\bar{X}(t)$, that depends only on cosmic time, and a spatially dependent perturbation

$$\delta X(t, x) = X(t, x) - \bar{X}(t) \quad (3.22)$$

Thanks to the symmetries of the spatially flat, homogeneous, and isotropic background we can perform the SVT decomposition, namely, we can decompose stress-energy tensor and metric perturbations into independent scalar, vector, and tensor components. This fact is possible thanks to the rotational invariance of the linear equations of motion for perturbations, while the translational invariance leads, in Fourier space, to the independence of different Fourier modes: for this reason, the SVT decomposition is easier if it is performed in Fourier space. On the other hand, in real space, this fact implies that tensors, vectors, and scalars have different transformation properties on spatial hypersurface; so, practically, for computations, this implies that each type of perturbation evolves independently, at the linear order, and so they can be analyzed separately.

Regarding the gauge choice, in fact, the splitting of Eq. (3.22) is not unique but depends on the choice of coordinates. The physics is independent of the choice of coordinates, but if one is not careful, this gauge dependence of perturbations can lead to confusion.

Let's start with an example: we consider as a starting point an unperturbed homogeneous and isotropic Universe with an energy density that is only a function of time $\rho(x, t) = \rho(t)$. Now we consider a new time coordinate $\tilde{t} = t + \delta t(x, t)$, in general, the energy density, with the new time coordinates, will not be homogeneous, $\tilde{\rho}(\tilde{t}, x) = \rho(t(\tilde{t}, x))$. So the $\delta\rho$ that appears is fictitious and is due to the new choice of coordinates. With analogous reasoning, we can remove a real perturbation with a "ad hoc" choice of coordinates. So, to avoid such ambiguities, a solution is considering the complete set of perturbations, i.e. switching from matter field to tensor perturbations by gauge transformations, and using gauge-invariant combinations of perturbations (by definition, fluctuations of gauge-invariant quantities cannot be removed by a coordinate transformation). An important gauge invariant quantity is the comoving curvature perturbation R , which is defined as

$$R \equiv \Psi - \frac{H}{\bar{p} + \bar{\rho}} \delta q = \Psi + \frac{H}{\dot{\phi}} \delta \phi \quad (3.23)$$

where Ψ is a spatially flat hypersurface and δq is the scalar part of the 3-momentum density $T_i^0 = \partial_i \delta q$. The second equality is valid only during inflation when $T_i^0 = -\dot{\phi} \partial_i \delta \phi$.

A statistical measure of the primordial scalar fluctuations is the power spectrum of R . If we consider R to be Gaussian, then all the statistical information is encoded in the power spectrum, while eventual primordial non-gaussianities will be contained in higher order correlation functions of R . Physically we can initially neglect non-gaussianities, since they are predicted to be small in single-field slow-roll inflation.

$$\langle R_k R_{k'} \rangle = (2\pi^3) \delta(k + k') P_R(k) \quad (3.24)$$

where $\langle \dots \rangle$ defines the ensemble average of fluctuations, k is the Fourier wavevector and

$$\Delta_R^2 = \frac{k^3}{2\pi^2} P_R(k) \quad (3.25)$$

The scalar spectral index defines the scale-dependence of the power spectrum:

$$n_s - 1 = \frac{d \ln \Delta_R^2}{d \ln k} \quad (3.26)$$

The trend of the power spectrum is often modeled as:

$$\Delta_R^2 = A_s(k^*) \left(\frac{k}{k^*} \right)^{n_s(k^*)} \quad (3.27)$$

with k^* as pivot scale.

We can proceed analogously for the tensor case defining, for each polarization of h_{ij} , i.e h_+ and h_x :

$$\langle h_k h_{k'} \rangle = (2\pi^3) \delta(k + k') P_h(k) \quad (3.28)$$

$$\Delta_h^2 = \frac{k^3}{2\pi^2} P_h(k) \quad (3.29)$$

And, we define the power spectrum of tensor perturbations as the sum of the power spectra for the two polarizations: $\Delta_t^2 = 2\Delta_h^2$.

Let's discuss how inflation connects microscopic physics to macroscopic observations. During inflation, quantum fluctuations are generated at all length scales, represented by different wavenumbers, or k values. However, the significant ones for cosmology are those within the Hubble radius (horizon), which is when $k \gg aH$. As inflation progresses, the comoving Hubble radius shrinks, and eventually, the fluctuations exit the horizon (for a constant k) when $k < aH$. On the other hand, gauge invariant objects, like R , remain constant after the horizon exit, so they are not impacted by post-inflation phenomena such as reheating. Once they re-enter the horizon due to the growth of the comoving Hubble radius, they can be used to

predict the universe's perturbations that lead to the CMB and LSS observations. This mechanism is well-summarized in the figure below:

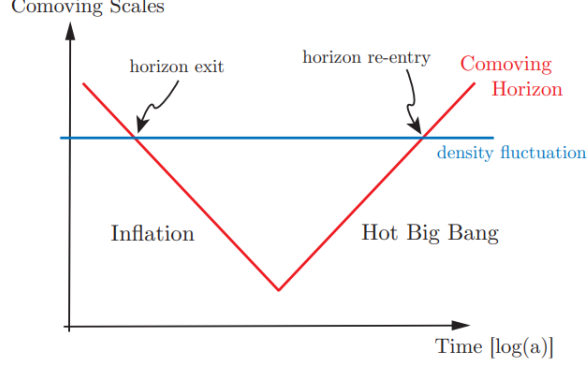


Figure 3.3. Creation and evolution of perturbations in the inflationary universe.

Now we want to illustrate the computation of the scalar and tensor power spectra from first principles but with the assumption of gaussianity of R and in the slow-roll approximation that allows us to treat the background spacetime as a slowly time-evolving quasi-de Sitter space. This fixes the mode functions completely and their large-scale limit is hence determined. Finally, we compute the power spectrum of curvature fluctuations at horizon crossing.

3.4.1 Scalar perturbations

Let's consider the single-field slow-roll models of inflation with Eq. (3.6) as the associated action. We choose a gauge such that the inflaton field is unperturbed (comoving gauge) and all scalar degrees of freedom are parameterized by the metric fluctuation $R(x, t)$, this translates in the three conditions:

$$\begin{aligned}\delta\phi &= 0 \\ g_{ij} &= a^2 [(1 - 2R)\delta_{ij} + h_{ij}] \\ \partial_i h_{ij} &= 0\end{aligned}\tag{3.30}$$

Given the invariance of R after horizon exit, we can limit the computation at the horizon crossing point. We expand the scalar part of the action for single-field slow-roll inflation, i.e. Eq. (3.6), to second-order fluctuations, obtaining⁴ the second-order expansion of the action in terms of R :

$$S_{(2)} = \frac{1}{2} \int d^4x \, a^3 \frac{\dot{\phi}^2}{H^2} \left[\dot{R} - a^{-2} (\partial_i R)^2 \right]\tag{3.31}$$

This approach ensures the correct normalization for the quantization of fluctuations. Now we perform the change of variable $v \equiv zR$ with $z^2 \equiv a^2 \frac{\dot{\phi}^2}{H^2}$ and switching to conformal time τ the action (3.31) becomes:

$$S_{(2)} = \frac{1}{2} \int d\tau \, d^3x \, a^3 \left[(v')^2 + (\partial_i v)^2 + \frac{z''}{z} v^2 \right]\tag{3.32}$$

⁴The derivation is in Appendix B of [1]

with prime denoting a partial derivative with respect to the conformal time τ . From the action (3.32), after Fourier expanding the field v , we derive the equation of motion

$$v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k = 0 \quad (3.33)$$

This is called the Mukhanov equation, it is in the harmonic oscillator form. However, the mode equation for v is hard to solve exactly since z depends on the background dynamics. So, we will discuss approximate analytical solutions in the de Sitter limit and the slow-roll approximation. To quantize the field v we follow the usual procedure for the harmonic oscillator: we promote v and v' to quantum operators

$$v \rightarrow \hat{v} = \int \frac{d^3k}{(2\pi)^3} \left[v_k(\tau) \hat{a}_k e^{ikx} + v_k^*(\tau) \hat{a}_k^\dagger e^{-ikx} \right] \quad (3.34)$$

and to satisfy the canonical commutation relation for creation and annihilation operators the following normalization condition for v has to hold:

$$\langle v_k, v_k \rangle \equiv \frac{i}{\hbar} (v_k^* v_k' - v_k'^* v_k) = 1 \quad (3.35)$$

Eq. (3.35) doesn't fix the mode function completely, and to define the vacuum state we match our solutions to the Minkowski vacuum in the ultraviolet, i.e. on small scales when the mode is deep inside the horizon. Mathematically, for $\tau \rightarrow -\infty$, or $k \gg aH$, Eq. (3.33) takes the form of a simple harmonic oscillator with time-independent frequency, namely

$$v_k'' + k^2 v_k = 0 \quad (3.36)$$

So, requiring that the vacuum is the minimum energy state and solving Eq. (3.36), we obtain the second boundary condition:

$$\lim_{\tau \rightarrow -\infty} v_k = \frac{e^{-ik\tau}}{\sqrt{2k}} \quad (3.37)$$

In the de Sitter background, $\frac{z''}{z} = \frac{2}{\tau^2}$ so if we solve now the Eq. (3.33) with conditions (3.35) and (3.37) we have:

$$v_k = \frac{e^{-ik\tau}}{\sqrt{2k}} \left(1 - \frac{i}{k\tau} \right) \quad (3.38)$$

Now we apply the Eqs. (3.24) and (3.25) to compute the power spectrum of $R = \frac{H}{\phi} \psi_k \equiv \frac{H}{\phi} a^{-1} \hat{v}_k$ at horizon crossing $a(t_*)H(t_*) = k$ obtaining

$$\Delta_R^2(k) = \frac{H_*^2}{(2\pi)^2} \frac{H_*^2}{\dot{\phi}_*^2} \quad (3.39)$$

The use of the "*" subscript indicates that the quantity is being evaluated at the time of horizon crossing.

In the slow-roll approximation Eq. (3.39) becomes

$$\Delta_R^2(k) \approx \frac{1}{12\pi^2} \frac{1}{M_{\text{Pl}}^6} \frac{V^3}{V'^2} \quad (3.40)$$

while the spectral index can be computed with Eq. (3.26) and can be expressed as a combination of slow-roll parameters, Eqs. (3.16) and (3.17)

$$n_s - 1 = 2\eta_V^* - 6\epsilon_V^* \quad (3.41)$$

3.4.2 Tensor perturbations

For this computation, we can start from the second-order expanded EH action:

$$S_{(2)} = \frac{M_{\text{Pl}}^2}{8} \int d\tau d^3x a^2 \left[(h'_{ij})^2 - (\partial_l h_{ij})^2 \right] \quad (3.42)$$

and, repeating the steps done in the scalar case, we arrive at the final result for the power spectrum for tensor modes:

$$\Delta_h^2 = \frac{2}{\pi^2} \frac{H_*^2}{(M_{\text{Pl}})^2} \quad (3.43)$$

In the slow-roll approximation Eq. (3.43) becomes

$$\Delta_h^2(k) \approx \frac{2}{3\pi^2} \frac{V}{M_{\text{Pl}}^6} \quad (3.44)$$

We can define also the tensor-to-scalar ratio r as:

$$r \equiv \frac{\Delta_h^2}{\Delta_R^2} = 8 \left(\frac{V'}{V} \right)^2 \quad (3.45)$$

The equal sign in Eq. (3.45) holds in the slow-roll approximation.

3.5 Experimental constraints

This section briefly summarises all the observational constraints for inflation and how they can be obtained by defining precise relations between the scalar and tensor power spectra, and the observed CMB anisotropies and the galaxy power spectrum. In section 3.4, we computed the primordial perturbations at the time of horizon exit τ_* , but to parametrize the quantity W_k , which is effectively measured by experiments, we have to define:

$$W_k(\tau) = T_W(k, \tau, \tau_*) R_k(\tau_*) \quad (3.46)$$

where T_W is the transfer function between R fluctuations at time τ_* and W fluctuations at some later time τ .

3.5.1 CMB

The cosmic microwave background is a microwave radiation that fills all space in the observable Universe, it can be considered a relic radiation of the early times. Let's now outline the mechanism that leads to the CMB. The Universe passed through a phase, after 10^{-6} seconds after the Big Bang, where it was filled with a uniform

glow of interacting plasma composed⁵ of photons, electrons, and quarks. Due to the expansion of the Universe, the energy density of the plasma decreased, by adiabatic cooling, until it became energetically favorable for electrons and protons to form Hydrogen atoms. At that point, since neutral nuclei could no longer interact with thermal radiation, through Thomson scattering, the Universe became transparent to electromagnetic radiation leading to the matter-radiation decoupling. So, the photons began to travel freely through the Universe and constitute what is called today CMB. The temperature when the decoupling happened was around 3000 K, but today, due to redshift, the CMB can be seen as an emission of a uniform black body at a temperature of 2.72548 ± 0.00057 K. According to the Big Bang model, the radiation from the sky we measure today comes from a spherical surface called the "surface of last scattering". This represents the set of locations in space at which decoupling occurred at a time such that the photons from that distance have just reached the observer.

The Fig. 3.4 shows the nine-year WMAP heat map of temperature fluctuations (which are of the order of 10^{-5}) in the cosmic microwave background:

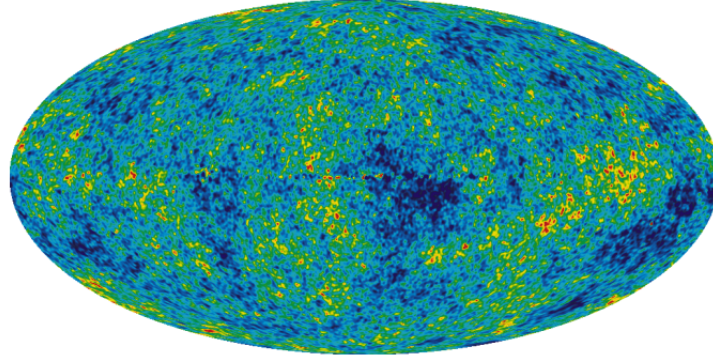


Figure 3.4. Temperature fluctuations in the CMB. Blue spots represent directions in the sky where the CMB temperature is $\sim 10^{-5}$ below the mean, $T_0 = 2.7K$. Yellow and red indicate hot regions. Figure taken from [1]

The statistical properties of these fluctuations contain important information about the background evolution and the initial conditions of the Universe. By performing a harmonic expansion⁶ of the quantity $\Delta T/T_0$, then combining the multipole coefficients in an invariant angular power spectrum C_l^{XY} , and, finally, assuming that CMB temperature fluctuations are dominated by the scalar modes R (for low values of the tensor-to-scalar ratio r), the following relation can be obtained [1]:

$$C_l^{XY} = \frac{2}{\pi} \int k^2 dk P(K) \Delta_{Xl}(k) \Delta_{Yl}(k) \quad (3.47)$$

with the transfer functions $\Delta_{Xl}(k)$ that respect the relation

⁵Plasma is a state of matter in which an ionized substance becomes highly electrically conductive to the point that long-range electric and magnetic fields dominate its behavior. It's the most abundant form of ordinary matter in the Universe, mostly in stars, but also dominates the intergalactic/interclusters mediums

⁶We use the standard spherical harmonics on a 2-sphere with $l = 0, 1, 2$ that correspond respectively to mono/di/quadrupole

$$\Delta_{XI}(k) = \int_0^{\tau_0} d\tau S_X(k, \tau) P_{XI}(k[\tau_0 - \tau]) \quad (3.48)$$

where we have defined the inflationary spectra as $P(k) \equiv \{P_R(k), P_h(k)\}$. X, Y can be the temperature T or the tensor modes B, E .

The E and B tensor modes are two scalar fields that completely specify the linear polarization field. E-mode polarization vectors are radial around cold spots and tangential around hot spots in the sky, while B -mode polarization vectors have vorticity around any given point in the sky.

We note that the transfer function can be parametrized as the product of a source term $S_X(k, \tau)$ and a geometric projection term $P_{XI}(k[\tau_0 - \tau])$.

The symmetries of temperature and polarization (E and B modes) allow the computation of four correlation functions $\Delta_{XI}(k)$: TT, EE, TE (these three combinations affect $P_R(k)$), and BB (which affects $P_h(k)$).

According to 2018 Planck[8] TT, TE, EE + lowE + lensing data, the amplitude of scalar perturbations, see Eq. (3.27), is known to be $A_s = 2.0989^{+0.0296}_{-0.0292} \cdot 10^{-9}$, the scalar spectral index is measured to be $n_s = 0.9649 \pm 0.0042$ at 68% C.L.

For the tensor to scalar ratio r (computed at the pivot scale of $k^* = 0.002 Mpc^{-1}$), combining 2018 Planck with BICEP2/Keck Array 2014 B-mode polarization data, we have $r_{0.002} < 0.064$ with 95% C.L.

3.5.2 LSS

The LSS of the Universe can give us information about early times, in particular, the galaxy (or dark matter) power spectrum is a measure of the spectrum of primordial curvature fluctuations if the effects of subhorizon evolution are taken into account. To study them, we define the density contrast $\delta \equiv \delta\rho/\bar{\rho}_0$, denoted as δ_g for galaxies and δ for dark matter. Usually, we assume that galaxies are (biased) tracers of the underlying dark matter distribution, so $\delta_g = b\delta$, and if we find an independent way to estimate b we can infer the dark matter distribution from the one of the galaxies. Generally, the late-time dark matter power spectrum can be related to the primordial spectrum of curvature fluctuations as:

$$P_\delta(k, \tau) = \frac{4}{25} \left(\frac{k}{aH} \right)^4 T_\delta^2(k, \tau) P_R(k) \quad (3.49)$$

The numerical and k prefactors are conventional.

T_δ is the dark matter transfer function, and intuitively we can consider

$$T_\delta(k) \approx \begin{cases} 1 & k < k_{eq} \\ \left(\frac{k_{eq}}{k} \right)^2 & k > k_{eq} \end{cases} \quad (3.50)$$

During the radiation dominated era, fluctuations are limited due to high radiation pressure, resulting in $\delta \sim \ln a$. However, in the matter dominated era, gravitational collapse becomes more effective as the background pressure becomes negligible, resulting in $\delta \sim a$. In most cases, the transfer function should be calculated numerically, because the transition in the spectrum at matter-radiation equality (i.e. $k \approx k_{eq}$) of Eq. (3.50) is not accurate enough.

Therefore, the dark matter transfer function can be calculated using the background cosmological parameters and used to relate to the inflationary spectrum $P_R(k)$ and the dark matter power spectrum $P_\delta(k, \tau)$.

3.6 String Theory conjectures and inflation

This section refers mainly to [2, 9, 10, 11].

In this section, we will focus on the constraints arising from String Theory conjectures, which would guarantee the compatibility of a generic theory (inflation in our case) with a quantum gravity theory. It is always assumed that inflation can be treated in the low-energy effective field theory (EFT) formalism, since the energy scale of inflation was below the Planck scale when the currently observable scale exited the Hubble horizon. We want to see if among these EFTs some can be embedded into a quantum theory of gravity, in our case String Theory. This shows the need for criteria that distinguish those theories that belong to the landscape of String Theory from the others that are in the Swampland; these criteria take the name of "Swampland conjectures". They are the first theoretical constraints arising from String Theory and have direct, non-trivial effects on cosmology and inflation. The String Theory Swampland is defined as the set of (apparently) consistent effective field theories that cannot be completed into quantum gravity in the ultraviolet, while the landscape consists of those theories compatible with String Theory, an illustrative representation is given in Fig. 3.5.

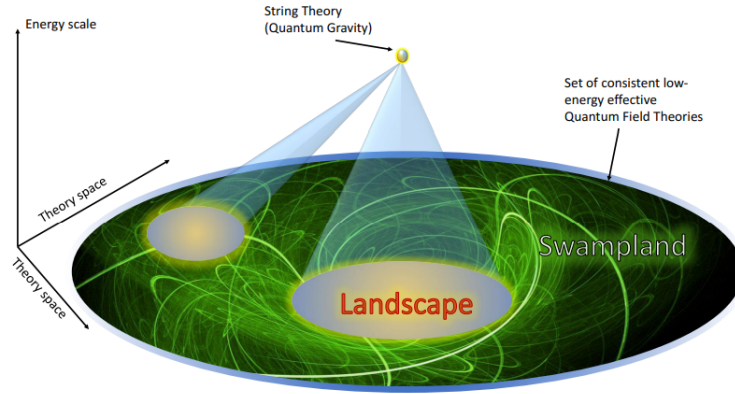


Figure 3.5. A schematic illustration of the space of (apparently) self-consistent effective quantum field theories. The sub-set which could arise from String Theory is the string Landscape, while all the other theories are in the string Swampland

Notice that the process of constraining inflation with String Theory conjectures is quite speculative. There are, in fact, three assumptions: to start we don't actually know if String Theory is the correct quantum theory of gravity. Secondly, the conjectures are not proven from a microscopic physics perspective, but derive from some examples concerning well-known limits within String Theory. Finally, their application to phenomenology often needs extra assumption to match known quantities with String Theory ones.

In addition to these top-down considerations, we can stress the fact that some of the implications arising from the Swampland conjectures overlap with requests that we need in a bottom-up approach. In particular, flat potentials are not easy to obtain from first-principles QFT model building, since quantum/thermal corrections that arise for masses or couplings tend to spoil the flatness of the potential itself. Furthermore, transplanckian field excursions can create problems due to higher-dimensional operators: if we consider that these $D > 4$ operators are suppressed

by a scale comparable to M_{Pl} , then they start to become important when the field excursion approaches M_{Pl} .

In this section, we want to briefly discuss the principal conjectures that have important implications for inflation, namely the Swampland distance conjecture, the de Sitter conjecture, and the Trans-Planckian censorship conjecture.

3.6.1 Swampland distance conjecture

The Swampland distance conjecture (SDC), once applied to inflation, tries to formalize and quantify the possibility of constructing large-field inflation⁷ (in String Theory). The (Refined) Swampland conjecture states [2]: "Consider a theory, coupled to gravity, with a moduli space M which is parametrized by the expectation values of some field ϕ_i which have no potential. Let be the geodesic distance between any point $P \in M$ and another point $Q \in M$ denoted as $d(P, Q)$. There exists an infinite tower of states⁸, with an associated mass scale M , such that $M(Q) < M(P)e^{-\alpha \frac{d(P, Q)}{M_{\text{Pl}}}}$ if $d(P, Q) \geq M_{\text{Pl}}$, where α is some positive constant. This statement holds even for fields with potential, not just for moduli, where the moduli space is replaced with the field space in the effective theory" (such as in the case of inflation). The term moduli is used, in physics, to refer to a scalar field with gravitational-strength couplings that has vanishing potential at some level of approximation. Some moduli have exactly vanishing potential before supersymmetry breaking, while others have vanishing classical potential but obtain a mass from quantum effects. The second part of the conjecture also impacts the type of potentials that may be allowed in quantum gravity. In particular, the cutoff Λ of an EFT sets the scale for energy densities so that the potential is bounded as $V(\phi)^{\frac{1}{4}} < \Lambda$.

Then it is believed that there is a finite radius in field space where the effective action (such as Eq. (3.6)) is valid. In particular, if we go a large distance D in field space, as stated before, a tower of light modes appears with mass scale $M \sim M_{\text{Pl}} e^{-\alpha D}$ which invalidates the above effective Lagrangian (where $\alpha \sim M_{\text{Pl}}^{-1}$). This means that any effective Lagrangian is valid for

$$\Delta\phi < \Delta \sim \mathcal{O}(M_{\text{Pl}}) \quad (3.51)$$

3.6.2 de Sitter conjecture

The Swampland de Sitter conjecture (SdSC) arises from the fact that constructing de Sitter vacua in String Theory is challenging. This might be because the theory is supersymmetric, whereas de Sitter space is not. Additionally, stabilizing all the moduli required for a de Sitter vacuum is a difficult problem, with few understood mechanisms to accomplish it. These technical difficulties may indicate a fundamental obstruction to creating exact de Sitter solutions in String Theory. Therefore, it is possible that de Sitter vacua are in the Swampland.

At first glance, this statement may contradict what we have observed about the Universe. We have strong evidence that the Universe is currently experiencing a late-time acceleration, which is believed to be caused by a positive cosmological constant. However, it is important to note that inflation, another phase of acceleration that the

⁷Models of large-field inflation are those where the inflaton ϕ undergoes a super-Planckian excursion in field space

⁸Usually the tower of states is thought of as a collection of particles, but this is not mandatory; in fact, an alternative could be the presence of light extended objects, as is the case in some String Theory examples.

universe likely went through, was not caused by a cosmological constant. Instead, it was driven by a scalar field rolling down a potential. Based on this idea, explaining the current late-time acceleration by a similar mechanism involving scalar fields is not unreasonable. The de Sitter conjecture proposes a lower bound on the gradient of potentials in positive regions. Quantitatively, the scalar potential of a theory coupled to gravity must satisfy a bound on its derivatives with respect to scalar fields:

$$|\nabla V| \geq \frac{c}{M_{\text{Pl}}} V \quad (3.52)$$

where $c > 0$ is a constant of order one. The ∇ indicates that one must consider the vector of derivatives with respect to all the scalar fields in the theory. To summarise, with this condition, the SdSC conjecture disallows flat potentials that resemble a positive cosmological constant, effectively ruling out de Sitter minima, but not excluding other types of critical points.

3.6.3 Trans-Planckian censorship Conjecture

The ideas of this subsection are borrowed by [12, 13]. The Trans-Planckian censorship Conjecture (TCC) tries to formalize the Trans-Planckian problem of inflationary models and to extrapolate some constraints. The Trans-Planckian problem consists in the fact that, for the majority of inflationary models, the physical wavelengths of the modes that correspond to the present large-scale structure in the Universe were much smaller than the Planck length at the initial time, since they underwent an exponential redshift during inflation. The issue arises because in the usual computation of the spectrum of fluctuations we extrapolate weakly self-coupled field theory coupled with classical gravity into a regime where these theories are known to break down. The problem explained above can be expressed in the following equation:

$$\frac{a_i}{a_f} l_{\text{Pl}} < \frac{1}{H_f} \quad (3.53)$$

where a_i and a_f are, respectively, the scale factors at the beginning and at the end of the evolution, H_f is the Hubble parameter at the end of that evolution, and l_{Pl} is a length scale of the order of the Planck scale.

Assuming a period of quasi-de Sitter inflation and an instant reheating phase at the end of inflation (where H can be considered constant), the TCC implies a model-independent upper bound on the energy scale of inflation:

$$V^{\frac{1}{4}} < 6 \cdot 10^8 \text{GeV} \sim 3 \cdot 10^{-10} M_{\text{Pl}} \quad (3.54)$$

that translates into an upper bound on the tensor-to-scalar ratio if we assume the standard inflation picture, namely $r < 10^{-30}$ ⁹. We anticipate that in the warm inflation picture, as Eq. (3.114) shows, there are possibilities to obtain a highly suppressed tensor-to-scalar ratio, in particular in the strong regime. Since this conjecture is recent, there are open discussions about how to relax the bounds arising from it, and in this thesis, we just state it without considering its implications for the "Minimal Warm Inflation" model.

⁹By accounting for a non-standard evolution after the end of inflation, it has been shown [14] that it can lead to a more accessible result for r , $r < 10^{-8}$

3.7 Models of inflation: two examples

In this section, we illustrate two examples of possible inflation realizations; we will analyze these two models, by using the notions introduced in sections 3.2, 3.3, and 3.4, and will highlight their compatibility with experimental constraints and if they satisfy the bounds arising from String Theory conjectures, seen in section 3.6.

3.7.1 The simplest case: $m^2\phi^2$ potential

In this example, inflation is driven by a mass term, and the potential is

$$V(\phi) = \frac{1}{2}m^2\phi^2 \quad (3.55)$$

We can compute the slow-roll parameters, using Eqs. (3.16) and (3.17), obtaining

$$\epsilon_V = \eta_V = 2 \left(\frac{M_{\text{Pl}}}{\phi} \right)^2 \quad (3.56)$$

If we require that the slow-roll conditions are satisfied, namely $\epsilon_V < 1$ and $\eta_V < 1$, we have to consider super-Planckian values for the inflaton: $\phi > \sqrt{2}M_{\text{Pl}} \equiv \phi_{\text{end}}$. We can also compute the number of e -folds as a function of ϕ obtaining

$$N_e(\phi) = \frac{\phi^2}{4M_{\text{Pl}}^2} - \frac{1}{2} \quad (3.57)$$

and from this, adding the condition (3.21), we can obtain the value of the field at the CMB horizon crossing time: $\phi_{\text{cmb}} \simeq 15M_{\text{Pl}}$.

If we substitute in Eq. (3.56) the expression of ϕ_{cmb} we obtain $\epsilon = \eta \sim \frac{1}{2N_{\text{CMB}}}$. For the calculation of the power spectrum of scalar perturbations, we use Eq. (3.40) to obtain:

$$\Delta_R^2 = \frac{m^2}{M_{\text{Pl}}^2} \frac{N_{\text{CMB}}^2}{3} \quad (3.58)$$

The value measured by Planck is $\Delta_R^2 \simeq 10^{-9}$. This fixes the inflaton mass to $m \simeq 10^{-6}M_{\text{Pl}}$.

The scalar spectral index evaluated at CMB scales is $n_s - 1 \simeq \frac{2}{N_{\text{CMB}}}$ and for the tensor-to-scalar ratio we have $r \approx \frac{8}{N_{\text{CMB}}}$.

We note that the experimental bound $r < 0.064$ (see section 3.5) is not satisfied by this model while the spectral index is compatible with the experimental estimate $|n_s - 1| = 0.0351 \pm 0.0042$ within a σ .

String Theory conjectures are violated, but rather weakly. The distance conjecture is not satisfied because, if we assume a minimum of 60 e -folds, we obtain a field excursion, $\Delta\phi = \phi_{\text{cmb}} - \phi_{\text{end}} > M_{\text{Pl}}$, which is super-Planckian. The refined De Sitter conjecture translates into the following condition

$$\phi \leq \frac{M_{\text{Pl}}}{2c} \quad (3.59)$$

that is weakly violated by the model. Given the unknown $\mathcal{O}(1)$ numbers in the conjectures, one cannot firmly exclude this model.

3.7.2 Hybrid inflation

The model described in the previous Section is the simplest possible model of inflation. Including more than one field can change the predictions. It is worth exploring what happens in a well-known two-fields model, where inflation is supported when field excursions are small compared to M_{Pl} .

In this section, we consider Hybrid Inflation, mainly following the original article by Linde [15]. The basic idea of this model consists of the presence of two interacting scalar fields, where the end of inflation coincides with a rapid rolling of the "waterfall" field σ triggered by the slow rolling of the inflaton ϕ to its critical value. The potential for this hybrid realization is:

$$V(\phi, \sigma) = \frac{1}{4\lambda}(M^2 - \lambda\sigma^2)^2 + \frac{1}{2}m^2\phi^2 + \frac{1}{2}g^2\phi^2\sigma^2 \quad (3.60)$$

The effective squared mass of the "waterfall" field can be obtained as the coefficient of the σ^2 term, after developing the different pieces of the potential, Eq. (3.60), and it is equal to:

$$M^2(\phi) = -M^2 + g^2\phi^2 \quad (3.61)$$

We can note that the potential (3.60) has three stationary points in σ , and they are $\sigma = 0$, and $\sigma = \pm\sqrt{-M^2(\phi)/\lambda} = \sqrt{(M^2 - g^2\phi^2)/2\lambda}$. But there is a critical point for ϕ , i.e. $\phi_c \equiv M/g$, so that the two latter minima become degenerate with the first for this value, and then stop to exist for $\phi > \phi_c$.

So, while the inflaton is bigger than its critical value $\phi_c \equiv M/g$, i.e. $\phi > \phi_c$, the potential $V(\phi, \sigma)$ only has one minimum at $\sigma = 0$. The curvature of the potential, (3.60), in the σ -direction is much greater than in the ϕ -direction. Thus we expect that at the first stages of the Universe's expansion, the field σ rolled down to $\sigma = 0$, whereas the field ϕ could remain large for a much longer time. So, given ϕ_c the critical value for the inflaton, the inflation phase is when $\phi > \phi_c$ and as soon as $\phi < \phi_c$, σ rolls to a deeper minimum triggering the end of inflation.

The term "hybrid" refers to the fact that inflation is not driven by the energy density of the inflaton field ϕ but by a constant, namely the minimum of the potential $V(0, 0) = \frac{M^4}{4\lambda}$.

For the constant term to dominate, we require

$$m^2\phi^2 \ll \frac{M^4}{4\lambda} \quad (3.62)$$

if we insert in place of ϕ its critical value, i.e. $\phi_c = M/g$, Eq. (3.62) can be rewritten as $M^2 \gg \frac{\lambda m^2}{g^2}$.

With these prescriptions, and in the slow-roll approximation, the equations of motion, i.e. Eqs. (3.11) and (3.12), take the form, respectively of Eqs. (3.18) and (3.19).

So, in our case, the Hubble constant, during inflation is obtained by applying Eq. (3.18) with $V = \frac{M^4}{4\lambda}$ as the value for the potential

$$H^2 \approx \frac{M^4}{12\lambda M_{\text{Pl}}^2} \quad (3.63)$$

If we consider at most Planckian field excursion ($\phi \leq M_{\text{Pl}}$), we can see that Eq. (3.62) combined with Eq. (3.18), implies that $m^2 \ll H^2$. If we add to the latter inequality the H^2 explicit expression, (3.63), we obtain

$$M^2 \gg m M_{\text{Pl}} \sqrt{12\lambda} \quad (3.64)$$

The condition (3.64), linked to $\phi > \phi_c$ is a sufficient condition for inflation.

Now, let's study the behavior of the two fields, ϕ and σ , after a time corresponding to 1 e -fold,

$$\Delta t = H^{-1} = \sqrt{12\lambda} \frac{M_{\text{Pl}}}{M^2} \quad (3.65)$$

from the moment t_c when the field ϕ becomes equal to its critical value ϕ_c . Using the Eq. (3.19) to evaluate the variation in ϕ (decreasing from ϕ_c), we obtain

$$\Delta\phi_{1\text{efold}} \sim \dot{\phi} H^{-1} = \frac{m^2 \phi_c}{3H^2} = \frac{4\lambda m^2 M_{\text{Pl}}^2}{gM^3} \quad (3.66)$$

The absolute value of the negative effective σ mass squared, after 1 e -fold, can be obtained, by using Eq. (3.61) with $(\phi_c - \Delta\phi_{1\text{efold}})$.

$$M^2(\phi)_{1\text{efold}} = \frac{8\lambda m^2 M_{\text{Pl}}^2}{M^2} \quad (3.67)$$

where, if we develop the square of $(\phi_c - \Delta\phi_{1\text{efold}})$, the ϕ_c^2 term cancels $-M^2$ and the double product term is the leading one.

In fact, in Eq. (3.67) we have neglected the $\Delta\phi_{1\text{efold}}^2$ term because it is suppressed. To be more quantitative, let's rewrite Eq. (3.64) as $X \equiv m M_{\text{Pl}} \sqrt{12\lambda}/M^2 \ll 1$, we note that, once we have factorised out $m \lambda^{\frac{1}{2}} M_{\text{Pl}}$, the term $2\phi_c \cdot \Delta\phi_{1\text{efold}}$ goes as $2X$ while the $\Delta\phi_{1\text{efold}}^2$ term goes as $0.4X^3$, so it is greater by two power of X and can be discarded.

As soon as ϕ reaches the threshold ϕ_c , the potential (3.60) re-acquires the two "lost" minima (discussed at the beginning of this section) and so the field σ can roll down to another minimum like $M(\phi)/\sqrt{\lambda}$. Requiring this process to last less than H^{-1} , we can explicitly impose

$$\frac{\Delta\sigma}{\dot{\sigma}} = \frac{M(\phi)}{\sqrt{\lambda}} \frac{3H}{V_{,\sigma}} \ll H^{-1} \quad (3.68)$$

where we have estimated the σ field excursion as the difference of the two minima $(M(\phi)/\sqrt{\lambda} - 0)$, and for $\dot{\sigma}$ we have assumed the slow-roll approximation and so the validity of $\dot{\sigma} \approx V_{,\sigma}/3H$, which is the analogous of Eq. (3.19) but for the σ field. So by differentiating V along the σ direction and substituting the minimum value $\sigma = \frac{M(\phi)}{\sqrt{\lambda}}$ we obtain that $M^2(\phi) \gg H^2$ has to hold. If we use Eq. (3.63) in the latter condition we obtain the following relation

$$M^3 \ll 4\sqrt{6}\lambda m M_{\text{Pl}}^2 \quad (3.69)$$

So, in this case, the field σ , within the time $\Delta t \sim H^{-1}$, rolls down to its minimum at $\sigma(\phi) = \frac{M(\phi)}{\sqrt{\lambda}}$, rapidly oscillates near it and loses its energy due to the expansion of the Universe.

However, the field cannot simply relax near this minimum, since the potential $V(\phi, \sigma)$ at $\sigma(\phi)$ has a non-vanishing partial derivative

$$\frac{\partial V}{\partial \phi} = m^2 \phi + \frac{g^2 \phi M^2(\phi)}{\lambda} \quad (3.70)$$

We can compute a new ϕ velocity starting from Eq. (3.70) as $\dot{\phi}' = \frac{\partial V}{\partial \phi} \cdot (3H)^{-1}$. Now, if we want the inflaton ϕ to roll, along this direction, to the minimum of the effective potential within a time much smaller than H^{-1} , we can impose that $\phi/\dot{\phi}' \ll H^{-1}$ (repeating the reasoning used, for the field σ , to arrive at Eq. (3.69)) obtaining the following relation

$$M^3 \ll \sqrt{\lambda} g m M_{\text{Pl}}^2 \quad (3.71)$$

Thanks to this condition inflation ends almost instantaneously, as soon as ϕ reaches the threshold ϕ_c .

If we compute the scalar power spectrum, considering $V \approx \frac{M^4}{4\lambda}$ and $V' = m^2 \phi$, and using Eq. (3.40), we have:

$$\Delta_R^2 = \frac{1}{768\pi^2} \frac{M^{10} g^2}{M_{\text{Pl}}^6 m^4 \lambda^3} \quad (3.72)$$

If we constrain Eq. (3.72) with the experimental value, i.e. $\Delta_R^2 \sim 10^{-9}$, and again use Eq. (3.64), we obtain the following inequality:

$$M^3 \ll 20\pi \cdot 10^{-5} \frac{\lambda}{g} m M_{\text{Pl}}^2 \quad (3.73)$$

This means that the "waterfall conditions", namely Eqs. (3.69) and (3.71), are automatically satisfied once we require $m^2 \ll H^2$ and impose the experimental constraints unless λ and g are extremely small; therefore the waterfall regime is realized in this model for a wide variety of values of the parameters, m , λ , g , and M .

Let's now discuss how this model relates to the bounds arising from String Theory. To estimate the field excursion we note that if we combine the expression of the field excursion for 1 e -fold, Eq. (3.66), with Eq. (3.64) we obtain that $\Delta\phi_{1\text{efold}} \gg M/3g$.

Since the previous relation holds for 1 e -fold, it is obvious that the total field excursion has to be $\Delta\phi > M/3g$ too. If now we impose the bound arising from the SDC we obtain:

$$1 > \frac{\Delta\phi}{M_{\text{Pl}}} \gg \frac{M}{3gM_{\text{Pl}}} \quad (3.74)$$

If now we substitute to M its lower bound of Eq. (3.64), Eq. (3.74) to be satisfied implies the following relation among the model parameters

$$\lambda^{\frac{1}{4}} m^{\frac{1}{2}} \ll 1.61 g M_{\text{Pl}}^{\frac{1}{2}} \quad (3.75)$$

Let's now focus on the condition of the potential flatness, arising from de Sitter conjecture. The bound obtained, imposing Eq. (3.52) translates in the following relation

$$L = \frac{4m^2 \frac{\lambda}{g} M_{\text{Pl}}}{M^3} \geq c \quad (3.76)$$

We can manipulate the condition (3.76) substituting to M its lower bound of relation (3.64). We would have a $L' > L \geq c$ which implies the following relation (assuming $c \sim 1$)

$$\lambda^{\frac{1}{4}} m^{\frac{1}{2}} \geq 1.61 g M_{\text{Pl}}^{\frac{1}{2}} \quad (3.77)$$

We note that condition (3.77) is in contrast to Eq. (3.75), to be more precise they cannot be both satisfied at the same time. This implies that, in the cold hybrid inflation model, we can achieve a sub-planckian field excursion as long as we drop the requirement for a flat potential, and vice-versa.

3.8 Warm inflation

In section 3.4 we have discussed the importance of quantum fluctuations during inflation. In this section, following the approach of [16], we illustrate another way to realize inflation, called warm inflation, where thermal fluctuations dominate the quantum ones. In warm inflation, the inflaton is able to dissipate its energy into other light degrees of freedom with a rate that is faster than the Hubble one. Then, these particles can thermalize and during this phase, we can model their contribution as a simple radiation fluid ρ_r :

$$\rho_r \equiv \tilde{g}_* T^4 \quad (3.78)$$

where $\tilde{g}_* = \frac{\pi^2}{30} g_*$ and g_* stands for the effective number of relativistic degrees of freedom of the produced particles. Energy conservation implies that all the energy lost by the inflaton field by dissipation must be transferred to the radiation fluid, so the evolution equations for the system can be evaluated from the conservation of the energy-momentum tensor:

$$\nabla_\mu T^{\mu\nu} = 0 \quad (3.79)$$

If we work in a spatially flat Friedmann-Lemaître-Robertson-Walker (FLRW) metric, i.e $ds^2 = -dt^2 + a(t)^2 \delta_{ij} dx^i dx^j$, Eq. (3.79) translates into a set of continuity equations for each component of the cosmological fluid, namely the inflaton ϕ and the radiation:

$$\dot{\rho}_\phi + 3H(\rho_\phi + p_\phi) = -\Upsilon(\rho_\phi + p_\phi) \quad (3.80)$$

$$\dot{\rho}_r + 3H(\rho_r + p_r) = \Upsilon(\rho_\phi + p_\phi) \quad (3.81)$$

where Υ is the dissipation coefficient and the right-hand side of the previous equations plays the role of a source term. In the most general case, the dissipation coefficient can be parametrized as follows¹⁰:

$$\Upsilon(\phi, T) = C_\Upsilon T^c \phi^p M^{1-p-c} \quad (3.82)$$

where C_Υ is a dimensionless constant that carries the details of the microscopic model used to derive the dissipation coefficient, M is some mass scale in the model and depends on its construction, while c and p are numerical powers, which can be either positive or negative.

We can also generalize the Friedmann equation, Eq. (3.12), in the presence of radiation obtaining:

$$H^2 = \frac{1}{3M_{\text{Pl}}^2} \left(V(\phi) + \frac{1}{2} \dot{\phi}^2 + \rho_r \right) \quad (3.83)$$

¹⁰We note that the dimension of the dissipation coefficient in Eq. (4.3) (is $[\Upsilon] = [\text{energy}]$)

where we use the dot to indicate a time derivative while the prime denotes a derivative with respect to ϕ .

Considering the energy density and pressure for a standard canonical inflaton field, Eqs. (3.9) and (3.10), and with $p_r = \rho_r/3$, Eq. (3.80) becomes:

$$\ddot{\phi} + 3H(1+Q)\dot{\phi} + V'(\phi) = 0 \quad (3.84)$$

where we have defined the dissipation ratio Q in warm inflation, which measures the strength of the dissipative particle production effects in comparison to the spacetime expansion:

$$Q = \frac{\Upsilon(T)}{3H} \quad (3.85)$$

while Eq. (3.81) becomes

$$\dot{\rho}_r + 4H\rho_r = \Upsilon\dot{\phi}^2 \quad (3.86)$$

We can note that Eq. (3.84) differs from the inflation dynamics equation, Eq. (3.11), only for the extra friction term, Υ , which is due to the dissipative particle production and the following thermalization.

Even if inflation takes place when the scalar potential dominates over the radiation energy density and the scalar kinetic term, i.e. $\rho_r, \dot{\phi}^2/2 \ll V$, the resulting ρ_r must satisfy the condition $\rho_r^{1/4} > H$, which, if we assume thermalization, namely Eq. (3.78), reduces to $T > H$, which is usually considered as a condition for warm inflation. This condition must hold because the thermal fluctuations of the inflaton become important when they dominate over the quantum fluctuations. These latter are characterized by the inflaton mass scale m_ϕ , which is of the order of the Hubble parameter H during inflation.

To solve the Eqs. (3.84) and (3.86), we can use the slow-roll approximation, introduced in section 3.3, which translates into dropping the highest derivative term in each equation. Assuming $\ddot{\phi} \ll \dot{\phi}^2/2$, $V'(\phi)$ and $\dot{\rho}_r \ll 4H\rho_r$, $\Upsilon\dot{\phi}^2$, the slow-roll equations are:

$$\dot{\phi} \approx -\frac{V'}{3H(1+Q)} \quad (3.87)$$

$$4H\rho_r \approx \Upsilon(T)\dot{\phi}^2 \quad (3.88)$$

In the slow-roll regime, the Friedman equation, Eq. (3.83), reduces to:

$$H^2 \approx \frac{V}{3M_{\text{Pl}}^2} \quad (3.89)$$

We can adapt the slow-roll parameters, Eqs. (3.16) and (3.17), to warm inflation as :

$$\epsilon_V^{\text{wi}} \equiv \frac{M_{\text{Pl}}^2}{2(1+Q)} \left(\frac{V'}{V} \right)^2 \quad (3.90)$$

$$\eta_V^{\text{wi}} \equiv \frac{M_{\text{Pl}}^2}{(1+Q)} \frac{V''}{V} \quad (3.91)$$

We note that, in warm inflation, to satisfy the slow-roll conditions, i.e. $\epsilon_V^{\text{wi}}, \eta_V^{\text{wi}} \ll 1$, Eqs. (3.17) and (3.16) don't need to be much smaller than one, as in the standard

inflation realization. In fact, provided a large enough dissipation ratio, the potential slow-roll parameters can be bigger than one. This fact relaxes the necessity for very flat potential.

When we use the slow-roll approximation we have to check its consistency. It can be shown [17], through a linear stability analysis, that sufficient conditions under which the system remains close to the slow-roll solution many Hubble times are:

$$\epsilon_V^{\text{wi}}, \eta_V^{\text{wi}}, \beta_V \ll 1 + Q \quad 0 < b < \frac{Q}{1+Q} \quad |d| < 4 \quad (3.92)$$

where we have defined $\beta_V \equiv M_{\text{Pl}}^2 \frac{V'}{V} \frac{\Upsilon'}{\Upsilon}$, $b = T \frac{V'_{,T}}{V'}$ and $c = T \frac{\Upsilon_{,T}}{\Upsilon}$.

There are two regimes in warm inflation, which are determined by the value of the dissipation ratio Q , Eq. (3.85). When $Q < 1$, we refer to the weak dissipative regime: in this case, the background dynamics are not significantly altered by dissipation and are similar to those in standard inflation. However, thermal fluctuations in radiation energy density can still have a strong impact on field fluctuations and on the primordial spectrum of perturbations, as long as $T > H$. The other regime in warm inflation is the strong dissipative regime, where $Q > 1$. Here, dissipation strongly influences both the background dynamics and primordial fluctuations. It is important to note that Q is not necessarily constant and can vary depending on the dissipation coefficient and inflaton potential. As such, a model can start in the weak dissipative regime and later transition into the strong dissipative regime, or vice versa.

3.8.1 Thermal fluctuations

In this section, we want to illustrate the main features of the generation of fluctuations in the warm inflation picture. As said before, when $T > H$, thermal fluctuations start to become larger than the quantum ones, and, for this reason, the source of density fluctuations can be identified with thermal fluctuations in the radiation field, which is then transferred to the inflaton field as adiabatic curvature perturbations. To study this process we use a computational strategy similar to the one highlighted in section 3.4. We can start writing the scalar perturbed FLRW metric:

$$ds^2 = (1 - 2\alpha)dt^2 - 2a\partial_i\beta dx^i dt + a^2 [\delta_{ij}(1 + 2\varphi) + 2\partial_i\partial_j\gamma] dx^i dx^j \quad (3.93)$$

where α , β , γ , and φ are the spacetime-dependent perturbed variables¹¹. Moreover, one needs to expand the inflaton field, the radiation energy density, and also the radiation pressure around their background values in the FLRW metric, such as Eq. (3.22) shows, so we have:

$$\phi(t, x) = \bar{\phi}(t) + \delta\phi(t, x) \quad (3.94)$$

$$\rho_r(t, x) = \bar{\rho}_r(t) + \delta\rho_r(t, x) \quad (3.95)$$

$$p_r(t, x) = \bar{p}_r(t) + \delta p_r(t, x) \quad (3.96)$$

¹¹to be precise φ is the intrinsic curvature of a constant time hypersurface

and since the dissipation coefficient is, in general, a function of T and ϕ we can proceed analogously as

$$\Upsilon(t, x) = \bar{\Upsilon}(t) + \delta\Upsilon(t, x) \quad (3.97)$$

Now, we expand at linear order the dynamics equations, i.e. Eqs. (3.80) and (3.81), and working in momentum space, defining the Fourier transform with respect to the comoving coordinates, the equations of motion for the fluctuations with comoving wavenumber k are given by¹²:

$$\delta\dot{\rho}_r + 3(1 + w_r)H\delta\rho_r = (1 + w_r)\rho_r(k - 3H\alpha) + \frac{k^2}{a^2}\Psi_r + \delta Q_r + Q_r\alpha \quad (3.98)$$

$$\dot{\Psi}_r + 3H\Psi_r = -w_r\delta\rho_r - (1 + w_r)\rho_r\alpha + J_r \quad (3.99)$$

where Ψ_r is the momentum perturbation and w_r is the equation of state parameter for radiation. We also have defined the shear $\chi = \alpha(\beta + a\dot{\gamma})$ and the perturbed expansion scalar of the comoving frame $-\kappa = -(3H\alpha - \dot{\phi}) - \frac{k^2\chi}{a^2}$. Furthermore, we have also the momentum source $J_r = -\Upsilon\dot{\phi}\delta\phi$ and the energy density source term, i.e. the projection of the four-vector source term along the direction of the fluid, $Q_r = \Upsilon\phi^2$. The associated perturbation can be written as:

$$\delta Q_r = \delta\Upsilon\dot{\phi}^2 + 2\Upsilon\dot{\phi}\delta\phi - 2\alpha\Upsilon\dot{\phi}^2 \quad (3.100)$$

On the other hand, we also have the evolution equation for the inflaton field fluctuations $\delta\phi$. Assuming the universe remains near thermal equilibrium during WI, the fluctuations of the inflaton field obey a fluctuation-dissipation relation. Therefore, the evolution of the inflaton fluctuations is achieved by perturbing the inflaton field equation and adding stochastic quantum and thermal white noise terms, following the fluctuation-dissipation theorem, as:

$$\delta\ddot{\phi} + 3h\delta\dot{\phi} + \left(\frac{k^2}{a^2} + V''\right)\delta\phi = \xi_q + \xi_T - \delta\Upsilon\dot{\phi} + \dot{\phi}(\kappa + \dot{\alpha}) + (2\ddot{\phi} + 3H\dot{\phi})\alpha - \Upsilon(\delta\dot{\phi} - \alpha\dot{\phi}) \quad (3.101)$$

where $\xi_{q,T}$ are stochastic Gaussian sources related to the quantum and thermal fluctuations with mean zero. and with appropriate amplitudes, which are defined by the two-point correlation functions:

$$\langle \xi_T(x, t)\xi_T(x', t') \rangle = \frac{2\Upsilon T}{a^3}\delta(t - t')(2\pi^3)\delta(k - k') \quad (3.102)$$

$$\langle \xi_q(x, t)\xi_q(x', t') \rangle = \frac{H^2(9 + 12\pi Q)^{\frac{1}{2}}(1 + 2n_*)}{\pi a^3}\delta(t - t')(2\pi^3)\delta(k - k') \quad (3.103)$$

where n denotes the inflaton statistical distribution due to the presence of the radiation bath; if we assume the thermal equilibrium during inflation, n can be considered as a Bose-Einstein distribution, i.e. $n = (e^{H/T} - 1)^{-1}$.

¹²For simplicity, we keep the same notation for the fluctuations $\delta f(x, t)$ and their Fourier transform $\delta f(k, t)$

To continue our manipulation of the evolution equations for perturbations, we have to specify the fluctuation of the dissipation coefficient, which according to Eq. (3.82), is:

$$\delta\Upsilon = \Upsilon \left[\frac{c\delta T}{T} + p \frac{\delta\phi}{\phi} \right] \quad (3.104)$$

If we want the calculation to hold, the system has to be close to equilibrium, therefore $p_r = \rho_r/3$, $\rho_r \propto T^4$ and so $\delta T/T = \delta\rho_r/4\rho_r$, with these considerations we can rewrite the variation of Q_r as, Eq. (3.100):

$$\delta Q_r = 3HQ\dot{\phi}^2 \left(\frac{c\delta\rho_r}{4\rho_r} + p \frac{\delta\phi}{\phi} \right) + 6HQ\dot{\phi}\delta\dot{\phi} - 6HQ\dot{\phi}^2\alpha \quad (3.105)$$

Using the previous results, we can rewrite Eqs. (3.99), (3.98) and (3.101) in the final set of dynamics equations:

$$\delta\ddot{\phi} = -3H\delta\dot{\phi}(1+Q) - \left(\frac{k^2}{a^2} + V'' + \frac{3pHQ\dot{\phi}}{\phi} \right) \delta\phi + \xi_q + \xi_T - \frac{cH}{\dot{\phi}} \delta\rho_r + \dot{\phi}(\kappa + \dot{\alpha}) + [2\ddot{\phi} + 3H(1+Q)\dot{\phi}]\alpha \quad (3.106)$$

$$\delta\dot{\rho}_r = -H \left(4 - \frac{3cQ\dot{\phi}^2}{4\rho_r} \right) \delta\rho_r + \frac{k^2}{a^2} \Psi_r + 6HQ\dot{\phi}\delta\dot{\phi} + \frac{3pHQ\dot{\phi}^2}{\phi} \delta\phi + \frac{4\rho_r}{3} \kappa - 3H \left(Q\dot{\phi}^2 + \frac{4\rho_r}{3} \right) \alpha \quad (3.107)$$

$$\dot{\Psi}_r = -3H\Psi_r - 3HQ\dot{\phi}\delta\phi - \frac{1}{3}\delta\rho_r - 4\rho_r \frac{\alpha}{3} \quad (3.108)$$

One can immediately realize from Eqs. (3.106) and (3.107) that the inflaton fluctuations $\delta\phi$ are coupled to the radiation fluctuations $\delta\rho_r$ when $c \neq 0$. Such coupling results in a growing mode if $c > 0$, or a decreasing mode for $c < 0$ in the curvature power spectrum as the dissipation ratio Q increases. So the basic idea is to compute the interesting quantity, in our case the scalar curvature power spectrum, in the simplest case $c = 0$, (recalling Eq. (3.82), it's when the dissipation coefficient doesn't depend on the temperature), i.e. when the dynamics equations decouple and they become solvable analytically, then we try to parametrize the other cases as

$$\Delta_R^2|_{c \neq 0} = \Delta_R^2|_{c=0} G(Q_*) \quad (3.109)$$

Let's choose a convenient gauge for computations as a "zero-shear" gauge that implies the following conditions:

$$\begin{aligned} \chi &= 0 \\ \kappa &= \frac{3}{2M_{\text{Pl}}^2} (\dot{\phi}\delta\phi - \Psi_r) \\ \alpha &= -\varphi \\ \dot{\varphi} &= -H\varphi - \frac{1}{3}\kappa \end{aligned} \quad (3.110)$$

In the case $c = 0$ we can find an analytical solution for Δ_R^2 with Green function methods, but, in this section, we are satisfied with its first-order approximation, where we drop the slow-roll coefficients obtaining:

$$\Delta_R^2|_{c=0} = \frac{H_*^2}{(2\pi)^2} \frac{H_*^2}{\dot{\phi}_*^2} \left(1 + 2n_* + \frac{2\sqrt{3}\pi Q_*}{\sqrt{3+4\pi Q_*}} \frac{T_*}{H_*} \right) \quad (3.111)$$

Neglecting both the metric perturbations and the slow-roll parameters and also the field dependence in Eq. (3.104), we can find that $G(Q)$ has an asymptotic behavior in the strong regime, i.e. for $Q \gg 1$, given by

$$G(Q) = \left(\frac{Q}{Q_c} \right)^{3c} \quad (3.112)$$

where Q_c is a constant depending on the value of c . After outlining the computation, it is important to note that we currently only have the ability to determine the growing function $G(Q)$, for a generic value of Q , by solving the complete set of background and perturbation equations numerically and then fitting the spectrum for a chosen dissipation coefficient.

Once $G(Q)$ is specified we can compute quantities such as the spectral tilt, defined in Eq. (4.11), so we obtain [16]

$$\begin{aligned} n_s - 1 = \frac{1+Q}{1+Q-\epsilon_V} & \left(\frac{d \ln \left(\frac{T}{H} \right)}{dN_e} + \frac{\frac{d \ln Q}{dN_e} (-3 + Q\{3 + 2\pi[-1 + Q(3 + \sqrt{9+12Q\pi})\})}{(1+Q)[3 + Q\pi(4 + \sqrt{9+12Q\pi})]} \right) \\ & + \frac{d \ln Q}{dN_e} \left(\frac{3 + 2\pi Q}{3 + 4\pi Q} + Q \frac{d \ln G(Q)}{dQ} \right) + \frac{-6\eta_V + 2\epsilon_V}{1+Q} \end{aligned} \quad (3.113)$$

As expected, this reduces to the spectral index for standard inflation, i.e. Eq. (3.41), when $Q \rightarrow 0$ and $T \rightarrow 0$.

Although the curvature power spectrum experiences notable changes due to dissipative effects, the tensor power spectrum remains unaffected due to the weak gravitational interactions. Hence, the expression for Δ_h^2 remains the same as that for cold inflation, which is given by Eq. (3.43).

If we compute the tensor-to-scalar ratio, by using the definition of Eq. (3.45) we obtain:

$$r = \frac{16\epsilon_V}{1+Q} \mathcal{F}^{-1} \quad (3.114)$$

So we can see that the inflation result is suppressed by $(1+Q)$ and $\mathcal{F}$, where $\mathcal{F}$ is defined as:

$$\mathcal{F} \equiv \left(1 + 2n_* + \frac{2\sqrt{3}\pi Q_*}{\sqrt{3+4\pi Q_*}} \frac{T_*}{H_*} \right) G(Q_*) \quad (3.115)$$

In particular, as a consequence of this, warm inflation can produce a much reduced tensor-to-scalar ratio as compared to cold inflation for the same type of inflaton potentials.

Another difference between standard and warm inflation is the necessity in the latter case of a distinct reheating phase, which is the process whereby the inflaton's energy density is converted back into conventional matter after inflation, marking the beginning of the standard Big Bang theory. In standard inflation, this phenomenon

cannot be incorporated in a single theory and therefore needs to be described separately. In WI, thanks to the presence of dissipation coefficients, it's possible to smoothly transition from the end of inflation to a radiation-dominated epoch. Fig. 3.6 summarizes the concept explained above.

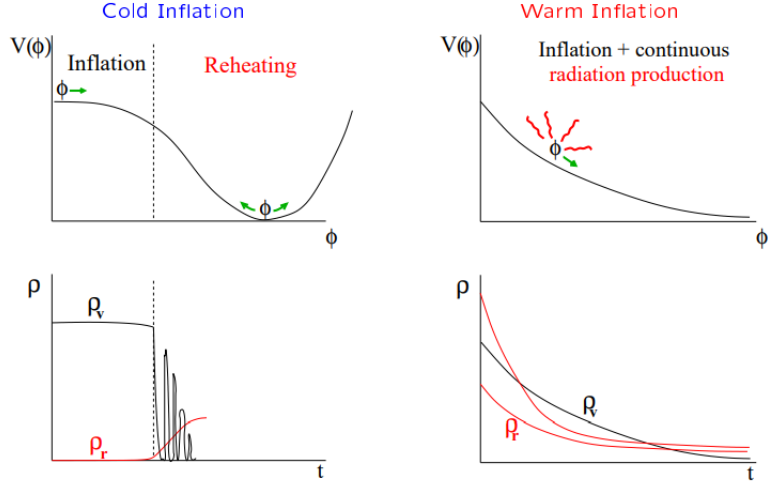


Figure 3.6. Comparison between standard and warm inflationary pictures. The top graphs show the scalar field evolution while the bottom ones show the vacuum and radiation energy density evolution, i.e. respectively ρ_v and ρ_r . Figure taken from [18].

Chapter 4

Minimal warm inflation model

In section 3.7.2, we have described the hybrid inflation model, which is characterized by a two-field interacting potential. Furthermore, we have applied the constraints arising from String Theory, and we have noted that the SDC and SdSC bounds can theoretically be satisfied but not at the same time. In this chapter, we want to illustrate a model that, starting from the same hybrid potential, Eq. (3.60), can overcome the limitations of the hybrid inflation model. This so-called "minimal warm inflation" [3] has the features of "slow-roll" warm inflation (3.8) in the strong regime, i.e. $Q \gg 1$.

In this case, warm inflation is realized by giving the inflaton an axion-like gauge coupling to a pure Yang-Mills group.

Through this mechanism, we can obtain a self-consistent thermal bath without significant backreaction on the inflaton potential. By backreaction, we mean the effects of matter field, radiation, or cosmological inhomogeneities on the cosmology degrees of freedom, which form the homogeneous dynamical background. In this specific case, the principal¹ source of possible backreaction is thermal but the overall contribution is negligible due to the axion shift symmetry².

To account for the axion-like interaction of the inflaton, besides the action associated with single slow-roll inflation models, Eq. (3.6), in the minimal warm inflation model, we add the following interaction lagrangian

$$\mathcal{L}_{\text{int}} = \frac{\alpha}{16\pi} \frac{\phi}{f} \epsilon^{\mu\nu\alpha\beta} G_{\mu\nu}^a G_{\alpha\beta}^a \quad (4.1)$$

where $G_{\mu\nu}^a$ is the standard field strength of the Yang-Mills group and we have defined the gauge coupling as $\alpha = \frac{g_{YM}^2}{4\pi}$.

At high temperatures, there are classical transitions between vacua with different topological charges that are no longer suppressed: this leads to fluctuations in topological charge. As a result, the friction experienced by the inflaton ϕ is not solely due to thermal factors, but also involves topological fluctuations. These fluctuations increase with temperatures because higher temperatures enhance the transition rate (known as the sphaleron³ rate). So the friction, due to interaction (4.1), can be

¹There is also a subleading contribution due to non-perturbative effects proportional to the axion mass.

²To be precise this symmetry is softly broken by giving the inflaton a UV-potential, but the authors of [3] assure that the backreaction due to the breaking term is negligible.

³A sphaleron is a static (time-independent) solution to the electroweak field equations of the Standard Model, and is involved in certain hypothetical non-perturbative processes that violate

estimated from the sphaleron rate Γ_{sphal} for sufficient small axion mass (less than $\alpha^2 T$) as

$$\Upsilon(T) = \frac{\Gamma_{\text{sphal}}(T)}{2f^2 T} \quad (4.2)$$

With a scaling of the sphaleron rate of the form $\Gamma_{\text{sphal}}(T) \sim \alpha^5 T^4$, supported by simulations within classical lattice gauge theory for pure SU(3), the friction coefficient Υ can be written as:

$$\Upsilon(T) = k(\alpha, N_c, N_f) \alpha^5 \frac{T^3}{f^2} \quad (4.3)$$

Where k is an $\mathcal{O}(100)$ number, which has a weak logarithmic dependence on α and whose exact value depends on the number of colors N_c and flavors N_f of the group.

We can note that the potential for a hybrid realization of warm inflation is the same used in the expression (3.60) while the dynamics is determined by the warm inflation equations (namely Eqs. (3.89), (3.87) and (3.88)), seen in chapter 3.8.

The potential has two fields, the inflaton ϕ , and the waterfall field σ , with the second one remaining constant during inflation at $\sigma = 0$. Then in that period, the effective potential for the inflaton takes the following form (which can be obtained setting $\sigma = 0$ in Eq. (3.60)):

$$V_{\text{eff}}(\phi) = V(\phi, 0) = \frac{M^4}{4\lambda} + \frac{1}{2}m^2\phi^2 \quad (4.4)$$

As we have done in the standard realization, we require that, after the phase transition, ϕ rolls to the minimum of its effective potential much faster than a Hubble time as long as:

$$M^3 \ll \frac{\sqrt{\lambda} g m M_{\text{Pl}}^2}{Q} \quad (4.5)$$

We can observe that condition (4.5) is analogous to (3.71) in the warm case, with the addition of a factor of Q at the denominator.

For the potential (4.4) the eta slow-roll parameter, Eq. (3.91), becomes:

$$\eta_V \equiv \frac{4\lambda m^2 M_{\text{Pl}}^2}{M^4 Q} \quad (4.6)$$

From the Eqs. (3.89), (3.87), (4.3) and (3.88), the temperature T and the dimensionless parameter Q can be re-expressed as follows:

$$T(\phi) \approx \left(\frac{f^2}{k\alpha^5} \frac{\sqrt{3} M_{\text{Pl}} V'^2}{4\tilde{g}_* \sqrt{V}} \right)^{\frac{1}{7}} \quad (4.7)$$

$$Q(\phi) \approx \left(\left(\frac{k\alpha^5}{f^2} \right)^4 \frac{M_{\text{Pl}}^{10} V'^6}{576 \tilde{g}_*^3 V^5} \right)^{\frac{1}{7}} \quad (4.8)$$

baryon and lepton numbers; geometrically, a sphaleron is a saddle point of the electroweak potential (in infinite-dimensional field space)

Let's now focus on the predictions of a warm inflation model in the presence of temperature-dependent friction, Eq. (4.3). In the strong regime thermal perturbations of the inflaton dominate over the quantum ones. The cubic temperature-dependent friction couples the inflaton dynamics to the radiation fluctuations, affecting the computation of the curvature power spectrum, by giving rise to a "growing mode". To determine it, we use Eq. (3.109), with the expression (3.111) for the temperature-independent power spectrum $\Delta_R^2|_{c=0}$ (where we neglect terms without Q since we are in the strong regime), and with Eq. (3.112), where we set $c = 3$. So we obtain:

$$\Delta_R^2 \approx \frac{\sqrt{3}}{4\pi^{\frac{3}{2}}} \frac{H^3 T}{\dot{\phi}^2} \left(\frac{Q}{Q_3} \right)^9 Q^{\frac{1}{2}} \quad (4.9)$$

with $Q_3 \approx 7.3$, this value is fixed by matching the boundary conditions for the solution of the inflaton perturbations in different regimes⁴.

Considering warm inflation with temperatures way below the Planck scale, the prediction for tensor perturbations is not affected and the result is the same as in standard inflation, Eq. (3.44), computed in section 3.4.

Substituting the expression of Q , (4.8), of H , (3.89) and of $\dot{\phi}$, (3.87) in Eq. (4.9) we have:

$$\Delta_R^2 \approx \frac{1}{4\pi^{\frac{3}{2}} 576^{\frac{23}{14}} Q_3^9} \left(\frac{3}{16} \right)^{\frac{1}{14}} \left(\frac{k\alpha^5}{f^2} \right)^{\frac{45}{7}} \frac{M_{\text{Pl}}^{\frac{81}{7}} V'^{\frac{57}{7}}}{\tilde{g}_*^{\frac{71}{14}} V^{\frac{81}{14}}} \quad (4.10)$$

The numerical prefactor is $2 \cdot 10^{-14}$.

Another parameter that must match observations is the spectral index, which is defined as

$$n_s - 1 = \frac{d \ln \Delta_R^2}{dN} \quad (4.11)$$

If we apply the definition to Eq. (4.9) and rewrite it with the expressions of warm inflation slow-roll parameters, Eqs. (3.91) and (3.90) we obtain:

$$n_s - 1 \approx \frac{3}{7} (27\epsilon_V^{\text{wi}} - 19\eta_V^{\text{wi}}) \quad (4.12)$$

During the final stage of inflation, namely when ϕ approaches its critical value $\phi_c = \frac{M}{g}$, the dominant term of the effective potential (4.4) is the constant one, $\frac{M^4}{4\lambda}$. This results in a suppression for ϵ_V^{wi} as we can see from the following expression

$$\epsilon_V^{\text{wi}} = \frac{M_{\text{Pl}}^2 (4\lambda)^2 m^4 \phi^2}{2Q M^8} \quad (4.13)$$

So if we neglect ϵ_V^{wi} and substitute η_V^{wi} with Eq. (4.6) in the expression of $|n_s - 1|$, i.e. Eq. (4.12), we obtain

$$|n_s - 1| \approx \frac{57}{7} \frac{4\lambda m^2 M_{\text{Pl}}^2}{M^4 Q} \approx \frac{228}{7} (576)^{\frac{1}{7}} \frac{\lambda m^2}{M^4} \left(\frac{k\alpha^5}{f^2} \right)^{-\frac{4}{7}} \frac{M_{\text{Pl}}^{\frac{4}{7}} V^{\frac{5}{7}} \tilde{g}_*^{\frac{3}{7}}}{V'^{\frac{6}{7}}} \quad (4.14)$$

⁴Eq. (4.9) is more accurate when $Q \gg Q_3$. But we can rely on the analytical solution, since, if we use numerical results, the change in the discussed phenomenology is negligible, as assured by [3]

The numerical prefactor is 80.76

By rewriting Eq. (4.10) as a function of the spectral index, Eq. (4.14), we obtain:

$$A_s(\phi_{\text{cmb}}) \approx 7.45 \cdot 10^{-41} \left(\frac{k\alpha^5}{f^2} \right)^{18} \left(\frac{\sqrt{|n_s - 1|}^{81} m^{21} \phi_{\text{cmb}}^{51}}{\sqrt{\tilde{g}_*}^{55}} \right)^{\frac{1}{2}} \quad (4.15)$$

The number of e -folds, namely the duration of inflation, can be formulated as:

$$N_e = \int H dt = \int_{\phi_{\text{end}}}^{\phi_{\text{start}}} \frac{1}{M_{\text{Pl}}^2} \frac{V}{V'} Q(\phi) d\phi \approx \frac{\Delta}{\eta_V} \quad (4.16)$$

with $\phi_{\text{end}} \approx \phi_c \equiv \frac{M}{g}$ and $\phi_{\text{start}} = (1 + \Delta)\phi_c$, so $\Delta \equiv \frac{g}{M} \Delta\phi$

We can derive the previous relation, $N_e \approx \frac{\Delta}{\eta_V}$, if we plug in Eq. (4.16) the expression of Q , Eq. (4.8), and the one of the effective potential, Eq. (4.4), and use the assumption $\frac{M}{g} \ll M_{\text{Pl}}$. Adding to this one the expression (4.6) for η , we obtain the following result for the number of e -folds:

$$N_e = (576)^{-\frac{1}{7}} \Delta\phi \left(\frac{k\alpha^5}{f^2} \right)^{\frac{4}{7}} \frac{V^{\frac{2}{7}}}{M_{\text{Pl}}^{\frac{4}{7}} V'^{\frac{1}{7}} \tilde{g}_*^{\frac{3}{7}}} \quad (4.17)$$

where we have considered valid the two following approximations: $\frac{V'}{m^2} = \phi \approx \frac{M}{g}$ and $V \sim \frac{M^4}{4\lambda}$.

If we rewrite the expressions of Q , Eq. (4.8), H , Eq. (3.89), m , T , Eq. (4.7) in terms of the quantities we can measure experimentally⁵, namely $|n_s - 1|$ and Δ_R^2 , we have (by considering that the condition $f \ll 10^{-8} \cdot \frac{M}{g}$ must hold to maintain valid the strong regime condition, $Q \gg 1$):

$$Q \approx 120 \left(\frac{\Delta_R^2}{2 \cdot 10^{-9}} \right)^{\frac{2}{21}} \left(\frac{|n_s - 1|}{0.035} \right)^{\frac{4}{7}} \left(\frac{g_*}{17} \right)^{-\frac{4}{21}} \left(\frac{k\alpha^5}{10^{-3}} \right)^{\frac{2}{7}} \left(\frac{gf}{10^{-8}} \right)^{-\frac{4}{7}} \quad (4.18)$$

$$H \approx 10^{-16} \left(\frac{\Delta_R^2}{2 \cdot 10^{-9}} \right)^{\frac{1}{21}} \left(\frac{|n_s - 1|}{0.035} \right)^{-\frac{19}{7}} \left(\frac{g_*}{17} \right)^{\frac{59}{42}} \left(\frac{k\alpha^5}{10^{-3}} \right)^{-\frac{13}{7}} \left(\frac{gf}{10^{-8}} \right)^{\frac{19}{7}} \frac{M}{g} \quad (4.19)$$

$$m \approx 10^{-16} \left(\frac{\Delta_R^2}{2 \cdot 10^{-9}} \right)^{\frac{2}{21}} \left(\frac{|n_s - 1|}{0.035} \right)^{-\frac{27}{14}} \left(\frac{g_*}{17} \right)^{\frac{55}{42}} \left(\frac{k\alpha^5}{10^{-3}} \right)^{-\frac{12}{7}} \left(\frac{gf}{10^{-8}} \right)^{\frac{24}{7}} \frac{M}{g} \quad (4.20)$$

$$T \approx 2 \cdot 10^{-9} \left(\frac{\Delta_R^2}{10 \cdot 10^{-9}} \right)^{\frac{1}{21}} \left(\frac{|n_s - 1|}{0.035} \right)^{-\frac{5}{7}} \left(\frac{g_*}{17} \right)^{\frac{17}{42}} \left(\frac{k\alpha^5}{10^{-3}} \right)^{-\frac{6}{7}} \left(\frac{gf}{10^{-8}} \right)^{\frac{12}{7}} \frac{M}{g} \quad (4.21)$$

⁵we use a pure SU(3) with $g^* = 17$ (two polarizations per eight gauge bosons plus one for the axion) and a gauge coupling $\alpha = 0.1$ as our default values.

An upper limit can be derived for $\frac{M}{g}$ from condition (4.5):

$$\frac{M}{g} \ll 3 \cdot 10^{-3} \left(\frac{\Delta_R^2}{2 \cdot 10^{-9}} \right)^{-\frac{1}{21}} \left(\frac{|n_s - 1|}{0.035} \right)^{\frac{3}{14}} \left(\frac{g_*}{17} \right)^{\frac{2}{21}} \left(\frac{k\alpha^5}{10^{-3}} \right)^{-\frac{1}{7}} \left(\frac{gf}{10^{-8}} \right)^{\frac{2}{7}} M_{\text{Pl}} \quad (4.22)$$

The condition (4.22) establishes an upper limit for $\frac{M}{g}$ equal to 10^{14} GeV, affecting Eqs. (4.18)-(4.21) by imposing an upper limit of 10^5 GeV for T , of $3 \cdot 10^5$ GeV for m and of 10^{-1} GeV for H . If we saturate the condition for the gauge group scale, i.e. $f \sim 10^{-8} \frac{M}{g}$, we obtain the value $f \sim 10^6$ GeV.

The quantities depend on parameters such as λ , M , g , and to break the degeneracy we require that the quantum corrections of the masses are naturally small compared to the cutoff Λ , implying the following conditions:

$$\frac{\lambda^2 \Lambda^2}{16\pi^2} < M^2 \quad (4.23)$$

$$\frac{g^2 \Lambda^2}{16\pi^2} < m^2 \quad (4.24)$$

Furthermore, the condition that the potential $V(\phi)$ is negligible compared to the vacuum energy during inflation requires:

$$\frac{M^4}{4\lambda} \gg \frac{1}{2} m^2 \phi_c^2 \longrightarrow \frac{2\lambda m^2}{g^2} \ll M^2 \quad (4.25)$$

Assuming a minimum cutoff value of $\Lambda = 4\pi M$, the following values for couplings and masses can be obtained by saturating the Eq. (4.24) and satisfying the condition (4.25) by one order of magnitude: $g = 10^{-8}$, $\lambda = 10^{-2}$, $m = 10^{-2}$ GeV, $M = 10^6$ GeV: these are the upper limits of the model parameters.

4.0.1 Reheating phase

For the sake of completeness, we describe also a possible realization for the reheating phase, which occurs after the end of inflation. We assume to have an abundance of waterfall fields σ before Big Bang nucleosynthesis, which dominates the energy density in the early Universe (after inflation). We need an interaction that allows the coupling between the field σ and Standard Model particles, which can be the mechanism to produce an early quark-gluon plasma. A possible example can be with a coupling of the standard model hypercharge:

$$\mathcal{L}_{\text{reheat}} = \frac{g'^2}{64\pi^2} \frac{\sigma}{f_B} \tilde{B}^{\mu\nu} B_{\mu\nu} \quad (4.26)$$

where g' is the Standard Model hypercharge gauge coupling. Since the values of the coupling g between the inflaton and the waterfall field are small, the decay of σ occurs mainly through the interaction (4.26). The decay rate needs to be large enough such that an abundance of σ particles has decayed into a quark-gluon plasma before the universe cools down to Big Bang nucleosynthesis temperatures, where the earliest cosmological precision constraints exist. Now we want to discuss the lower limits of the model parameters: since the waterfall field σ couples directly to the Standard Model there exist cooling bounds from supernovae as well as detection constraints from high-energy colliders. To avoid them, we can

conservatively assume $f_B > 1$ TeV and $M > 10$ GeV. Finally, we resume, in the following table, the possible range for the values of the model parameters, such as M , m , f , and of other important derived quantities, like H , T (which is the temperature during slow-roll maintained by the pure Yang-Mills radiation).

Min [GeV]	Quantity	Max [GeV]
10^{-7}	m	10^{-2}
10	M	10^6
0.3	f	$3 \cdot 10^6$
0.2	T	$2 \cdot 10^5$
10^{-7}	H	10^{-2}

Table 4.1. Summary table of values' range of model parameters (above the line and derived quantities (under the line). Min: lower bound, Max: upper bound.

4.1 Model with bounds of String Theory

In this section, the aim is to impose the conditions from String Theory, seen in section 3.6, to the minimal warm inflation model to find the maximum number of e -folds allowed by the model.

Let's start by imposing only the condition due to the "Swampland distance conjecture", i.e. Eq. (3.51), to Eq. (4.16), to obtain the following upper bound (with $\Delta = \frac{g}{M} \Delta\phi$):

$$N_e = 2.3 \cdot 10^2 \left(\frac{|n_s - 1|}{0.035} \right)^{-1} \Delta\phi \frac{g}{M} < 2.3 \cdot 10^2 \left(\frac{|n_s - 1|}{0.035} \right)^{-1} g \frac{M_{\text{Pl}}}{M} \quad (4.27)$$

If we consider the maximum values for model parameters (right column of table 4.1) for M and g , we obtain that the numerical prefactor amounts to $5.5 \cdot 10^6$. If we consider the minimum values for the masses (left column of table 4.1) we obtain the prefactor equal to $5.5 \cdot 10^{11}$.

If we now apply the de Sitter conjecture, Eq. (3.52), to Eq. (4.17) for the number of e -folds, we obtain the following upper bound (and $f \ll 10^{-8} \frac{M}{g}$):

$$N_e < (576)^{-\frac{1}{7}} \Delta\phi \left(\frac{k\alpha^5}{f^2} \right)^{\frac{4}{7}} \frac{V^{\frac{1}{7}}}{M_{\text{Pl}}^{\frac{3}{7}} c^{\frac{1}{7}} \tilde{g}_*^{\frac{3}{7}}} \quad (4.28)$$

If we add the SDC bound, Eq. (3.51), assuming $f \ll 10^{-8} \frac{M}{g}$, $V \sim M^4/4\lambda$ and $c \simeq 1$, we have the following inequality:

$$N_e \leq 2.3 \cdot 10^6 \left(\frac{k\alpha^5}{10^{-3}} \right)^{\frac{4}{7}} \left(\frac{g_*}{17} \right)^{-\frac{3}{7}} \left(\frac{gf}{10^{-8}} \right)^{-\frac{8}{7}} \left(\frac{M}{g} \right)^{-\frac{8}{7}} \left(\frac{M^4}{\lambda} \right)^{\frac{1}{7}} M_{\text{Pl}}^{\frac{4}{7}} \quad (4.29)$$

We can rewrite Eq. (4.29) with more observable quantities such as

$$N_e \leq 2.8 \cdot 10^2 \left(\frac{|n_s - 1|}{0.035} \right)^{-1} \lambda^{\frac{1}{7}} g \frac{M_{\text{Pl}}^{\frac{8}{7}}}{M^{\frac{11}{7}}} \quad (4.30)$$

If we substitute the example values for M , g , λ , we obtain a purely numerical prefactor that sets the scale for the upper limit of the number of e -folds: $N_e \leq 5.5 \cdot 10^5$. While, if we consider the minimal masses allowed by the model (with $g \sim 10^{-8}$ and $\lambda \sim 10^{-2}$ remain unchanged), i.e. $M \sim 10$ GeV and $m \sim 10^{-7}$ GeV, the scale of the upper limit of the number of e -folds amounts to $N_e \leq 5.2 \cdot 10^{10}$. This value is also the absolute maximum number of e -folds allowed by the minimal warm inflation model, with the assumption of a minimum cutoff value $\Lambda = 4\pi M$ and considering the condition (4.24) as equality.

As a consistency check, we can observe that the upper limit, for N_e , with only the SDC bound is bigger than the one constrained by both conjectures.

We postpone the discussion of the results, obtained in this section, for the number of e -folds in the conclusions. To do a comprehensive analysis, we first have to introduce the concepts of false vacuum decay and Multiverse.

Chapter 5

The false vacuum decay

In this chapter we follow the Coleman-De Luccia approach [19, 20]. A classical field theory may have more than one stable equilibrium state with different energy densities. This fact changes if we consider the quantum analogous of the theory, in this case, the state with the higher energy density will become unstable through barrier penetration, and for this reason, it is called a "false vacuum". The state with the minimum energy of the theory, by contrast, is called a "true vacuum". Consider now a quantum field theory of a single scalar field in four-dimensional space-time with the following associated lagrangian:

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \quad (5.1)$$

One of the simplest examples of such a potential is shown in the figure 5.1:

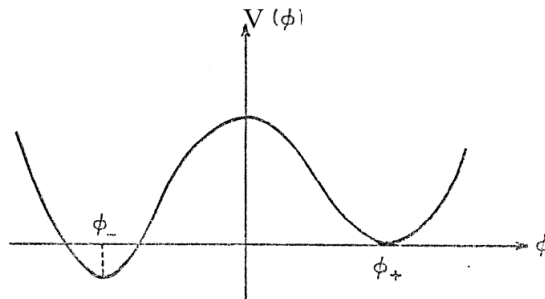


Figure 5.1. The potential $V(\phi)$ for a theory with a false vacuum (only two minima). Image taken from [20]

The transition happens via bubble nucleation of the true vacuum, thanks to quantum fluctuations, and when a bubble finds the favorable energetic conditions to grow, it expands through the Universe converting false vacuum to true vacuum, with a speed approaching the speed of light. In order to establish how likely such an event is, we have to compute the rate per unit volume of false vacuum decay, Γ/V . In the semiclassical limit (small $\hbar$) the rate can be expanded as¹:

$$\Gamma/V = A e^{-\frac{B}{\hbar}} (1 + \mathcal{O}(\hbar)) \quad (5.2)$$

¹There are algorithms for computing the coefficients A and B ; indeed, for the "thin wall" approximation, namely, in the limit of small energy-density difference between the two vacua, it is possible to compute B in closed form.

5.1 Vacuum decay with gravity

Now we want to include in the previous lagrangian, Eq. (5.1), the gravitational effects, so the new action takes the form of Eq. (3.6): this addition may appear superfluous because the bubble nucleation happens at scales where gravitation is negligible. However, this reasoning does not apply for the growth of the bubble, since the energy released by the conversion of the vacuum ("false to true") is proportional to the bubble volume, and this statement holds also for the Schwarzschild radius associated to that energy, so during the growth, the Schwarzschild radius may become comparable with the bubble radius. Another important point that has to be stressed is that for the lagrangian (5.1) the addition of a constant to $V(\phi)$ ² does not affect the physics of the system because in this case, there is no absolute zero of energy density; while in the gravitation case adding a constant to V is equivalent to adding a term proportional to $\sqrt{-g}$, namely, we are introducing a cosmological constant. So, once the false vacuum decays the cosmological constant inside the bubble will be different from the one outside; in practice we have to specify the initial condition for the absolute minimum of the energy density.

5.2 Computation of B coefficient

In this section, we want to show the computation of the coefficient B in the thin wall approximation.

5.2.1 ... in flat spacetime

Let's consider the Euclidean equation of motion

$$\left(\frac{\partial^2}{\partial \tau^2} + \nabla^2 \right) \phi = V'(\phi) \quad (5.3)$$

Given ϕ a solution of the Euler-lagrangian equation associated to the lagrangian (5.1), that respects the following properties:

1) ϕ approaches the false vacuum, ϕ_+ , at Euclidean infinity, i.e.

$$\lim_{|\vec{x}| \rightarrow \infty} \phi(\tau, x) = \phi_+ \quad (5.4)$$

2) ϕ is not a constant, and 3) ϕ has Euclidean action less than or equal to that of any other solution obeying 1) and 2), then the coefficient B can be expressed as³

$$B = S_E(\phi) - S_E(\phi_+) \quad (5.5)$$

Moreover, by Eq. (5.3), the minimum energy state ϕ_- can be reached only asymptotically, as

$$\lim_{\tau \rightarrow \pm \infty} \phi(\tau, x) = \phi_- \quad (5.6)$$

By time translation invariance, we might choose $\tau = 0$ when the field reaches ϕ_- , and as we know that $V(\phi_-) = 0$, Eq. (5.3) implies the following condition

²The addition of a constant is how the false vacuum decay to the true one manifests its effect on the potential

³The subscript "E" indicates that the action is the Euclidean one

$$\frac{\partial \phi}{\partial \tau}(0, \vec{x}) = 0 \quad (5.7)$$

Eqs. (5.4), (5.6), and (5.7) form the set of boundary conditions for the solution ϕ of the equation of motion (5.3).

From Eq. (5.7), we note that the motion of the particle for positive τ is just the time reversal of its motion for negative τ , so the particle bounces off at $\tau = 0$ and returns in ϕ_- for $\tau \rightarrow \infty$. For this reason, the solution ϕ is commonly called the "bounce": it is $O(4)$ invariant, basically depends only on the Euclidean distance (i.e. $\rho = \sqrt{\tau^2 + |\vec{x}|^2}$), once an appropriately center of coordinates is chosen.

Under this assumption, the equation of motion, Eq. (5.3) simplifies to

$$\left(\frac{d^2}{d\rho^2} + \frac{3}{\rho} \frac{d}{d\rho} \right) \phi = V'(\phi) \quad (5.8)$$

and the Euclidean action can be rewritten as

$$S_E = 2\pi^2 \int_0^\infty \rho^3 d\rho \left[\frac{1}{2} \left(\frac{d\phi}{d\rho} \right)^2 + V(\phi) \right] \quad (5.9)$$

Thin wall approximation

The thin wall approximation consists in assuming the form of the potential as the sum of a symmetric part (under inversion $\phi \rightarrow -\phi$) plus a symmetry-breaking term, so it can be expressed as.

$$V(\phi) = V_0 + \frac{\varepsilon}{\phi_- - \phi_+} (\phi - \phi_+) \quad (5.10)$$

with ϕ_+ as the false vacuum and ϕ_- as the true one, such that $V_0(\phi_-) = V_0(\phi_+)$ and assuming ε small and defined as

$$\varepsilon = V(\phi_-) - V(\phi_+) \quad (5.11)$$

With this approximation, Eq. (5.3) becomes

$$\frac{d^2 \phi}{d\rho^2} = \frac{dV_0}{d\phi} \quad (5.12)$$

Where we have discarded the ε term because we consider it small, and also the $\frac{d\phi}{d\rho}$ one. If we proceed to include in the Eq. (5.12) the 1) condition we obtain:

$$\frac{1}{2} \left(\frac{d\phi}{d\rho} \right)^2 - V_0 = -V_0(\phi_\pm) \quad (5.13)$$

Eq. (5.13) can be solved by finding an expression for ϕ , that depends only on one integration constant, which we call $\bar{\rho}_0$, namely the point at which ϕ is the average of its two extremal values. In other words, we have:

$$\int_{\frac{\phi_+ - \phi_-}{2}}^{\phi} d\phi [2(V_0 - V_0(\phi_\pm))]^{-\frac{1}{2}} = \rho - \bar{\rho}_0 \quad (5.14)$$

Now, we have to determine $\bar{\rho}_0$, and we assume that $\bar{\rho}_0$ is large if compared to the length scale of the field motion. To do this we compute the coefficient B ,

with Eq. (5.5), and then impose its stationarity under variations of $\bar{\rho}_0$. Doing the calculation we note that the integration domain is naturally split into three parts: inside, outside (the bubble), and the wall.

Outside the bubble $\phi = \phi_+$ so Eq. (5.5) leads to the result

$$B_{\text{outside}} = 0 \quad (5.15)$$

Inside the bubble $\phi = \phi_-$, so if we do the integral, by using the definition of ε , Eq. (5.11), we obtain:

$$B_{\text{inside}} = -\frac{\pi^2}{2}\bar{\rho}_0^4\varepsilon \quad (5.16)$$

Finally, in the wall region, we have, by assuming $\rho = \bar{\rho}_0$

$$B_{\text{wall}} = 2\pi^2\bar{\rho}_0^3S_1 \quad (5.17)$$

where we have defined the quantity S_1 as

$$S_1 = \int_{\phi_-}^{\phi_+} d\phi [2(V_0 - V_0(\phi_+))]^{\frac{1}{2}} = 2 \int d\rho [V_0(\phi) - V_0(\phi_+)] \quad (5.18)$$

Summing all these contributions we obtain:

$$B = -\frac{\pi^2}{2}\bar{\rho}_0^4\varepsilon + 2\pi^2\bar{\rho}_0^3S_1 \quad (5.19)$$

Imposing the stationarity, we find an expression for $\bar{\rho}_0$:

$$\bar{\rho}_0 = \frac{3S_1}{\varepsilon} \quad (5.20)$$

We note that the initial assumption on $\bar{\rho}_0$ is justified, in fact, $\bar{\rho}_0$ increases when ε decreases.

Substituting Eq. (5.20) in Eq. (5.19), we obtain the general result for the coefficient B :

$$B_0 = \frac{27\pi^2S_1^4}{2\varepsilon^3} \quad (5.21)$$

5.2.2 ... in curved spacetime

Let's now sketch the computation for the B coefficient but in the presence of gravity, i.e. in curved spacetime. As done in the flat spacetime case, we want to construct an $O(4)$ invariant Euclidean solution, because we assume that gravitation doesn't break the symmetry of the pure scalar problem treated before. We start writing the most general rotationally invariant Euclidean metric. The orbits of the rotation group are three-dimensional manifolds with the geometry of three-spheres. So, by choosing ξ as the radial coordinate, we can write the element of length as:

$$ds^2 = (d\xi)^2 + \rho(\xi)^2(d\Omega)^2 \quad (5.22)$$

where $(d\Omega)^2$ is the element of distance on a unit three-sphere, and ρ gives the radius of curvature of each three-sphere. From Eq. (5.22), manipulating the Christoffel symbols we can obtain the Euclidean equations of motion, in particular for the scalar field we have:

$$\left(\frac{d^2}{d\xi^2} + \frac{3}{\rho} \frac{d\rho}{d\xi} \frac{d}{d\xi} \right) \phi = V'(\phi) \quad (5.23)$$

On the other hand, if we consider Einstein's equations $G_{\xi\xi} = -kT_{\xi\xi}$ (where $k = 8\pi G$), they simply reduce to the following equation:

$$\left(\frac{d\rho}{d\xi} \right)^2 = 1 + \frac{k\rho^2}{3} \left(\frac{1}{2} \left(\frac{d\phi}{d\xi} \right)^2 - V(\phi) \right) \quad (5.24)$$

Finally, the action S_E takes the form

$$S_E = 2\pi^2 \int d\xi \left[\rho^3 \left(\frac{1}{2} \left(\frac{d\phi}{d\xi} \right)^2 + V(\phi) \right) + \frac{3}{k} \left(\rho^2 \frac{d^2\rho}{d\xi^2} + \rho \frac{d\rho}{d\xi} \right)^2 - \rho \right] \quad (5.25)$$

We observe that Eq. (5.23) is equal to the one for the flat spacetime, Eq. (5.3) except for the independent variable, which now is ξ instead of ρ , and for the coefficient of the term with one derivative of ϕ . So we can proceed in the same way adopted for the pure scalar case, starting by replicating Eq. (5.14)

$$\int_{\frac{\phi_+ - \phi_-}{2}}^{\phi} d\phi [2(V_0 - V_0(\phi_{\pm}))]^{-\frac{1}{2}} = \xi - \bar{\xi} \quad (5.26)$$

Once we have ϕ , we can find a solution for ρ by solving Eq. (5.24) given a choice for the integration constant $\bar{\rho} = \rho(\bar{\xi})$. We note that $\bar{\rho}$, unlike $\bar{\xi}$, has a convention-independent meaning, namely it is the curvature radius of the wall separating the false vacuum from the true one.

The computation procedure to find $\bar{\rho}$ is the same as shown in section 5.2.1. To simplify the B computation we manipulate the Euclidean action, firstly eliminating the second-derivative term, by integration by parts, and then using Eq. (5.24) to rewrite the ρ derivative; so Eq. (5.25) becomes:

$$S_E = 4\pi^2 \int d\xi \left[\rho^3 V(\phi) - \frac{3\rho}{k} \right] \quad (5.27)$$

From this action, we can evaluate the B coefficient in the thin-wall approximation. The results for the outside, Eq. (5.15), and wall part, Eq. (5.17), don't change from the pure scalar case. While, inside the wall, where ϕ is constant we obtain:

$$B_{\text{inside}} = \frac{12\pi^2}{k^2} \left\{ V(\phi_-)^{-1} \left(\left[1 - \frac{1}{3} k \bar{\rho}^2 V(\phi_-) \right]^{\frac{3}{2}} - 1 \right) - (\phi_- \rightarrow \phi_+) \right\} \quad (5.28)$$

The previous expression is algebraically quite involved, so we analyze a simpler case when $V(\phi_-) = 0$ and $V(\phi_+) = \varepsilon$. We can observe that, in this case, B is stationary for the value of $\bar{\rho} = \frac{12S_1}{4\varepsilon + 3kS_1^2}$ and we have for B

$$B = \frac{B_0}{\left[1 + \left(\frac{\bar{\rho}_0}{2\Lambda} \right)^2 \right]^2} \quad (5.29)$$

where the subscript 0 stands for the variables evaluated in the absence of gravitation and having defined $\Lambda = (k\varepsilon/3)^{-\frac{1}{2}}$.

So we can note that the effect of gravitation, in this special case, is a decrease in B and $\bar{\rho}$ that leads to an increase of the rate Γ for the vacuum decay.

5.3 Computation of A coefficient

The coefficient A is less crucial than B because the exponential factor is the most important term in the bubble nucleation rate. But for the sake of completeness, we want to show the result. In quantum field theory, the explicit 1-loop formula for the false vacuum decay rate can be calculated using the path integral formulation, so the coefficient A takes the following form:

$$A = \left(\frac{B}{2\pi\hbar} \right)^2 \left| \frac{\text{Det}' \mathcal{O}(\phi)}{\text{Det} \mathcal{O}(\phi_+)} \right|^{-\frac{1}{2}} \quad (5.30)$$

where the operator $\mathcal{O}$ is the fluctuation operator $\partial^2 + V''(\phi)$ and the prime to the determinant denotes that the zero modes associated with the translational symmetry of the bounce are removed⁴. A first discussion of this topic was by [21], but more precise calculations for coefficient A are possible [22] but they involve computing fluctuations around the bounce solution of all fields coupled to the scalar. This requires renormalization of the loop corrections, and also, to avoid double counting, expansion around the level bounce (rather than the bounce in the loop-corrected potential). In the gravitational case, we have to calculate the fluctuations of the scalar and gravitation too, and, since we are not able to renormalize the gravitational loops, we settle for a rough estimation, using dimensional analysis.

5.4 Bubble growth

Until now we have only talked about the materialization of the bubble, so in this section, we want to briefly discuss also the growth of the bubble.

5.4.1 ... in flat spacetime

We can describe the classical growth of a bubble using the bounce, denoted by ϕ . The intersection of Euclidean space (imaginary time) and Minkowski space (real-time) forms the surface $t = 0$. The value of ϕ on this surface represents the field configuration at the moment the bubble materializes, moreover, we know that its time derivative was zero. With these initial conditions and the classical field equations in Minkowski space, we can determine the bubble's growth without explicitly solving the equations. Analytically continuing the Euclidean solution, i.e. by substituting $\rho = \sqrt{|x|^2 - t^2}$, turns Euclidean $O(4)$ invariance into a Minkowskian $O(3,1)$ invariance. In the thin-wall approximation, the bubble materializes at rest with radius $\bar{\rho}_0$ and grows along the hyperboloid $\rho = \bar{\rho}_0$. From a macroscopic perspective, since $\bar{\rho}_0$ is usually at subnuclear scales, the bubble accelerates to nearly the speed of light and continues to grow indefinitely.

5.4.2 ... with gravity

The basic idea to realize the growth of the bubble is the same as the pure scalar case, but now we have to continue analytically the metric, turning an $O(4)$ invariant Euclidean manifold into an $O(3,1)$ invariant Minkowskian one. For the space-like points of the manifold, the analytical continuation consists of a simple

⁴These are responsible for the formula giving a decay rate per unit volume, as in the first paper of Coleman[19]

reinterpretation of $\rho(\xi)$ as a Minkowskian spacelike separation rather than Euclidean radial distance, this leads to the following metric

$$ds^2 = -d\xi^2 - \rho(\xi)^2(d\Omega_S)^2 \quad (5.31)$$

While, inside the bubble, to avoid pure coordinate singularity, we have to continue to the time-like region; we choose $\xi(0) = 0$ and continue $\xi = i\tau$ with τ real, obtaining for the metric

$$ds^2 = d\tau^2 - \rho(-i\tau)^2(d\Omega_T)^2 \quad (5.32)$$

where $(d\Omega_{T/S})$ is the element of length for a unit hyperboloid with time-like/space-like normal in Minkowski space. The equation above is telling us that the inside of the bubble can be described as a Robertson-Walker universe of open type. For the special case $V(\phi_+) = \varepsilon$, $V(\phi_-) = 0$, by following the just shown prescription, we can compute the metric inside/outside the bubble by solving the Eq. (5.24) remembering that, in the thin-wall approximation, ϕ is constant and assumes the value ϕ_+ outside the bubble and ϕ_- inside. So, since inside the bubble V vanishes, we obtain $\rho = \xi$ and an ordinary Minkowski space, while, outside the bubble, Eq. (5.24) reduces to

$$\left(\frac{d\rho}{d\xi}\right)^2 = 1 - \frac{\rho^2}{\Lambda^2} \quad (5.33)$$

that has as a solution $\rho = \Lambda \sin\left(\frac{\xi}{\Lambda}\right)$, which can be demonstrated to be an ordinary de Sitter space in an unconventional set of coordinates.

5.5 Example: "Mexican hat" potential

Let's now consider V_0 as a "Mexican hat" potential, like:

$$V_0 = \frac{\lambda}{8} \left(\phi^2 - \frac{\mu^2}{\lambda} \right)^2 \quad (5.34)$$

If we put this potential in Eq. (5.14) we obtain the following expression for ϕ :

$$\phi = \frac{\mu}{\sqrt{\lambda}} \tanh \left[\frac{1}{2} \mu (\rho - \bar{\rho}_0) \right] \quad (5.35)$$

A sketch of the ϕ behavior and its relation with the bubble is well represented by the figure below:

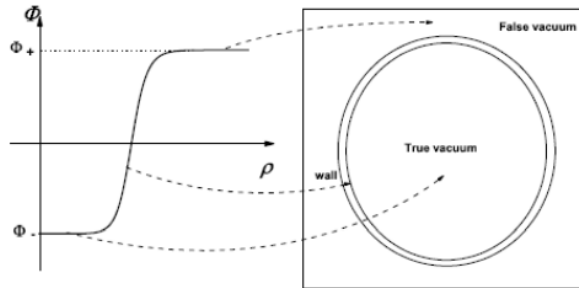


Figure 5.2. Typical vacuum bubble with the transitional region (wall) in the middle.
Figure taken from [23]

So we will assume $\phi_{\pm}$ to be respectively $\pm \frac{\mu}{\lambda}$ such that automatically $V(\phi_{\pm}) = 0$ holds. From Eq. (5.20) we can derive a more precise range of validity of the thin-wall approximation for the "Mexican hat" case, i.e.

$$\frac{2\mu^4}{\lambda \varepsilon} \gg 1 \quad (5.36)$$

If we compute the B coefficient in the pure scalar case, we have to perform the integral (5.18) with the potential (5.34) obtaining the following result

$$B_0^{\text{MH}} = \frac{8\pi^2 \mu^{12}}{3\varepsilon^3 \lambda^4} \quad (5.37)$$

We observe that the Mexican-hat potential has the features of the special case analyzed in section 5.2.2, so by using Eq. (5.28), we obtain the coefficient B :

$$B_{\text{MH}} = \frac{8\pi^2 \mu^{12}}{3\varepsilon^3 \lambda^4} \left[1 + 3 \frac{k}{3} \frac{\mu^6}{\varepsilon \lambda^2} \right]^{-2} \quad (5.38)$$

As explained in section 5.3, we estimate the A coefficient, as μ^4 .

So the rate for unit volume of the false vacuum decay for a "Mexican hat" potential is, by using Eq. (5.2) with the B coefficient computed in Eq. (5.38):

$$\Gamma_{\text{MH}}/V \sim \mu^4 \exp \left(-\frac{1}{\hbar} \left[\frac{8\pi^2 \mu^{12}}{3\varepsilon^3 \lambda^3} \right] \cdot \left[1 + 3 \frac{k}{3} \frac{\mu^6}{\varepsilon \lambda^2} \right]^{-2} \right) \quad (5.39)$$

If we assume an exponential expansion background (such as inflation), we can compute the number of e -folds needed for a bubble nucleation event to happen. We want to perform this computation for the "Mexican hat" potential, Eq. (5.34). To do so, firstly we evaluate the volume of an expanding Universe to turn Eq. (5.39) into a pure rate, we can write the following relation

$$\Gamma(N_e) = H^{-3} e^{3N_e} (\Gamma/V) \quad (5.40)$$

Then if we consider the bubble nucleation event as a Poissonian process, the probability can be written⁵ as

$$P(t) \sim e^{-\Gamma(N_e)t} \quad (5.41)$$

If we want to express it as a function of the number of e -folds N_e we obtain

$$P(N_e) \sim e^{-\frac{\Gamma(N_e)}{H} N_e} \quad (5.42)$$

Now, if we impose the expectation value equal to N_B , i.e. the wanted number of bubbles (nucleation events), we can infer the number of e -folds needed for N_B events to occur. From Poissonian distribution properties, we know that the expectation value is the parameter at the exponent, in this case, $\frac{\Gamma(N_e)}{H} N_e$. Let's now make this computation for the "Mexican hat" rate, Eq. (5.39), with the procedure described above, we obtain the following relation:

⁵The probability is not normalized, but for the scope of this computation this fact is not important

$$\left(\frac{\mu}{H}\right)^4 N_e \exp\left(3N_e - \frac{1}{\hbar} \left[\frac{8\pi^2 \mu^{12}}{3\varepsilon^3 \lambda^3}\right] \cdot \left[1 + \frac{k}{3\varepsilon} \frac{\mu^6}{\lambda^2}\right]^{-2}\right) = N_B \quad (5.43)$$

By putting $\hbar = 1$ and $k = M_{\text{Pl}}^{-2}$, we can rewrite the Eq. (5.43) obtaining the following transcendent equation for the number of e -folds

$$3N_e + \ln(N_e) = 4 \ln\left(\frac{H}{\mu}\right) + \left[\frac{8\pi^2 \mu^{12}}{3\varepsilon^3 \lambda^3}\right] \cdot \left[1 + \frac{1}{3M_{\text{Pl}}^2 \varepsilon} \frac{\mu^6}{\lambda^2}\right]^{-2} + \ln(N_B) \quad (5.44)$$

To solve the Eq. (5.44) we note that for a known term bigger than one (this case is always achieved even if we don't consider the square bracket term), the dominant term will be the linear one and the number of e -folds can be approximately evaluated by dividing by three the known term.

The number of bubble nucleation events needed for the Multiverse to solve the apparent unnaturalness of the cosmological constant is⁶ $N_B = 10^{120}$. So we sketch some values for the number of e -folds for different values of ε and λ (we assume that the coupling is at most 1), to be more precise, we consider the condition (5.36) satisfied by different orders of magnitude ($10 - 10^2$). For the value of μ we use M_{Pl} scale while for H we use a typical value, like 10^{15} GeV. The obtained results are summarised in the following table

λ	$2\mu^4/\lambda \varepsilon$	N_e
1	10	400
1	10^2	$1.27 \cdot 10^3$
10^{-1}	10	283.8
10^{-1}	10^2	236.2
10^{-2}	10	82.2
10^{-2}	10^2	83.5

Table 5.1. Characteristic number of e -folds for 10^{120} events and $H = 10^{15}$ GeV

5.6 Application to inflation and the Multiverse

The major interest of this process is linked to cosmology: there is the possibility that, during the inflationary process, the inflaton rolls in a false vacuum, i.e. a local minimum of the potential (which can have lots of local minima). If this happens, a false vacuum decay can occur leading to a bubble nucleation of a true vacuum. This mechanism is illustrated in the following figure

⁶We will motivate this choice in the following section

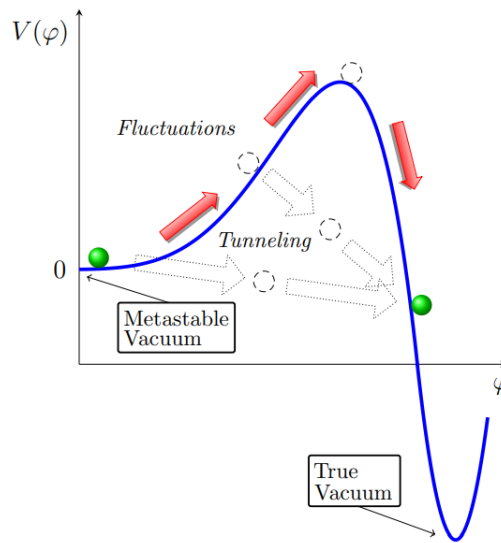


Figure 5.3. Illustration of vacuum decay for a potential with a metastable vacuum at the origin. Figure taken from [24]

The true vacuum bubble corresponds to a "new universe" within the starting one, that starts expanding at the speed of light while the "starting Universe" is still inflating. This process has a rate that depends on the form of the inflaton potential and can be estimated with the procedure discussed in chapter 5. How many times the bubble nucleation process could happen can be determined as the ratio of the maximum number of e -folds allowed by the chosen inflation model and the characteristic time of one vacuum decay (translated in e -folds).

This mechanism allows the birth of a Multiverse: with this word⁷, in this thesis, we mean a set of connected patches, separated by bubble walls, that can have different values of physical constants but the same laws of physics.

5.6.1 The Multiverse: strengths and weaknesses

This section was inspired by [25].

The Multiverse idea is interesting but at the same time very speculative, in this section we want to analyze why it's so discussed hypothesis, which breaks the scientific community in half. If we talk about the strength of the Multiverse framework, we have to cite the fact, that if linked with anthropic arguments, it can be an elegant solution to the complexity and naturalness problems. In fact, if we consider the fact that there are a multitude of Universes with different physical constants, it is not so difficult to believe that we are, among all, in a Universe that supports our life. So the variables that seem highly fine-tuned for our existence, from the cosmological constant to the masses of particles, where their slightest change can have dramatic consequences such as the loss of stability by life-important atoms and nuclei, are in that way only by accident. Despite this fact seems to be very unlikely, it's reasonable that such a configuration occurs, given such a great number of possible universes.

⁷We have to specify the meaning of the word, because there are 4 types of Multiverse, following Max Tegmark classification, Level I: an extension of our universe, Level II: universes with different physical constants, Level III: many-worlds interpretation of quantum mechanics (by Hugh Everett III), and Level IV: ultimate ensemble (Tegmark mathematical proposal)

The idea is that, provided a sufficiently large number of universes, the probability of an event that seems to us very unlikely reaches a probability near to one. Let's take the cosmological constant as a quantitative example: in the previous section, we have chosen 10^{120} as the number of bubble nucleation events because it is, more or less, the difference of the order of magnitude between the expected theoretical value and the measured one (as we have seen in chapter 2). It's important to stress that Multiverse idea is encouraged by String Theory, in fact, it was shown that a huge number of vacua can be constructed, like 10^{500} , creating in this way the so-called "String Theory landscape"⁸, and there is no principle to select a privileged vacuum among all.

Its principal weaknesses are the extreme difficulties in the testability and falsifiability of this idea, for example, we cannot test directly if we are in a patch of a Multiverse because the other domains are causally disconnected from us. Furthermore, since in the Multiverse hypothesis, all the theory parameters are not fixed uniquely, the only way that we can make predictions is probabilistic: we would like to estimate the distributions of these parameters. Finally, it is interesting to ask how likely our patch of the universe could be in a universe that has other patches that continue to inflate: here, the "measure problem"[26] occurs. In a few words, assuming an infinite amount of expanding Universes, the standard frequentist approach becomes ill-defined. In the frequency evaluation of an event, a ratio of infinities appears, and if we try to regularize this ratio, we don't succeed because we obtain different results according to the regularization choice, without knowing which approach is the correct one.

⁸The number of string vacua is enormous because the string ground states do not depend only on the way of extra dimensions' compactification but also on how the field components are arranged within these internal spaces.

Chapter 6

From thermal field theory to a dissipation coefficient computation

6.1 Field theory at finite temperature

This section follows the approach of [27]. The formalism used in standard quantum field theory applies to the computation of observables, like cross-sections, measured in empty space-time, as it may be the case for a particle interaction in an accelerator. However, in the early stages of the universe, the environment had a significant matter and radiation density, making the assumptions of conventional field theories impractical. Therefore, under those circumstances, the methods of QFT are no longer applicable and should be replaced by others that are closer to thermodynamics. In thermal field theory, the background state is a thermal bath, making it extremely useful to study phenomena that occurred in the early universe, such as phase transitions or inflationary cosmology. In this chapter, we review the Matsubara formalism, which is also called imaginary-time formalism.

It is important to point out that Matsubara's formalism needs the context of equilibrium systems to be developed completely. The idea is to give up the time variable in favor of the equilibrium temperature. This method is ideal for studying static, equilibrium properties of a quantum system. So, this approach is excellent for calculating equilibrium thermodynamic quantities, but it is unsuitable for studying non-equilibrium phenomena such as phase transitions.

The statistical behavior of a quantum mechanical system, in thermal equilibrium, is normally studied through an appropriate ensemble. An ensemble is a collection of an infinitely large number of identical and independent systems. There are three main ensembles that are used depending on the characteristics of the system, the first one is the microcanonical ensemble, which is used to describe an isolated system with fixed energy E , particle number N , and volume V . Then, there is the canonical ensemble, which describes a system in contact with a heat source at temperature T : the energy can be exchanged between them and T , N , and V are fixed. Finally, the grand canonical ensemble is used when the system can exchange energy and particles with the thermal bath: T , V , and the chemical potentials are fixed. In a generic ensemble, we can define a density matrix as

$$\rho(\beta) = \frac{e^{-\beta\mathcal{H}}}{Z(\beta)} = \frac{e^{-\beta\mathcal{H}}}{\text{Tr}(e^{-\beta\mathcal{H}})} \quad (6.1)$$

where β represents the inverse of the equilibrium temperature, i.e. $\beta = 1/T$ and $\mathcal{H}$ is an appropriate Hamiltonian, once the ensemble is chosen. The factor $Z(\beta)$ is the partition function of the ensemble, it is a scalar function of temperature and other equilibrium state properties. The second "=" of Eq. (6.1) arises from the fact that the density matrix can be also defined (in the energy basis) by quantum mechanics as

$$\rho = \sum_n p_n |n\rangle \langle n| \quad (6.2)$$

and since p_n s have to satisfy the normalization condition of probabilities, this, linked to the orthonormality of energy basis, leads to the following condition

$$\text{Tr } \rho(\beta) = 1 \quad (6.3)$$

where Tr stands for a sum over the expectation values of the operator on any complete basis.

To study equilibrium statistical mechanics, the main approach is to evaluate the partition function of the system. In general, however, the partition function of an interacting system cannot be evaluated exactly. In fact, we can see from Eq. (6.1), that a perturbative expansion in powers of a coupling constant does not seem to be easy, because we have a sum over the expectation values in all possible states, which turns out to be infinite for any QFT. The Matsubara formalism provides a way of evaluating the partition function perturbatively using a diagrammatic method which is analogous to that is used in conventional QFTs at zero temperature. In this chapter, we introduce this formalism through the path integral method.

We can recall the transition amplitude in a quantum field theory at zero temperature in Minkowski space, which is

$$\langle \phi(x_f, t_f) | \phi(x_i, t_i) \rangle = \langle \phi_f | e^{-iH(t_f - t_i)} | \phi_i \rangle = N' \int D\phi e^{iS} \quad (6.4)$$

where ϕ is the basic quantum field variable, N' is an irrelevant normalization constant and the subindices i, f stand for the initial and final coordinates of the path. In this case, the path integral is defined over a path that satisfies $\phi(x_i, t_i) = \phi_i$ and $\phi(x_f, t_f) = \phi_f$.

We note that if we define $t_f - t_i = -i\beta$, identify (up to a phase) the field variables at the endpoint as $\phi_f = \pm \phi_i$ and then integrate over them (which is equivalent to taking the trace), we can rewrite the partition function as

$$Z(\beta) = \text{Tr}(e^{-\beta\mathcal{H}}) = \int d\phi_f \langle \phi_f | e^{-\beta\mathcal{H}} | \phi_f \rangle = N' \int D\phi e^{-S_E} \quad (6.5)$$

where now $D\phi$ includes also the integration $d\phi_f$ and the action is the Euclidean one, if the chemical potential is zero¹.

$$S_E = \int_0^\beta d\tau \int d^3x \mathcal{L}_E \quad (6.6)$$

¹In general, $\mathcal{S}_E = S_E + \beta\mu N$, it can be derived considering the chemical potential as arising from a constant electrostatic potential for charged particles.

We note here that the contour of integration in the complex time plane, in the present formalism, is along the negative imaginary axis, since $t \rightarrow -i\tau$ and $0 \leq \tau \leq \beta$. Let's now concentrate on the two-point Green's functions at finite temperature. They can be defined as

$$\mathcal{G}(\tau, \tau') = Z^{-1}(\beta) \text{Tr} \left(e^{-\beta \mathcal{H}} \mathcal{T} \phi(\tau) \phi(\tau') \right) \quad (6.7)$$

where ϕ can represent either a bosonic or a fermionic field in the Heisenberg picture.²

The two-point Green's functions satisfy the following properties

$$\begin{aligned} \mathcal{G}_\beta(\tau, \tau') &= \mathcal{G}(\tau - \tau') \\ \mathcal{G}_\beta(\tau + \beta) &= \pm \mathcal{G}(\tau) \end{aligned} \quad (6.8)$$

The first one tells us that Green's functions depend only on the time difference, while the second shows that $\mathcal{G}_\beta(\tau)$ respects a periodicity property, for bosons, and an anti-periodicity in the fermion case.

Since the Greens functions are defined within a finite time interval, i.e. $-\beta \leq \tau \leq \beta$, it can be written as a Fourier series involving only discrete frequencies, namely

$$\mathcal{G}_\beta(\tau) = \frac{1}{\beta} \sum_n e^{-i\omega_n \tau} \mathcal{G}_\beta(\omega_n) \quad (6.9)$$

By imposing the (anti-)periodicity conditions to Eq. (6.9) we obtain the so-called Matsubara frequencies

$$\begin{cases} \omega_n = \frac{2n\pi}{\beta} & \text{for bosons} \\ \omega_n = \frac{(2n+1)\pi}{\beta} & \text{for fermions} \end{cases} \quad (6.10)$$

In contrast to the imaginary time variable τ at finite temperature, which takes values in a finite interval, the spatial coordinates have an infinite range just as in the case of zero temperature field theory, so we can generalize Eq. (6.9) as

$$\mathcal{G}_\beta(\tau, x) = \frac{1}{\beta} \sum_n \int \frac{d^3k}{(2\pi)^3} e^{-i\omega_n \tau + k \cdot x} \mathcal{G}_\beta(\omega_n, k) \quad (6.11)$$

Now we want to determine the free propagator for the bosonic/fermionic case, the simplest way to do so is going from Minkowski space to finite temperature Euclidean space³.

For a relativistic theory of a fermion, the two-point Green's function is

$$S(\omega_n, \vec{p}) = -\frac{\gamma_0 \omega_n - \vec{\gamma} \cdot \vec{p} + m}{\omega_n^2 + \vec{p}^2 + m^2} \quad (6.12)$$

While for a boson we have

$$D(\omega_n, \vec{p}) = -\frac{1}{\omega_n^2 + \vec{p}^2 + m_\phi^2} \quad (6.13)$$

with the prescription for the frequencies summarised in Eq. (6.10).

²Heisenberg picture is one, where all the time dependence is inside the operators, as opposed to Schrodinger's one where the time dependence is in the states.

³we have $P = (\omega_n, \vec{p})$

The partition function's path integral formulation is interesting because it shows how zero temperature and finite temperature descriptions of quantum field theories are similar at the qualitative level. The diagrams' definitions are the same, and the respective graphs can be obtained from an expansion of the path integral. The only difference is that the field variables now have to satisfy (anti-)periodicity conditions and correspondingly should be expanded in an appropriate basis. This leads to the energy values being discrete, as shown in Eq. (6.10) and the propagators acquire a non-trivial temperature dependence. Despite this fact, in this formalism, the propagator does not have a simple decomposition into a zero temperature part and a genuinely temperature-dependent part, although graphs calculated with them do. We stress that the quantization of the frequencies is in complete analogy with a quantum mechanical system in a one-dimensional box in the τ direction, so in this description, the overall space-time topology can be parametrized as $\mathbb{R}^3 \times S^1$.

6.1.1 Evaluation of Matsubara sums

In this section, we deal with the Matsubara sums, whose calculation is crucial in the computation of Feynman diagrams at finite temperatures in the imaginary-time formalism. We briefly review the standard contour integral method and the Saclay method, by following respectively the approach of Pisarski [28] and of [29].

Standard contour integral method

We want to evaluate sums of the following form

$$\sigma(T) \equiv T \sum_{n=-\infty}^{\infty} f(q_n) \quad (6.14)$$

where q_n can be the fermionic or bosonic Matsubara frequencies, i.e. Eq. (6.10).

The idea is to translate the Matsubara sum into a sum of the residues of a new function, $g \cdot f$, and then evaluate this one as an integral in the complex plane. To do so we need some mathematical constraints necessary when we extend f to complex variables. Now let $f : \mathbb{C} \rightarrow \mathbb{R}$ be meromorphic⁴ and regular on $\mathbb{R}$. Then let $g_{B/F} : \mathbb{C} \rightarrow \mathbb{C}$ be meromorphic with poles only on $z = q_n$ for all $n \in \mathbb{Z}$. Also, we assume that the poles are simple and the residue of $g_{B/F}$ at each of these poles is T . Then if $f(z)g_{B/F}(z) \rightarrow 0$ as $\text{Re}(z) \rightarrow \infty$ fast enough and $\text{Im}(z) \in (-\epsilon, \epsilon)$ for small enough $\epsilon > 0$, we have

$$\sigma_{B/F}(T) = \text{Res}(f \cdot g_{B/F}, q_n) = \frac{1}{2\pi i} \oint_{\Gamma_\epsilon} f(z)g_{B/F}(z) dz \quad (6.15)$$

where the contour Γ_ϵ consists of the line $\text{Im}(z) = -i\epsilon$ from $-\infty$ to ∞ travelled forward and back along the line $\text{Im}(z) = i\epsilon$ and it is shown, for the bosonic case, in figure 6.1.

⁴A meromorphic function, defined in an open subset D of the complex plane $\mathbb{C}$ is a function that is holomorphic on all of D except for a set of isolated points, which are the poles of the function. A function is holomorphic if it is complex differentiable for each point of the domain, so this means that it is infinitely differentiable and locally equal to its own Taylor series.

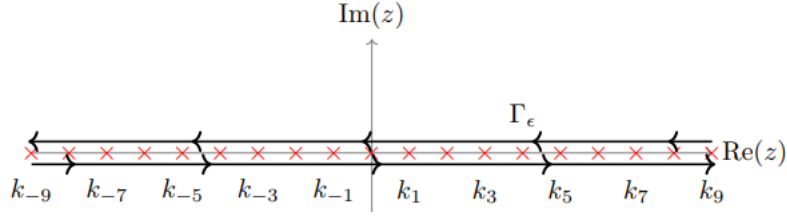


Figure 6.1. The integration contour Γ_ϵ for bosonic thermal sums. The "x" marks the Matsubara frequencies, which are only labeled at odd frequencies to avoid confusion. Figure taken from [29].

The mathematical conditions listed previously allow us to close the contour at infinity and thus achieve the last equality of Eq. (6.15) by the residue theorem.

We have to be careful in the application of the residue theorem because it holds only for compact contours, and it is not the case of Γ_ϵ , but Eq. (6.15) becomes exact for the path, shown in figure 6.2, $\Gamma \equiv \bigcup_{n \in \mathbb{Z}} \Gamma_\eta^{(n)}$, where $\Gamma_\eta^{(n)}$ is a circular contour with infinitesimal radius η around each Matsubara frequency. To bring back to our original contour we can link the circular contours along the real axis since integral over these patches cancel and then deform Γ in Γ_ϵ by the Cauchy theorem.

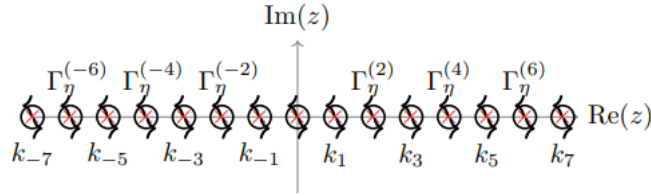


Figure 6.2. The circles intermediate method to obtain the contour in figure 6.1. Figure taken from [29].

A natural choice for the function $g_{B/F}(z)$ is $\pm n_{B/F}(iz)$, where $n_{B/F}$ is the Bose-Einstein/Fermi-Dirac distribution defined in Eq. (6.29). This function has residue equal to T for $z = q_n$ and enjoys the property $n_{B/F}(-iz) = \mp 1 - n_{B/F}(iz)$. So we obtain for σ

$$\begin{aligned} \sigma_{B/F} &= \pm \int_{-\infty-i\epsilon}^{\infty-i\epsilon} \frac{dz}{2\pi} f(z) n_{F/B}(iz) \pm \int_{\infty+i\epsilon}^{-\infty+i\epsilon} \frac{dz}{2\pi} f(z) n_{F/B}(iz) = \\ &= \int_{\mathbb{R}} \frac{dz}{2\pi} f(z) \pm \int_{-\infty-i\epsilon}^{\infty-i\epsilon} \frac{dz}{2\pi} [f(z) + f(-z)] n_{F/B}(iz) \end{aligned} \quad (6.16)$$

where we returned, by Cauchy theorem, to the real axis in the first term, thanks to the lack of singularities there and we made a change of variable, $z \rightarrow -z$ in the second term.

Saclay method

The basic idea of this method is to use propagators that are in a momentum representation in space, but in a coordinate representation in time. So for a scalar, we have

$$\mathcal{G}_\beta(\tau, p) = \frac{1}{\beta} \sum_n e^{-i\omega_n \tau} \frac{1}{P^2 + m_\phi^2} \quad (6.17)$$

To solve for $\mathcal{G}_\beta(\tau, p)$, instead of doing the sum over n it is preferable to Fourier transform Eq. (6.17) to

$$\frac{1}{P^2 + m_\phi^2} = \int_0^\beta d\tau e^{i\omega_n \tau} \mathcal{G}_\beta(\tau, p) \quad (6.18)$$

This equation, linked to periodicity boundary conditions, Eq. (6.8), has the following solution

$$\begin{aligned} \mathcal{G}_\beta(\tau, p) &= \frac{1}{2E_p} \sum_{s=\pm} f_s^B(p) \exp(-sE_p |\tau|) \\ \text{with } E_p &= \sqrt{p^2 + m_\phi^2} \end{aligned} \quad (6.19)$$

and with $f_+^B(p) = 1 + n_B(p)$, $f_-^B(p) = n_B(p)$.

We can repeat the same reasoning for the fermion case. By considering again $\mathcal{G}_\beta$ but this time with the fermionic Matsubara frequency and anti-periodicity boundary conditions, we have

$$\mathcal{G}_\beta(\tau, p) = \frac{1}{2E_p} \sum_{s=\pm} f_s^F(p) \exp(-sE_p |\tau|) \quad (6.20)$$

and with $f_+^F(p) = 1 - n_F(p)$, $f_-^F(p) = -n_F(p)$.

The fermion propagator can be written as

$$\frac{1}{i\not{P} + m} = \int_0^\beta d\tau e^{i\omega_n \tau} \mathcal{G}_F(\tau, p) \quad (6.21)$$

where the fermionic green function can be derived as

$$\mathcal{G}_F(\tau, p) = \left(-\frac{\partial}{\partial \tau} \gamma_0 + ip \cdot \gamma + m \right) \mathcal{G}_\beta(\tau, p) \quad (6.22)$$

Given the mixed propagators, the method of computation is straightforward: starting with an arbitrary loop diagram in momentum space, Eqs. (6.18) and (6.21) are used to Fourier transform each propagator in a loop with respect to time. After that, it is easy to do the energy sums, and then the integrals over time. Two relations useful in computations are the Matsubara representation of the Dirac delta

$$(2\pi)^3 \tau \delta^{(4)}(p - q) = \int_0^\tau e^{i\vec{p} - \vec{q} \cdot \vec{x} - i(\omega_n - \omega_l)x_0} \quad (6.23)$$

and how to re-express the factor $e^{-\beta E_p}$ as a function of $n_{B/F}$

$$e^{-\beta E_p} = \frac{f_-^B(p)}{f_+^B(p)} = -\frac{f_-^F(p)}{f_+^F(p)} \quad (6.24)$$

6.2 Computation of a dissipation coefficient: Yukawa interaction

Let's consider as interaction lagrangian a simple scalar Yukawa interaction

$$\mathcal{L}_{\text{int}} = y\phi\bar{\psi}\psi \quad (6.25)$$

So the resulting total lagrangian is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m_\phi^2\phi^2 + \bar{\psi}(i\not{\partial} - m)\psi + y\phi\bar{\psi}\psi \quad (6.26)$$

where m stands for the fermion mass and m_ϕ for the boson mass.

We want to compute, with means of thermal field theory, the dissipation coefficient experienced by a scalar from a Yukawa interaction with fermions. The idea is then to apply this calculation to a warm inflation model, where the inflaton plays the role of the scalar and the thermal bath is composed of SM fermions.

We know from [30] that the friction coefficient can be expressed as

$$\Upsilon = \frac{\rho(m_{\phi,\text{eff}}, 0)}{m_{\phi,\text{eff}}} \quad (6.27)$$

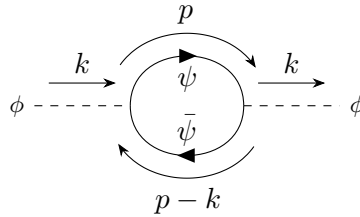
where ρ is the spectral function, i.e. the imaginary part of the two-point correlator, and $m_{\phi,\text{eff}}$ is the effective boson mass, to its computation we will devote the next subsection.

6.2.1 One-loop boson self-energy

The initial steps of the computation of self-energies follow [31]. The one-loop boson self-energy, i.e. the Euclidean correlator, can be expressed as

$$\Pi_E(\omega_a, \vec{k}) = y^2 \mathcal{T} \sum_{b=-\infty}^{\infty} \int \frac{d^3p}{(2\pi)^3} \text{Tr} \left[S(\omega_b, \vec{p}) S(\omega_b - \omega_a, \vec{p} - \vec{k}) \right] \quad (6.28)$$

and if we want to represent it through a Feynman diagram⁵ we have



Fermions follow the Fermi-Dirac distribution, while bosons the Bose-Einstein one, which, for $\mu = 0$, are

$$n_{F/B} = \left(e^{(E/T)} \pm 1 \right)^{-1} \quad (6.29)$$

⁵In this thesis, the Feynman diagrams were realized thanks to the Latex package "tikz-feynman" [32]

By doing the Matsubara sum and the trace, we can rewrite Eq. (6.28) as

$$\Pi_E(\omega_a) = y^2 \int \frac{d^3p}{(2\pi)^3} \left\{ \frac{\tanh(E_1/T)}{4E_1} \left[\frac{N}{(\omega_a - E_1)^2 - E_2^2} + \frac{N}{(\omega_a + E_1)^2 - E_2^2} + 2 \right] + (E_1 \longleftrightarrow E_2) \right\} \quad (6.30)$$

where $E_1 \longleftrightarrow E_2$ denotes the same expression as before but with index 1 and 2 interchanged. We have defined $N = 2s - 2(m_1 + m_2)^2$, with $s = \omega_a^2 - \vec{k}^2$, while the energies are $E_1 = \sqrt{m_1^2 + \vec{p}^2}$ and $E_2 = \sqrt{m_2^2 + (\vec{p} - \vec{k})^2}$. By separating in partial fractions and using the relation $\tanh(E/T) = 1 - 2n_F$, we can rewrite Eq. (6.31) as

$$\begin{aligned} \Pi_E(\omega_a) = y^2 \int \frac{d^3p}{(2\pi)^3} \frac{N}{4E_1E_2} & \left[\frac{1 - n_1 - n_2}{\omega_a - E_1 - E_2} + \frac{n_2 - n_1}{\omega_a + E_1 - E_2} \right. \\ & \left. + \frac{n_1 - n_2}{\omega_a - E_1 + E_2} + \frac{-1 + n_1 + n_2}{\omega_a + E_1 + E_2} \right] + \mathcal{C} \end{aligned} \quad (6.31)$$

Where $\mathcal{C}$ is a real constant independent of ω_a and $\vec{k}$, its explicit form is not useful because we are interested in the imaginary part of $\Pi(\omega_a)$ and so a real constant doesn't contribute.

When the discrete, imaginary frequency ω_a , is extended to an arbitrary complex variable ω , then Eq. (6.31) is again hermitian analytic with cuts along the real ω axis

If we take the discontinuity across the cuts of the computed correlator, Eq. (6.31), we obtain the spectral function $\rho(\omega, k)$

$$\begin{aligned} \rho(\omega, k) = \text{Im } \Pi_E = -\frac{1}{2i} \text{Disc } \Pi_E = \frac{y^2}{2} \int \frac{d^3p}{(2\pi)^3} \frac{2\pi N}{4E_1E_2} & [\delta(\omega - E_1 - E_2) ((1 - n_1)(1 - n_2) - n_1n_2) \\ & - \delta(\omega + E_1 - E_2) (n_1(1 - n_2) - n_2(1 - n_1)) \\ & - \delta(\omega - E_1 + E_2) (n_2(1 - n_1) - n_1(1 - n_2)) \\ & + \delta(\omega - E_1 - E_2) (n_1n_2 - (1 - n_1)(1 - n_2))] \end{aligned} \quad (6.32)$$

If we consider the fermion-antifermion interaction, then $m_1 = m_2 = m$. What we have to compute is $\rho(\omega, 0)$, so for $\vec{k} = 0$ we have $E_1 = E_2 = E$ and then for the Fermi distribution $n_1 = n_2 = n_F$, $s = \omega_a^2$, $N = 2(\omega_a^2 - 4m^2)$. This leads to

$$\rho(\omega, 0) = \frac{y^2}{4} \int \frac{d^3p}{(2\pi)^2} \frac{\omega^2 - 4m^2}{E^2} (1 - 2n_F) [\delta(\omega - 2E) - \delta(\omega + 2E)] \quad (6.33)$$

If we pass to spherical coordinates, then change the variable from $|p|$ to E (the differentials are related as follows $p dp = E dE$), rewrite the delta functions as $\delta(\omega \pm 2E) = \frac{1}{2}\delta(E \pm \omega/2)$, we obtain

$$\rho(\omega, 0) = \frac{y^2}{8\pi} (\omega^2 - 4m^2) \int_m^\infty dE \frac{\sqrt{E^2 - m^2}}{E} (1 - 2n_F) \left[\delta\left(E - \frac{\omega}{2}\right) - \delta\left(E + \frac{\omega}{2}\right) \right] \quad (6.34)$$

By performing the integral we have to be careful with the ω sign, in fact, since E is positively defined, it is integrated along the positive real axis, so, only one δ term contributes to the integral for each case. Defining $f(E)$ the integrand function, the result of the integral I , in general, can be written as

$$\begin{aligned} I &= \int_0^\infty f(E) \left[\delta \left(E - \frac{\omega}{2} \right) \pm \delta \left(E + \frac{\omega}{2} \right) \right] = \\ &= f \left(\frac{\omega}{2} \right) \theta(\omega) \pm f \left(-\frac{\omega}{2} \right) \theta(-\omega) = \\ &= \begin{cases} f \left(\frac{\omega}{2} \right) \text{sgn}(\omega) & \text{for } f \text{ even and " - " / } f \text{ odd and " + "} \\ f \left(\frac{\omega}{2} \right) & \text{for } f \text{ odd and " - " / } f \text{ even and " + "} \end{cases} \end{aligned} \quad (6.35)$$

If we explicit the $1 - 2n_F$ term, using the definition of the Fermi distribution, Eq. (6.29), we have

$$1 - 2n_F = \frac{e^{E/T} - 1}{e^{E/T} + 1} \quad (6.36)$$

By following the "delta rule" defined in Eq. (6.35), and noting that, in our case, $f(E)$ is even, we can rewrite Eq. (6.34) as

$$\rho(\omega, 0) = \frac{y^2}{8\pi} \frac{(\omega^2 - 4m^2)^{\frac{3}{2}}}{\omega} \left(\frac{e^{\omega/2T} - 1}{e^{\omega/2T} + 1} \right) \text{sgn}(\omega - 2m) \quad (6.37)$$

The effective mass of the scalar is defined as

$$m_{\phi, \text{eff}}^2 = m_\phi^2 + \delta m_\phi^2 \quad (6.38)$$

and the δm_ϕ^2 can be shown to be equal to the real part of the self-energy[30]

$$- \text{Re}\Pi_E = -\delta m_\phi^2 = \int_{-\infty}^{\infty} \frac{d\omega}{\pi} \frac{\rho(\omega, 0)}{\omega} \quad (6.39)$$

Thanks to the sign function of Eq. (6.37), the integrand of Eq. (6.39) becomes even and so we can rewrite

$$\delta m_\phi^2 = \frac{y^2}{4\pi^2} \int_{2m}^{\infty} d\omega \frac{(\omega^2 - 4m^2)^{\frac{3}{2}}}{\omega^2} \left(\frac{e^{\omega/2T} - 1}{e^{\omega/2T} + 1} \right) \quad (6.40)$$

Due to the integrand function domain, we note that the lower end of the integral is $2m$ and no more 0. Given a cutoff Λ , the integral of Eq. (6.40), diverges as Λ^2 : this is not so surprising. In general, the two-point correlator in Eq. (6.39) has a temperature-independent divergent part, because the correlator diverges at short distances. This amounts to a renormalization of the (bare) mass parameter m_ϕ^2 . In addition, there can be a finite T -dependent correction, which can be interpreted as a thermal mass, we call it $m_{\phi, s}^2$. So we can re-express the effective mass as

$$m_{\phi, \text{eff}}^2 = m_\phi^2 + \delta m_{\phi, T=0}^2 + m_{\phi, s}^2 \quad (6.41)$$

To set up the ingredients to compute the temperature-dependent part we can start with the following dispersion relation[28], which is a manipulation of Eq. (6.39) for a bosonic type of self-energy, to which corresponds an even real part (and an odd imaginary part)

$$\text{Re} \left(\Pi_E(-i\omega, p) - \Pi_E^{T=0}(-i\omega, p) \right) = \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{\text{Im} \left(\Pi_E(-i\omega', p) - \Pi_E^{T=0}(-i\omega', p) \right)}{\omega' - \omega} d\omega' \quad (6.42)$$

where we have written the integral only over positive ω' because the imaginary part is positive semi-definite (since we are assuming that only physical particles contribute to the discontinuity). We refer to the Cauchy principal value prescription with $\mathcal{P}$.

The spectral function at null temperature can be simply obtained by substituting $T = 0$ in Eq. (6.37), so it takes the form

$$\rho(\omega, 0)_{T=0} = \frac{y^2}{8\pi} \frac{(\omega^2 - 4m^2)^{\frac{3}{2}}}{\omega} \text{sgn}(\omega - 2m) \quad (6.43)$$

If we consider the renormalized inverse propagator of a scalar field we have $\sim p^2 - m_\phi^2 - \Pi_E$. So, for a thermal bath to screen, the self-energy at zero momentum, subtracted at $T = 0$, must be negative:

$$\Pi_E(2m, 0) - \Pi_E^{T=0}(2m, 0) = -m_{\phi,s}^2 \quad (6.44)$$

Since $\rho(2m, 0) = 0$, the left-hand side of Eq. (6.44), is equivalent to its real part.

So, by using Eqs. (6.41), (6.42), (6.43) and (6.44) we obtain the following relation for the thermal mass correction

$$\begin{aligned} m_{\phi,s}^2 &= -\frac{y^2}{4\pi^2} \mathcal{P} \int_{2m}^\infty d\omega' \frac{(\omega'^2 - 4m^2)^{\frac{3}{2}}}{\omega'(\omega' - 2m)} \left(\frac{2}{e^{\omega/2T} + 1} \right) = \\ &= -\frac{y^2}{2\pi^2} (2T)^2 \mathcal{P} \int_0^\infty \frac{\left(\omega^2 - \frac{2m}{T}\omega \right)^{\frac{3}{2}}}{\omega \left(\omega + \frac{m}{T} \right)} (e^{m/T} \cdot e^\omega + 1)^{-1} d\omega \end{aligned} \quad (6.45)$$

Thanks to the principal value prescription, the integration domain was restricted directly to $(2m, \infty)$, since for $\omega < 0$ the integral diverges.

If we make the assumption that the ratio m/T tends to zero, by solving the integral keeping only the finite part, we obtain an explicit expression for $m_{\phi,s}^2$, i.e.

$$m_{\phi,s}^2 = \frac{y^2}{6} T^2 \quad (6.46)$$

In this approximation, the thermal correction, Eq. (6.46), is the same as we would have obtained considering the fermion massless in the Yukawa interaction, Eq. (6.26).

Renormalization procedure

Let's now concentrate on the divergent part, which is in common with the 0-temperature case, so we can compute with standard means of QFT [33] and with dimensional regularization technique.

$$-i\Pi(k^2)_{T=0} = -y^2 \int \frac{d^d p}{(2\pi)^d} \text{Tr} \left(\frac{i}{\not{p} - \not{k} - m} \frac{i}{\not{p} - m} \right) \quad (6.47)$$

Here, the overall minus sign is there because of the closed fermion loop, we can rewrite the fermion propagators and combine them by Feynman parametrization, obtaining

$$-i\Pi(k^2)_{T=0} = y^2 \int \frac{d^d p}{(2\pi)^d} \int_0^1 dz \frac{\text{Tr}[(\not{p} + \not{k} + m)(\not{p} + m)]}{(p + zk)^2 + z(1-z)k^2 - m^2} \quad (6.48)$$

Developing the trace at the numerator, keeping only terms quadratic and zeroth order in p (since the integration volume $d^d p$ and the denominator are even in p , the odd powers in p in the numerator yield vanishing integrals), and performing the shift $p \rightarrow p - zk$ we have

$$-i\Pi(k^2)_{T=0} = 4y^2 \int \frac{d^d p}{(2\pi)^d} \int_0^1 dz \frac{p^2 - z(1-z)k^2 + m^2}{(p + zk)^2 + z(1-z)k^2 - m^2} \quad (6.49)$$

By doing the d -integral in p and by defining $\epsilon = 2 - d/2$, we have

$$\begin{aligned} -i\Pi(k^2)_{T=0} &= 4y^2 \int_0^1 dz \left[\frac{-i\Gamma(-1+\epsilon)}{(4\pi)^{2-\epsilon}} [m^2 - z(1-z)k^2]^{1-\epsilon} \right. \\ &\quad \left. - 2z(1-z)k^2 \frac{i\Gamma(\epsilon)}{(4\pi)^{2-\epsilon}} [m^2 - z(1-z)k^2]^{-\epsilon} \right] \end{aligned} \quad (6.50)$$

Let's now introduce a counterterm lagrangian and a renormalization prescription

$$\mathcal{L}_{\text{ct}} = \frac{1}{2} \delta_{Z_\phi} (\partial_\mu \phi)^2 - \frac{1}{2} \delta_{m_\phi^2} \phi^2 + \delta_{Z_\psi} \bar{\psi} i \not{\partial} \psi - \delta_m \bar{\psi} \psi + \delta_y \phi \bar{\psi} \psi \quad (6.51)$$

Considering both the tree-level lagrangian and the counterterm one, Eqs. (6.26) and (6.51), the two-point (connected) function is

$$i\Gamma(k^2) = i(k^2 - m_\phi^2) - i\Pi(k^2)_{T=0} + (i\delta_{Z_\phi} k^2 - \delta_{m_\phi^2}) \quad (6.52)$$

With the zero-momentum subtraction scheme, we require

$$\begin{aligned} \Gamma(0) &= -m_\phi^2 \\ \left. \frac{d\Gamma}{dk^2} \right|_{k^2=0} &= 1 \end{aligned} \quad (6.53)$$

which translates in the following conditions

$$\begin{aligned} \Pi_{T=0}(0) + \delta_{m_\phi^2} &= 0 \\ \left. \frac{d\Pi_{T=0}}{dk^2} \right|_{k^2=0} - \delta_{Z_\phi} &= 0 \end{aligned} \quad (6.54)$$

If we apply conditions of Eq. (6.54) to the one-loop expression of the scalar two-point function, Eq. (6.50), we can obtain the explicit form for the mass and kinetic counterterms

$$\delta_{m_\phi^2} = 4y^2 \int_0^1 dz \frac{-i\Gamma(-1+\epsilon)}{(4\pi)^{2-\epsilon}} (m^2)^{1-\epsilon} \quad (6.55)$$

$$\delta_{Z_\phi} = 4y^2 \int_0^1 dz \, 3z(1-z) \frac{-i\Gamma(\epsilon)}{(4\pi)^{2-\epsilon}} (m^2)^{1-\epsilon} \quad (6.56)$$

In this renormalization scheme, the counterterms subtract to $\Pi(k^2)_{T=0}$ expression, i.e. Eq. (6.50), the integrand at $k^2 = 0$ and the piece at first order in k^2 . By expanding the Euler function and logarithmic terms, and then coming back to 4 dimensions sending $\epsilon \rightarrow 0$, we can rewrite the full two-point function, Eq. (6.52), showing that is manifestly finite.

$$\begin{aligned} i\Gamma(k^2) &= i(k^2 - m_\phi^2) + 4y^2 \frac{i}{(4\pi)^2} \int_0^1 dz \left[[-m^2 + 3z(1-z)k^2] \ln \left(\frac{m^2 - z(1-z)k^2}{m^2} \right) - z(1-z)k^2 \right] = \\ &= i(k^2 - m_\phi^2) + i \frac{y^2}{4\pi^2} \left(k^2 - k\sqrt{4m^2 - k^2} \arctan \left(\frac{k}{\sqrt{4m^2 - k^2}} \right) \right) \end{aligned} \quad (6.57)$$

The location of the pole mass $m_{\phi,\text{pole}}^2 = m_\phi^2 + \delta m_{\phi,T=0}^2$ can be determined by imposing $\Gamma(m_{\phi,\text{pole}}^2) = 0$, from this we can obtain the zero-temperature correction as

$$\delta m_{\phi,T=0}^2 = -\frac{y^2}{4\pi^2} \int_0^1 dz \left[[-m^2 + 3z(1-z)m_\phi^2] \ln \left(\frac{m^2 - z(1-z)m_\phi^2}{m^2} \right) - z(1-z)m_\phi^2 \right] \quad (6.58)$$

In the integral, we have substituted only m_ϕ^2 to k^2 , neglecting the $\delta m_{\phi,T=0}^2$ because it would be a term at $\mathcal{O}(y^4)$ order. So, if we perform the integral in z , we have

$$\delta m_{\phi,T=0}^2 = \frac{y^2}{4\pi^2} \left(m_\phi^2 - m_\phi \sqrt{4m^2 - m_\phi^2} \arctan \left(\frac{m_\phi}{\sqrt{4m^2 - m_\phi^2}} \right) \right) \quad (6.59)$$

The residue at the pole is $Z_B \equiv \left(d\Pi_{T=0}(m_{\phi,\text{pole}}^2)/dk^2 \right)^{-1} \neq 1$, so any N -point function needs to be multiplied by $Z_B^{-N/2}$ to work out the S -matrix elements according to the LSZ reduction formulae.

$$\begin{aligned} Z_B^{-1} &= \frac{d\Gamma}{dk^2} \Big|_{k^2=m_{\phi,\text{pole}}^2} = 1 + \frac{y^2}{4\pi^2} \int_0^1 dz \, z(1-z) \left[3 \ln \left(\frac{m^2 - z(1-z)m_\phi^2}{m^2} \right) + \frac{m^2 - 3z(1-z)m_\phi^2}{m^2 - z(1-z)m_\phi^2} - 1 \right] \\ &= 1 + \frac{y^2}{4\pi^2} \left[\sqrt{4m^2 - m_\phi^2} \left(\frac{2m^2 - m_\phi^2}{4m_\phi m^2 - m_\phi^3} \right) \arctan \left(\frac{m_\phi}{\sqrt{4m^2 - m_\phi^2}} \right) - \frac{1}{2} \right] \end{aligned} \quad (6.60)$$

The renormalized lagrangian, which is the sum of the tree-level lagrangian plus the counterterm one, respectively Eqs. (6.26) and (6.51), can be re-written in terms of renormalized fields⁶

⁶For the masses we omit the subscript R to point that are renormalized quantities.

$$\begin{aligned}
\mathcal{L}_R &= -\frac{1}{2}\phi_B(\partial^2 + m_{\phi,B}^2)\phi_B + \bar{\psi}_B(i\cancel{\partial} - m_B)\psi_B + y_B \phi_B \bar{\psi}_B \psi_B \\
&= -\frac{1}{2} Z_\phi \phi_R(\partial^2 + Z_{m_\phi} m_\phi^2)\phi_R + Z_\psi \bar{\psi}_R(i\cancel{\partial} - Z_m m)\psi_R + Z_y Z_\psi Z_\phi^{\frac{1}{2}} y_R \phi_R \bar{\psi}_R \psi_R =
\end{aligned} \tag{6.61}$$

The renormalization factors of kinetic terms are defined, as always, $Z_i = 1 + \delta_i$, while for the mass ones, we have $Z_{m_\phi} Z_\phi = 1 + (\delta_{m_\phi^2}/m_\phi^2)$ and $Z_m Z_\psi = 1 + (\delta_m/m)$. The second equality of Eq. (6.61) is satisfied by expressing the bare fields as a function of renormalized ones, in particular for the scalar field we have $\phi_B = Z_\phi^{\frac{1}{2}} \phi_R$, while for the fermion field $\psi_B = Z_\psi^{\frac{1}{2}} \psi_R$. The vertex renormalization factor Z_y can be obtained by computing the one-loop vertex correction, i.e. the fermion-fermion-scalar three-point function, but, since this is not crucial for our purposes, we will skip its derivation.

If we want to rewrite the renormalized and the bare two-point connected function, by explicating the physical quantities, we obtain

$$\begin{aligned}
\Gamma^{\text{Ren}} &= Z_\phi(k^2 - Z_{m_\phi} m_\phi^2) - \Pi_{T=0} \\
\Gamma^{\text{bare}} &= Z_B^{-1}(k^2 - m_\phi^2 - \delta m_{\phi,T=0}^2)
\end{aligned} \tag{6.62}$$

and by substituting $k^2 = m_{\phi,\text{pole}}^2$ in Eq. (6.52) we obtain the relation between the mass correction and the renormalized parameters of the lagrangian

$$\begin{aligned}
\delta m_{\phi,T=0}^2 &= Z_\phi^{-1} \Pi_{T=0}(m_\phi^2) + Z_{m_\phi} m_\phi^2 - m_\phi^2 \\
Z_B^{-1} &= Z_\phi - \left. \frac{d\Pi_{T=0}}{dk^2} \right|_{k^2=m_\phi^2}
\end{aligned} \tag{6.63}$$

So the effective boson mass, Eq. (6.41), summing up all the pieces (till $\mathcal{O}(y^2)$), takes the following form:

$$\begin{aligned}
m_{\phi,\text{eff}}^2 &= (m_\phi^2 + \delta m_{\phi,T=0}^2) + m_{\phi,s}^2 = \\
m_\phi^2 + \frac{y^2}{4\pi^2} &\left(m_\phi^2 - m_\phi \sqrt{4m^2 - m_\phi^2} \arctan \left(\frac{m_\phi}{\sqrt{4m^2 - m_\phi^2}} \right) \right) + \frac{y^2}{6} T^2
\end{aligned} \tag{6.64}$$

Now that we have completed the calculation of the boson effective mass, we can insert this result into the spectral value expression to obtain the final result for the dissipation coefficient. We can obtain the dissipation coefficient, for the Yukawa interaction, by substituting the expression of the spectral function, Eq. (6.37), in Eq. (6.27). So we have

$$\Upsilon_Y = \frac{y^2}{8\pi} \frac{(m_{\phi,\text{eff}}^2 - 4m^2)^{\frac{3}{2}}}{m_{\phi,\text{eff}}^2} \left(\frac{e^{m_{\phi,\text{eff}}/2T} - 1}{e^{m_{\phi,\text{eff}}/2T} + 1} \right) \tag{6.65}$$

If we consider the limit where $yT \gg m_\phi, m$, the effective boson mass is dominated by its thermal part, i.e. $y^2 T^2/6$. So, the dissipation coefficient takes the form

$$\Upsilon_Y \xrightarrow{yT \gg m_\phi, m} \frac{y^3 T}{8\sqrt{6}\pi} \frac{e^{y/2\sqrt{6}} - 1}{e^{y/2\sqrt{6}} + 1} \sim y^3 T \quad (6.66)$$

Let's consider, in general, this theory as an effective field theory valid until a specific energy scale M_{UV} . Physically speaking, this scale can be the typical one of a new symmetry that will emerge at higher energies and could protect the mass of the scalar ϕ . Let's now assume that the quantum correction to the scalar mass (which, in the EFT, is the M_{UV} term) dominates the scalar mass, so we can write

$$m_\phi^2 \sim y^2 \frac{M_{UV}^2}{16\pi^2} \quad (6.67)$$

This fact implies the condition $M_{UV} \gg T$. In this regime, we can expand the dissipation coefficient Υ_Y , i.e. Eq. (6.65), as

$$\Upsilon_Y \xrightarrow{M_{UV} \gg T} \frac{y^2}{8\pi} m_\phi \quad (6.68)$$

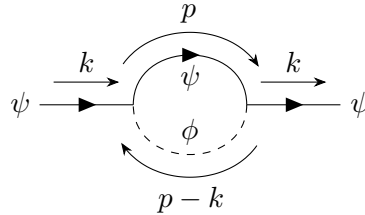
Fermion self-energy

To have a complete discussion of the topic, we have also to check that the corrections (thermal and not) to the fermion mass are not dominant and do not affect the results for the dissipation coefficient, i.e. Eq. (6.65). Since we have done already the full computation for the boson self-energy, in this case, we outline only the main steps to follow and discuss the results. To do so, we can start writing the correlator of a bosonic propagator and a fermionic one

$$\Sigma(\omega_a, \vec{k}) = -y^2 \mathcal{T} \sum_{b=-\infty}^{\infty} \int \frac{d^3 p}{(2\pi)^3} S(\omega_b, \vec{p}) D(\omega_b - \omega_a, \vec{p} - \vec{k}) \quad (6.69)$$

where both ω_a and ω_b are in the fermionic form of Eq. (6.10) and we have defined the bosonic propagator as in Eq. (6.13)

Its relative Feynman diagram is



By performing the fermionic Matsubara sum, we can rewrite the fermion self-energy, Eq. (6.69), as

$$\begin{aligned} \Sigma(\omega_a, \vec{k}) = & -y^2 \int \frac{d^3 p}{(2\pi)^3} \frac{1}{4E_1 E_2} \left[(1 - n_{F1} + n_{B2}) \left(\frac{\not{p}_1 + m}{\omega_a - E_1 - E_2} + \frac{\not{p}_1 - m}{\omega_a + E_1 + E_2} \right) \right. \\ & \left. + (n_{F1} + n_{B2}) \left(\frac{\not{p}_1 - m}{\omega_a + E_1 - E_2} + \frac{\not{p}_1 + m}{\omega_a - E_1 + E_2} \right) \right] \quad (6.70) \end{aligned}$$

where $p_1^0 = E_1 > 0$, but now $E_1 = \sqrt{m^2 + \vec{p}^2}$ and $E_2 = \sqrt{m_\phi^2 + (\vec{p} - \vec{k})^2}$.

Now, by following the steps done for Π_E and taking the imaginary part of Eq. (6.70), we obtain

$$\begin{aligned} \text{Im } \Sigma(\omega, \vec{k}) = \frac{1}{2i} \text{Disc } \Sigma(\omega, \vec{k}) = & -\frac{y^2}{2} \int \frac{d^3 p}{(2\pi)^3} \frac{2\pi}{4E_1 E_2} \left[\delta(\omega - E_1 - E_2)(\not{p}_1 + m)(1 - n_{F1} + n_{B2}) \right. \\ & + \delta(\omega + E_1 - E_2)(\not{p}_1 - m)(n_{F1} + n_{B2}) \\ & + \delta(\omega - E_1 + E_2)(\not{p}_1 + m)(n_{F1} + n_{B2}) \\ & \left. + \delta(\omega + E_1 + E_2)(\not{p}_1 - m)(1 - n_{F1} + n_{B2}) \right] \end{aligned} \quad (6.71)$$

Also, in this case, we can split the mass correction as a thermal part plus the zero-temperature correction. So the effective fermion mass can be written as

$$m_{\text{eff}} = (m + \delta m_{T=0}) + m_s \quad (6.72)$$

where $\delta m_{T=0}$ can be determined with standard dimensional regularization, as done for the boson self-energy at $T = 0$, and takes the following form

$$\delta m_{T=0} = \frac{y^2}{(4\pi)^2} m \int_0^1 dz (2 - z) \ln \left(\frac{(1 - z)m_\phi^2 + z^2 m^2}{(1 - z)m_\phi^2 + z m^2} \right) \quad (6.73)$$

In general, to compute the thermal fermion mass we can use the following equations, which are analogous to Eqs. (6.44) and (6.39) but for the fermion.

$$\Sigma(0, 0) - \Sigma_{T=0}(0, 0) = -m_s \quad (6.74)$$

$$m_s = \int_{-\infty}^{\infty} \frac{d\omega}{\pi} \frac{\text{Im } \Sigma(\omega, 0)_{T \neq 0}}{\omega} \quad (6.75)$$

In the case where temperature dominates, we can find the result in a more straightforward way. In fact, if we assume $T \gg m_\phi, m$, we note that we will re-acquire the result of a massless Yukawa theory. To obtain the massless result we can expand for hard loop momenta Eq. (6.70), once we have set $m = 0$. So the fermion self-energy takes the form[34]

$$\Sigma(\omega, \vec{k}) = i \frac{m_s^2}{k} \gamma_0 Q_0 \left(\frac{\omega}{k} \right) + \frac{m_s^2}{k} \gamma_i \hat{k}_i \left[1 - Q_0 \left(\frac{\omega}{k} \right) \right] \quad (6.76)$$

where the thermal fermion squared mass is

$$m_s^2 = \frac{y^2 T^2}{16} \quad (6.77)$$

and $Q_0(x)$ is the Legendre function of the second kind, defined as

$$Q_0(x) = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right) \quad (6.78)$$

6.2.2 Validity of the perturbation expansion

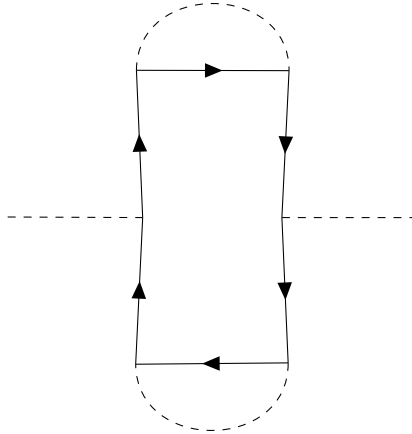
This section is inspired by [35, 36].

Until now, we have used the imaginary-time formalism to compute temperature-dependent quantities; and as we have said, this formalism keeps the diagrammatic perturbative expansion used in standard QFT. But, in general, the usual perturbative expansion in powers of the coupling constant can fail from temperatures beyond a critical one. It has to be replaced by an improved perturbative expansion where an infinite number of diagrams are resummed at each order in the new expansion: so, in this section, we try to understand when the ordinary one-loop approximation needs to be improved. Furthermore, for the improved perturbation theory to hold, we have to check that the higher loop resummed diagrams give subleading contributions. To do this, we have to estimate, by power counting arguments, the behavior of resummed loop diagrams, allowed by Yukawa theory, and see which approximation works in our case.

Before starting the discussion on the perturbativity, let us remember that the boson/fermion self-energy diagrams go as $y^2 T^2$, so the thermal masses for boson and fermion go as $m_s \sim yT$ and $m_{\phi,s}^2 \sim y^2 T^2$, as shown respectively in Eqs. (6.46) and (6.75).

For the boson

The one-loop daisy diagram is the boson self-energy diagram with attached all along n fermion self-energy diagrams. The following diagram represents an example of this type of diagram, one with $n = 2$ attached fermion self-energies



To estimate this diagram, we consider only the leading contribution for each order. Practically, for each loop, we insert the term⁷ $T \int k^2 dk$, for the fermion propagators, we have a factor k^{-1} , while for the boson ones k^{-2} . Moreover, we use the temperature T as the UV cutoff, while for the IR cutoff the mass M . We stress the fact that the mass M has a different meaning for the improved and the unimproved theories. More precisely, for the unimproved theory, M is the mass in

⁷This term accounts for the momentum integration plus the Matsubara sum. Furthermore, we note that in each integration we put k , because in this first approximation, we are assuming a unique scale

the lagrangian, while in the improved theory, M is given by the thermal mass. We will refer to the boson case with M_ϕ while we will use M_ψ for the fermion. So, the daisy diagram for the boson can be evaluated as

$$\begin{aligned}
 \text{daisy boson diagram} &\sim (y^2 T)^{1+n} \sum_{n=1}^{\infty} \int \left(\frac{1}{k^2}\right)^n \left(\frac{(k^2 dk)}{k^2}\right)^{n+1} \\
 &\sim y^2 T^2 \left(y^2 \ln \left(\frac{T}{M_\phi}\right) + \sum_{n=2}^{\infty} (y)^{2n} \left(\frac{T}{M_\phi}\right)^{n-1} \right) \\
 &\sim y^2 T^2 \left(y^2 \ln \left(\frac{T}{M_\phi}\right) + y^2 \sum_{n=2}^{\infty} \left(y^2 \frac{T}{M_\phi}\right)^{n-1} \right) \tag{6.79}
 \end{aligned}$$

Where we have brought out the factor $y^2 T^2$, which is the behavior of the boson self-energy, to highlight the expansion parameter. We can define it as $\alpha_\phi = y^2 \frac{T}{M_\phi}$, and for the 1-loop approximation to be valid we have to require that

$$y^2 \frac{T}{m_\phi} \ll 1 \tag{6.80}$$

along the standard requirement for ordinary perturbation expansion, i.e. $y^2 \ll 1$. If condition (6.80) is satisfied, also the first term in the expansion of Eq. (6.79) becomes negligible, since $\ln x < x$. And so, every term turns out to be $\ll 1$, i.e. we can stop the resummation at the 1-loop boson self-energy.

We note that if condition (6.80) is not satisfied, we have to use the improved expansion (also called daisy-resummation). But to check its validity, we have to understand if the contribution of the two-loop non-daisy diagram is subleading to the daisy diagram one.

The two-loop diagrams are those in the figure below

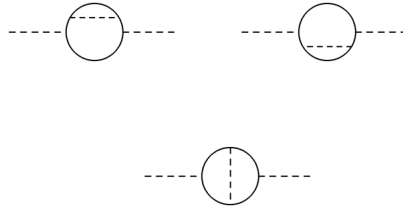


Figure 6.3. Two-loop self-energy diagrams from Yukawa theory. Figure taken from [34]

We observe that (excluding the third one since it vanishes in dimensional regularization[34]) the other two diagrams are topological equivalent to the one-loop diagram with attached 1-fermion self-energy. If we proceed to attach other fermion self-energies to these diagrams, we would realize that their behavior is the same as the daisy one, Eq. (6.79), but without the first logarithmic term. So we can conclude that, in the boson case, the daisy diagram resummed all the contributions without the necessity of other conditions.

It is important to stress that, to remain in the one-loop perturbative approximation in the expansion which leads to Eq. (6.66), the condition (6.80) has to be

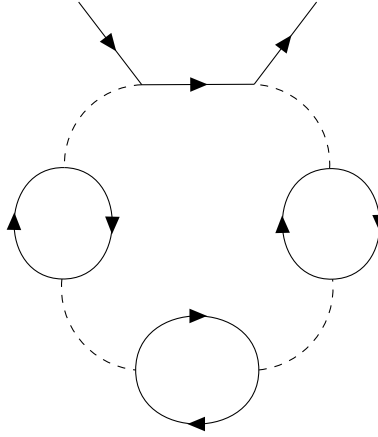
combined with $y^2 T^2 \gg m_\phi^2$ giving

$$y \ll \frac{m_\phi}{yT} \ll 1 \quad (6.81)$$

For the fermion

We can repeat the process for the fermion and the general considerations, made before, continue to hold also in the fermion case.

The one-loop daisy diagram is the fermion self-energy diagram with all along the boson propagator n fermionic loops (i.e. boson self-energy diagrams). We can see an example of such a diagram with $n = 3$



If we want to evaluate the fermion daisy diagram, we have

$$\begin{aligned} \text{daisy fermion diagram} &\sim (y^2 T)^{1+n} \sum_{n=1}^{\infty} \int \left(\frac{1}{k}\right)^{1+2n} \left(\frac{(k^2 dk)}{k^2}\right)^{n+1} \\ &\sim y^2 T \sum_{n=1}^{\infty} \left(y^2 \frac{T}{M_\psi}\right)^n \end{aligned} \quad (6.82)$$

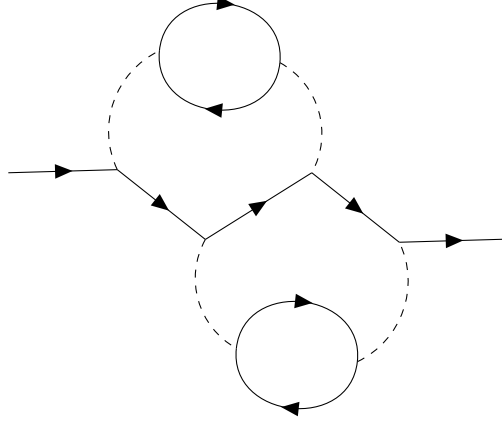
Analogously, we define the expansion parameter as $\alpha_\psi = y^2 \frac{T}{M_\psi}$, and so we require

$$y^2 \frac{T}{m} \ll 1 \quad (6.83)$$

As for the boson case, to remain in the one-loop perturbative approximation in the expansion which leads to Eq. (6.66), the condition (6.83) has to be combined with $y^2 T^2 \gg m^2$ giving

$$y \ll \frac{m}{yT} \ll 1 \quad (6.84)$$

An example of a two-loop non-daisy diagram (with $n = 2$) is the diagram represented below, in general, they are two-loop fermionic diagrams where there are n fermionic loops all along the bosons propagator.



It can be evaluated as

$$\begin{aligned}
 \text{non-daisy fermion diagram} &\sim (y^2 T)^{2+n} \sum_{n=1}^{\infty} \int \left(\frac{1}{k}\right)^{3+2n} \left(\frac{k^2 dk}{k^2}\right)^{n+2} \\
 &\sim y^4 \frac{T^2}{M_\psi} \sum_{n=1}^{\infty} \left(y^2 \frac{T}{M_\psi}\right)^n
 \end{aligned} \tag{6.85}$$

We can see that the non-daisy fermion diagram is suppressed with respect to the daisy one by a factor $\beta_\psi = y^2 \frac{T}{M_\psi}$. The validity of improved approximation is guaranteed if

$$y^2 \frac{T}{m_s} \ll 1 \tag{6.86}$$

So, in our case $m_s \sim yT$, and we obtain $\beta_\psi \sim \frac{y^2 T}{yT} = y$. So the improved perturbation theory is sufficient for the Yukawa theory because the non-daisy diagrams are subleading with respect to the daisy diagrams, and so they can be neglected in the resummation.

6.2.3 Inflation model with Yukawa interaction

In this section, we take a model with a mass term plus constant potential, namely

$$V = \frac{1}{2} m_\phi^2 \phi^2 + \Lambda^4 \tag{6.87}$$

It has a form analogous to Eq. (4.4), but, in this case, we add to the inflaton lagrangian the Yukawa interaction. In this way, we obtain a warm inflation model where the constant part of the potential V dominates over the mass term ($\Lambda^4 \gg \frac{1}{2} m_\phi^2 \phi^2$), and where the dissipation coefficient is the one computed in section 6.2, i.e. Eq. (6.68). We want to study this inflation model in the warm case in the strong regime, $Q \gg 1$, but also in the absence of the thermal fermions bath (i.e. without Yukawa interaction).

We require that in the potential V , the constant term, Λ^4 , dominates over the mass term (if this did not happen the constant would not play any role), this translates into the following condition

$$\frac{m_\phi^2 \phi^2}{\Lambda^4} \ll 1 \quad (6.88)$$

For this potential, we want to compute the slow-roll parameters in the case of both standard inflation and warm inflation.

In the standard inflation case, we have

$$\epsilon_V = \frac{M_{\text{Pl}}^2}{2} \frac{m_\phi^4 \phi^2}{\Lambda^8} \quad (6.89)$$

$$\eta_V = M_{\text{Pl}}^2 \frac{m_\phi^2}{\Lambda^4} \quad (6.90)$$

if we require the slow-roll conditions, we have

$$\begin{aligned} \epsilon_V < 1 &\longrightarrow \phi < \sqrt{2} \frac{\Lambda^4}{m_\phi^2 M_{\text{Pl}}} \\ \eta_V < 1 &\longrightarrow M_{\text{Pl}}^2 \frac{m_\phi^2}{\Lambda^4} < 1 \end{aligned} \quad (6.91)$$

If we compute the dissipation ratio for this model, by assuming Eq. (6.67) valid (while it was not in the standard case) and so a dissipation coefficient like Eq. (6.68), we obtain

$$Q_Y = \frac{\Upsilon_Y}{3H} = \frac{1}{\sqrt{3}} \frac{y^2 m_\phi M_{\text{Pl}}}{\Lambda^2} \quad (6.92)$$

If we compute the slow-roll parameters in the warm case, we have

$$\epsilon_V^{\text{wi}} \simeq \frac{1}{Q_Y} M_{\text{Pl}}^2 \frac{m_\phi^4 \phi^2}{\Lambda^8} \ll \frac{M_{\text{Pl}}}{y M_{\text{UV}}} \quad (6.93)$$

where the $\ll$ symbol holds after we have used Eq. (6.88)

$$\eta_V^{\text{wi}} \simeq \frac{1}{Q_Y} \frac{m_\phi^2 M_{\text{Pl}}^2}{\Lambda^4} \simeq \frac{M_{\text{Pl}}}{y M_{\text{UV}}} \quad (6.94)$$

where in the both expressions we have used that $m_\phi^2 \sim y^2 M_{\text{UV}}^2 / 16\pi^2$, as discussed at the end of section 6.2, and taken $\Lambda^4 \sim M_{\text{UV}}^4 / (16\pi^2)$ at its natural value. We see immediately that inflation cannot be sustained long enough, in fact, for $M_{\text{UV}} < M_{\text{Pl}}$, $\eta_V^{\text{wi}} > 1$ in the perturbative regime (where $y \ll 1$).

Chapter 7

Conclusions

In this Masters thesis we have assessed the compatibility between selected models of inflation and the requirement of 1) No trans-Planckian field excursions 2) No flat potentials ($V'/V > 1/M_{\text{Pl}}$). These requirements are inspired by String Theory, but when satisfied also simplify bottom-up model building. We then studied in which cases one can satisfy 1) and 2) and still have enough e -folds of inflation to generate a Multiverse.

More specifically, the principal results are the study of the hybrid inflation model, in the standard inflation picture, which turned out to be unable to satisfy both bounds arising from String Theory at the same time. So we have moved on to study the Minimal Warm Inflation model [3], which has the same form for the potential as the hybrid inflation model, but is developed in the warm inflation picture, where the thermal bath is generated by giving the inflaton an axion-like gauge coupling. For this model, we have done the computation of the maximum number of e -folds, once the bounds arising from the Swampland Distance conjecture and the Swampland de Sitter conjecture were imposed. The idea behind this calculation is to see if this specific model, after the imposition of compatibility conditions with String Theory, supports the Multiverse idea. In fact, if the number of e -folds, namely the duration of inflation, is sufficient to allow a great number of vacuum decays, then, each bubble nucleation event will correspond to a different casually disconnected Universe (in turn related to a different vacuum).

Furthermore, we performed the calculation of the bubble nucleation rate, by following the Coleman - De Luccia procedure, for a Mexican hat potential and then obtained the number of e -folds corresponding to 10^{120} events, assuming an exponential expansion background (as in the inflation case). This part is complementary to the section about the e -folds number computation for the inflation model and is oriented toward the understanding of the e -folds number required for the Multiverse realization. In this thesis, we considered the Mexican hat potential as a generator of a spacetime landscape, opening the way, in the future, to repeat the same process for other landscape realizations or inflationary models.

To conclude this reasoning, we have to say that even though the "Minimal Warm inflation" model supports a sufficient number of e -folds to solve the fine-tuning problem for the cosmological constant within the Multiverse framework, the effective Multiverse realization cannot be achieved since the typical Hubble scale of the model, i.e. Eq. (4.19), is too low ($10^{-7} - 10^{-2}$ GeV), if compared to the M_{Pl} scale. This is an issue because we expect that a high-energy theory, like Multiverse, which tries to describe new physics, would become valid at an energy scale beyond the explored ones, such as the Planck scale.

Then, we worked on the calculation of a dissipation coefficient by using thermal

quantum field theory for a Yukawa interaction: this computation has the aim of exploring a warm inflation model where the dissipation coefficient has a behavior proportional to temperature. This can be achieved, as shown in Eq. (6.66), if we assume that the thermal mass correction dominates over the masses of the fermion/scalar. But, in general, in an EFT valid until a certain scale, as M_{UV} , the temperature never dominates, in fact, if $T \gg M_{UV}$ the EFT would lose its validity, otherwise we come back to the case $M_{UV} \gg T$, so the mass of the scalar is dominated by the M_{UV} contribution. It will be left to future work to investigate, in a complete UV theory, such as a supersymmetric model, whether temperature can dominate the masses of the particles involved. If that is the case, it will be possible to construct a warm inflationary model with a Yukawa interaction, and the temperature expanded dissipation coefficient, computed in section 6.2 of this thesis, may prove useful. Finally, we considered an inflation model with a known potential, in both standard and warm pictures. To realize warm inflation we allowed the inflaton to have a Yukawa interaction like the one studied in chapter 6. We computed the slow-roll parameters in both cases and we noted that the addition of the dissipation coefficient Υ_Y (computed in the case when the inflaton mass is dominated by M_{UV} corrections) does not allow us to gain anything in terms of e -folds with respect to the standard inflation case.

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